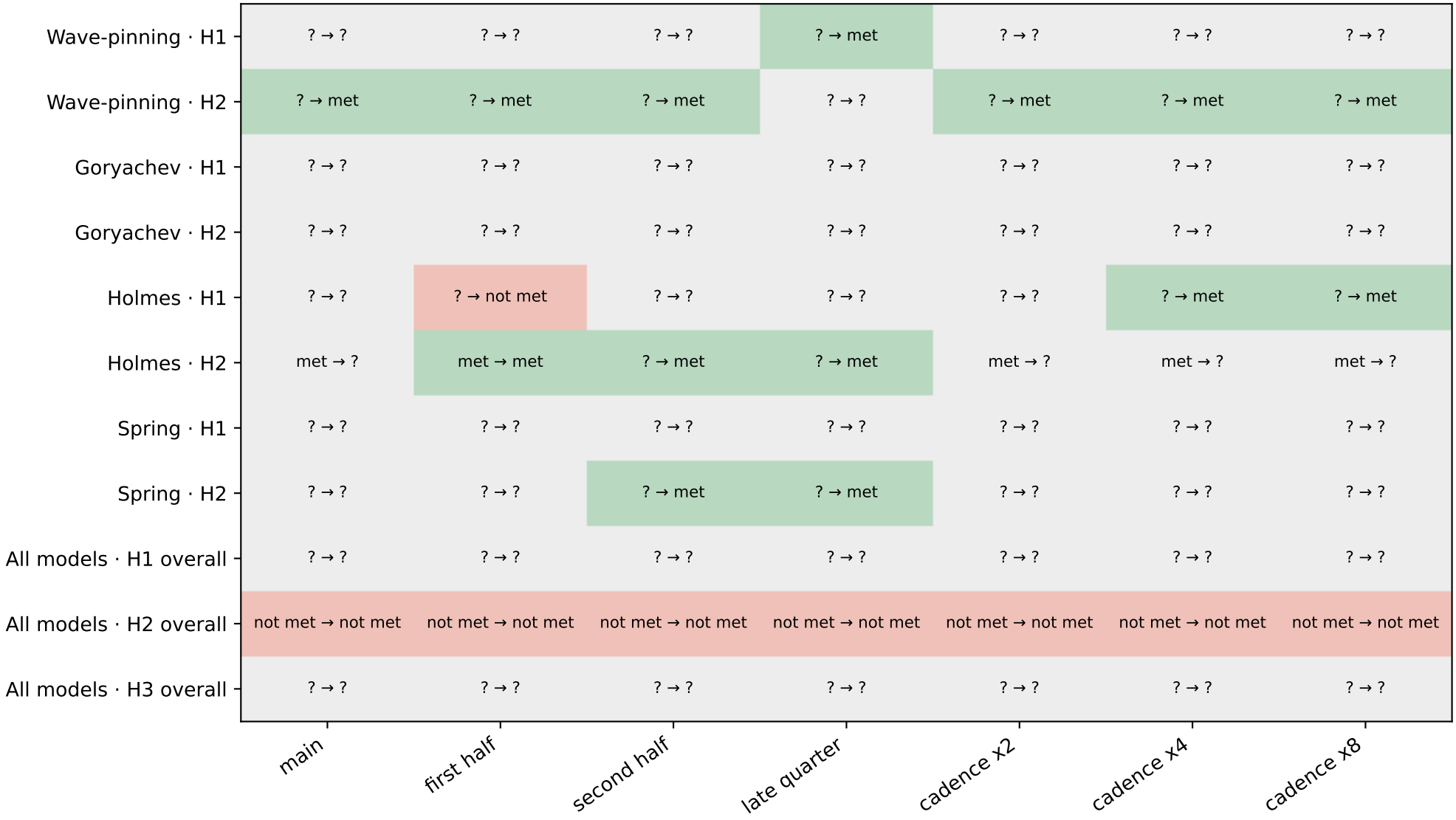
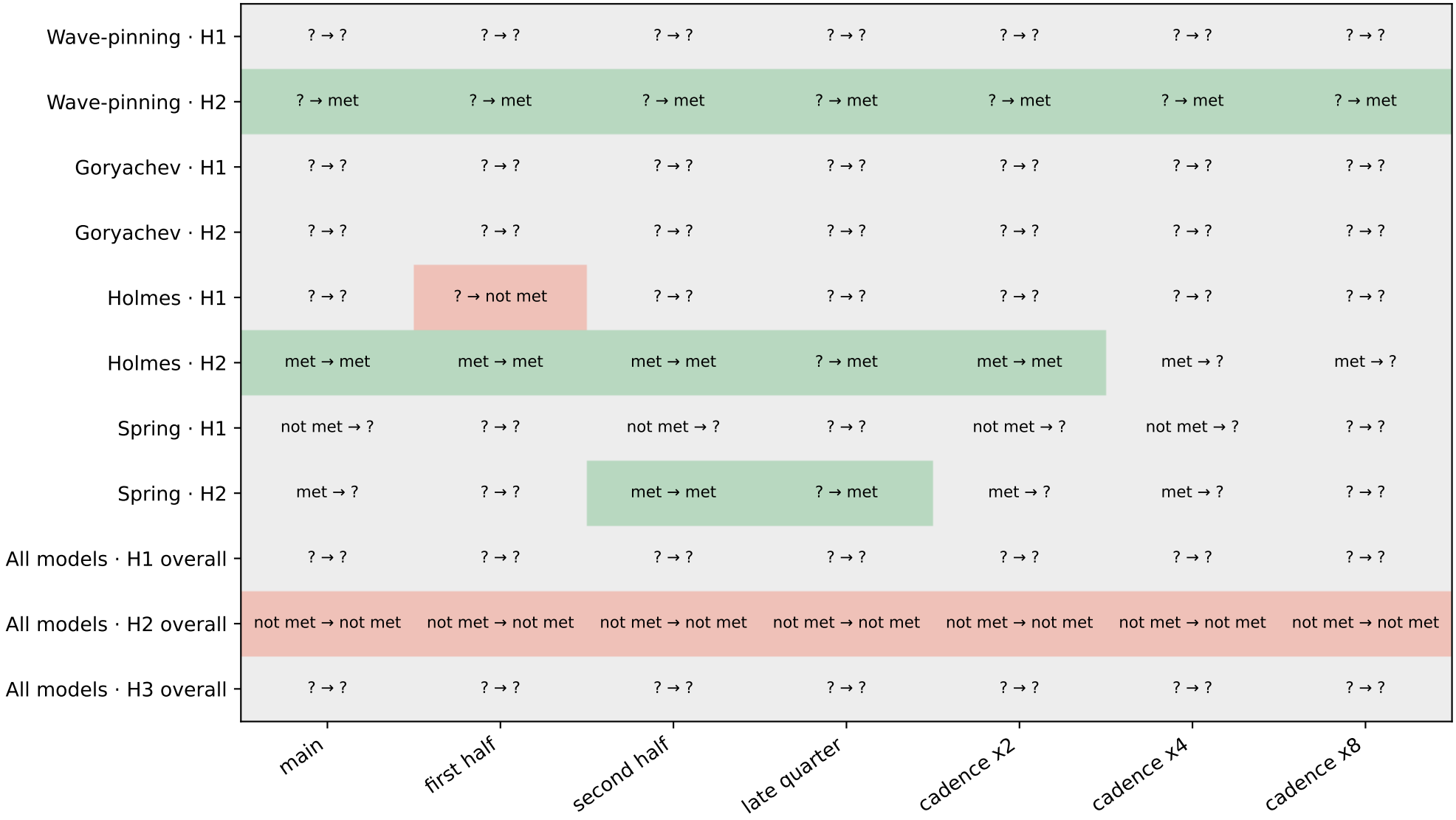
S0 · common · without forced states
Old persistence-time call → new rate/noise-tolerance call



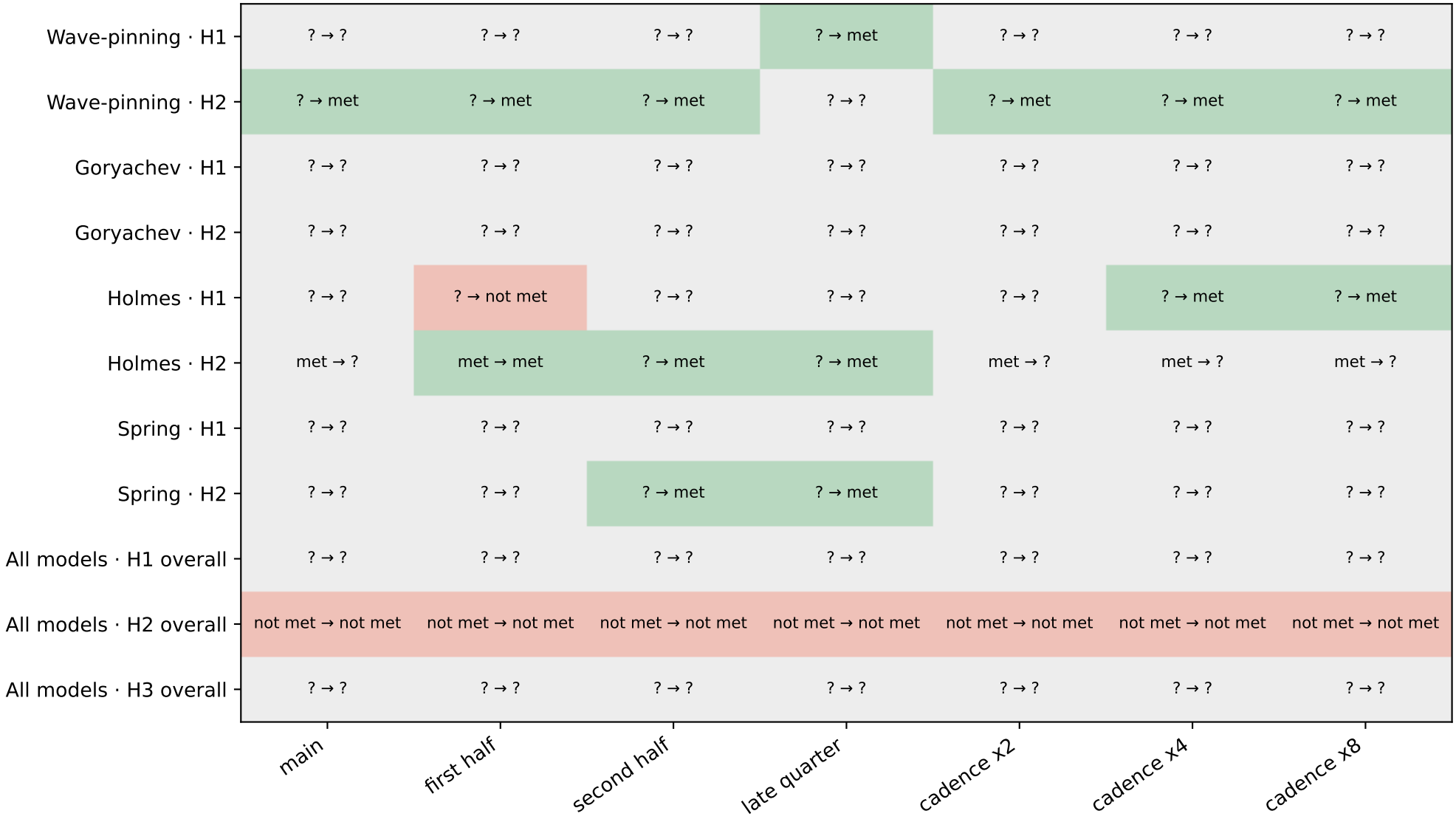
Colour shows new call. met: criterion met; not met: criterion not met; ?: inconclusive.
Frozen criteria text is unchanged. H3 and the A6 predictor gate retain their explicitly direct-τ definitions.
Geometric normalization is a reported outcome, not a gate. Full components and old records are in the CSV.

S0 · common · with forced states
Old persistence-time call → new rate/noise-tolerance call



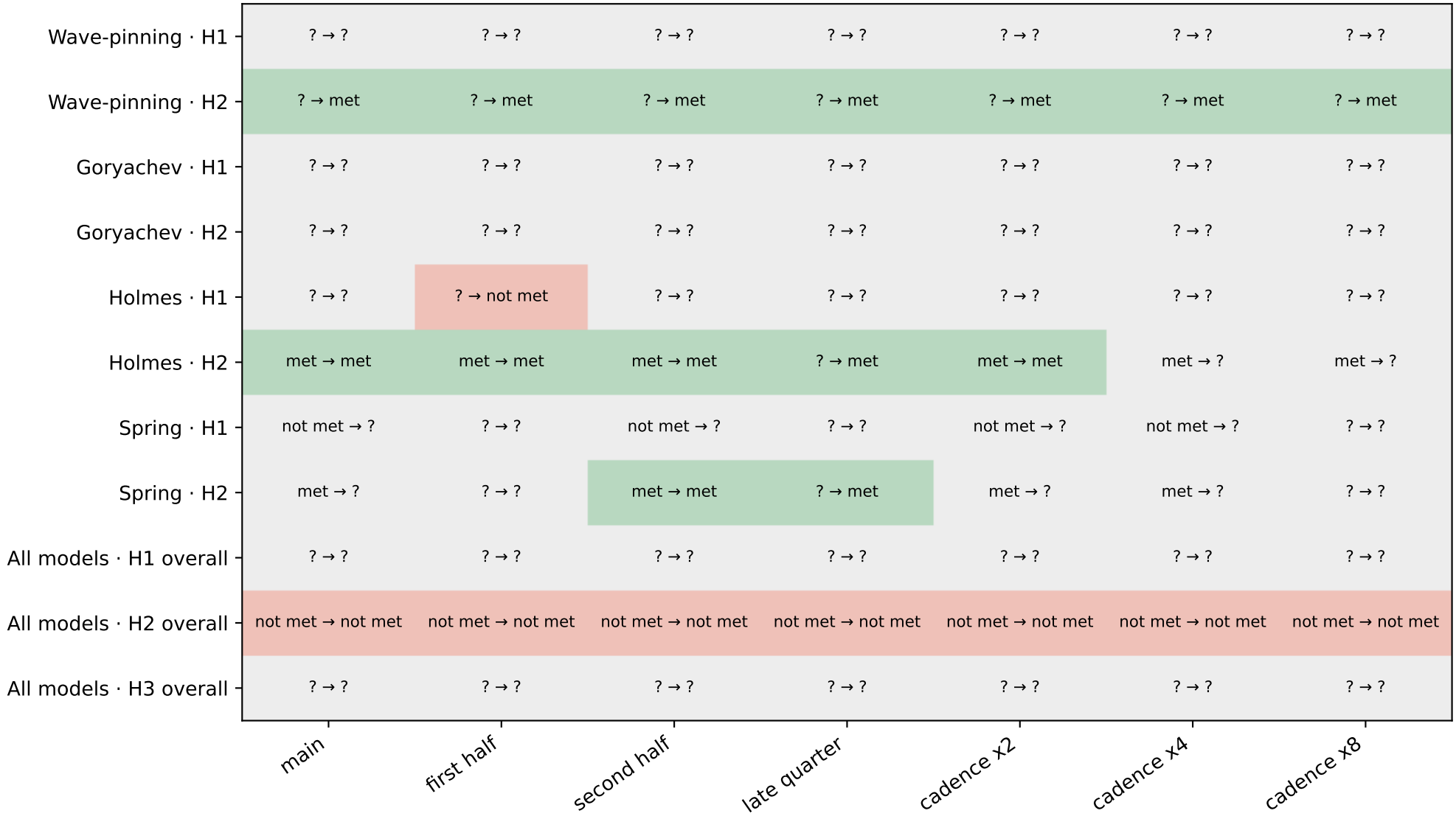
Colour shows new call. met: criterion met; not met: criterion not met; ?: inconclusive.
Frozen criteria text is unchanged. H3 and the A6 predictor gate retain their explicitly direct-τ definitions.
Geometric normalization is a reported outcome, not a gate. Full components and old records are in the CSV.

S0 · longest · without forced states
Old persistence-time call → new rate/noise-tolerance call

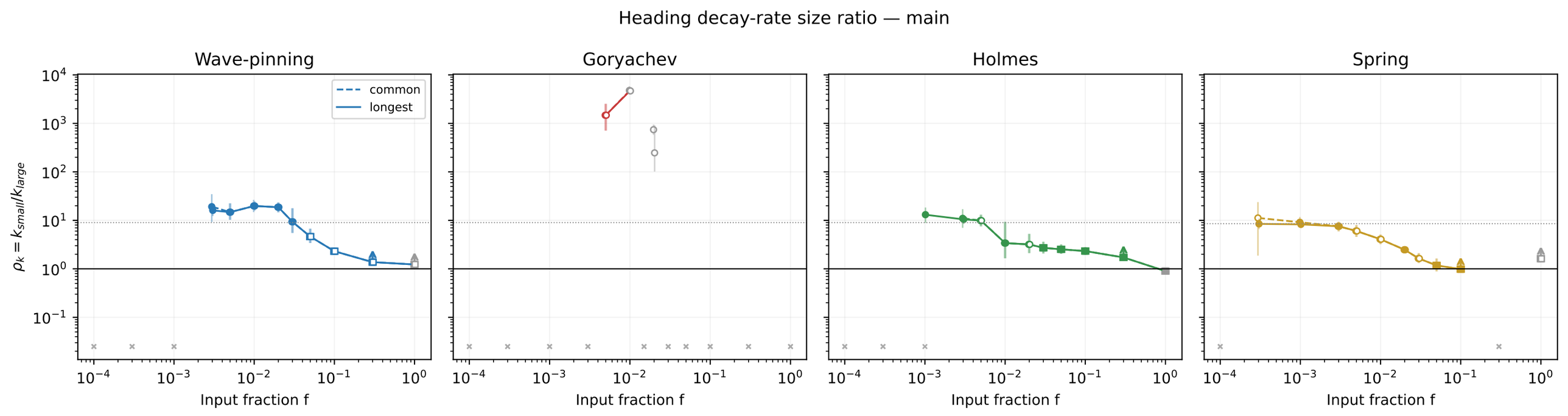


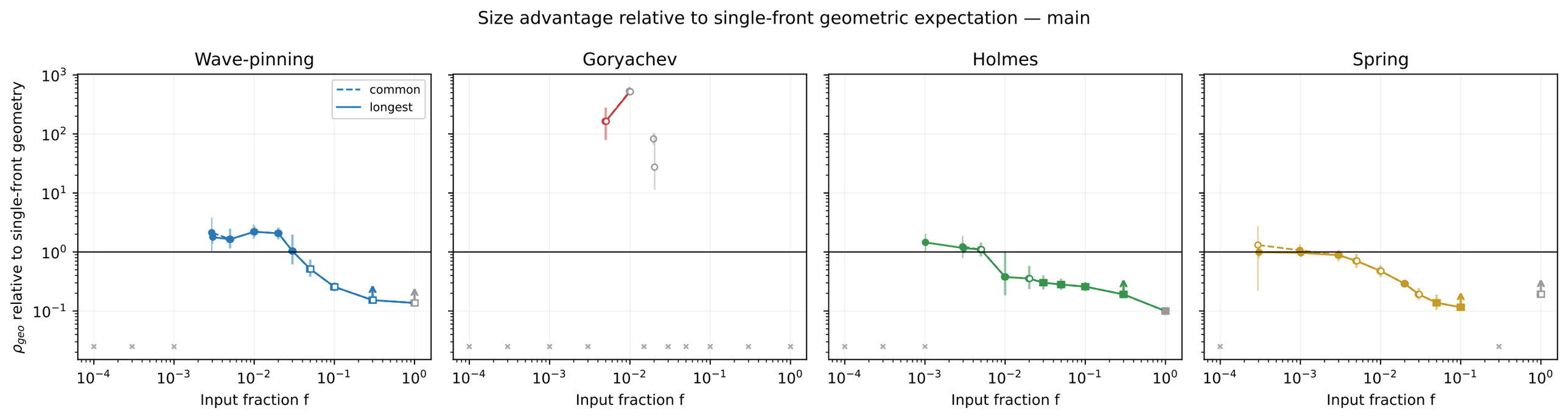
Colour shows new call. met: criterion met; not met: criterion not met; ?: inconclusive.
Frozen criteria text is unchanged. H3 and the A6 predictor gate retain their explicitly direct-τ definitions.
Geometric normalization is a reported outcome, not a gate. Full components and old records are in the CSV.

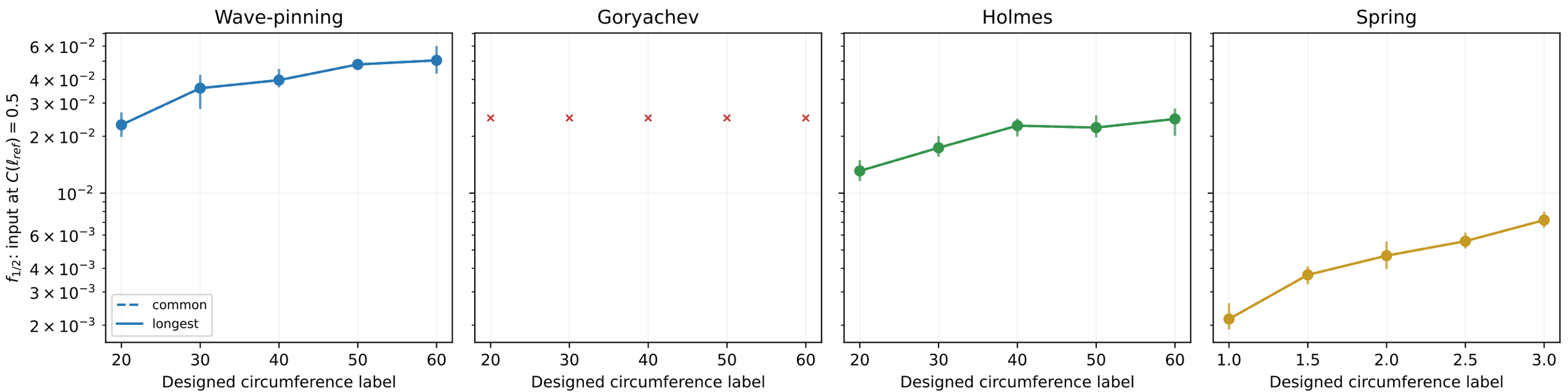
S0 · longest · with forced states
Old persistence-time call → new rate/noise-tolerance call

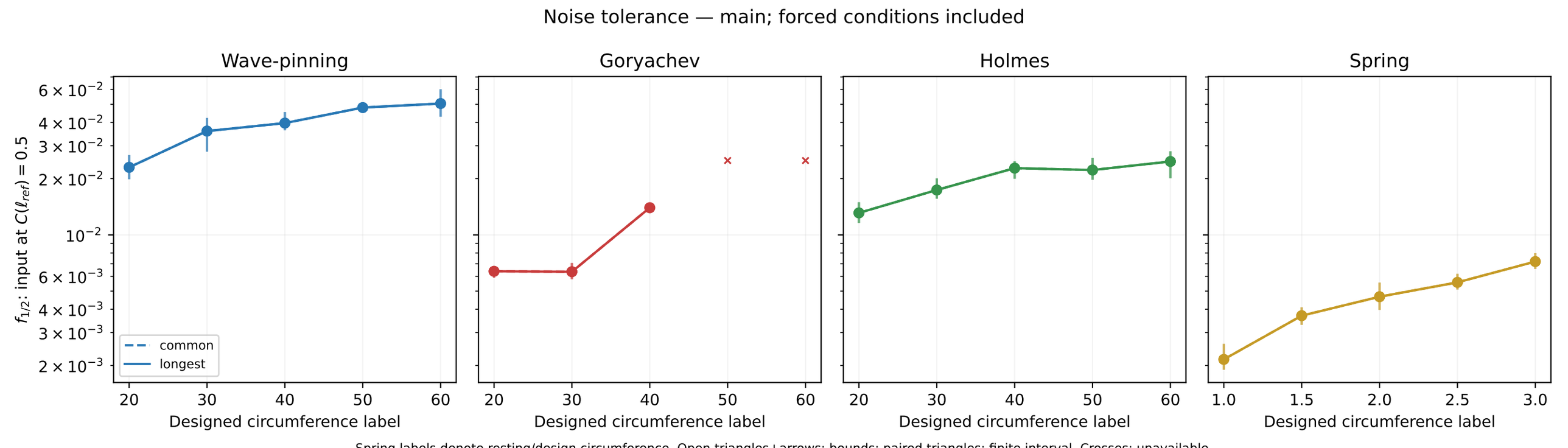


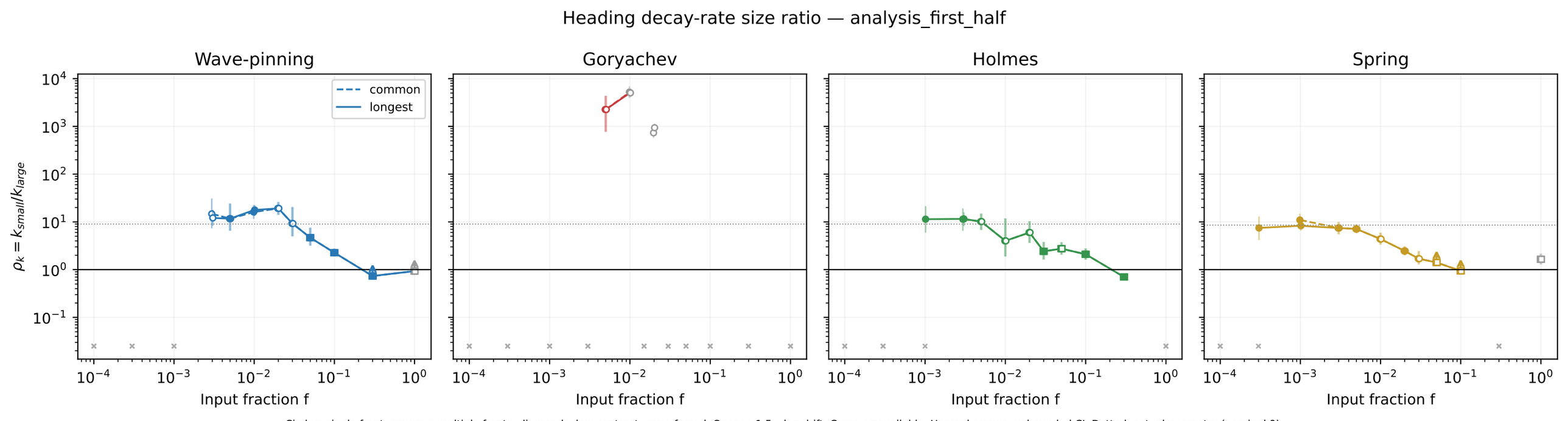
Colour shows new call. met: criterion met; not met: criterion not met; ?: inconclusive.
Frozen criteria text is unchanged. H3 and the A6 predictor gate retain their explicitly direct-τ definitions.
Geometric normalization is a reported outcome, not a gate. Full components and old records are in the CSV.

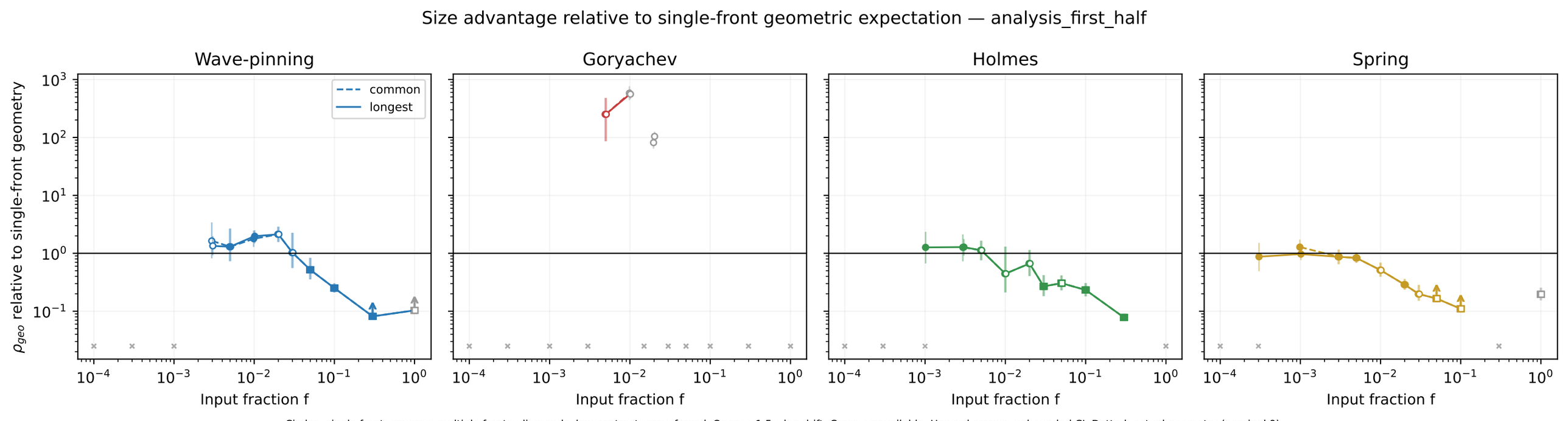

Heading decay-rate size ratio — main
Wave-pinning
Goryachev
Holmes
Spring
common
longest
$\rho_k = k_{small}/k_{large}$
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^0
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

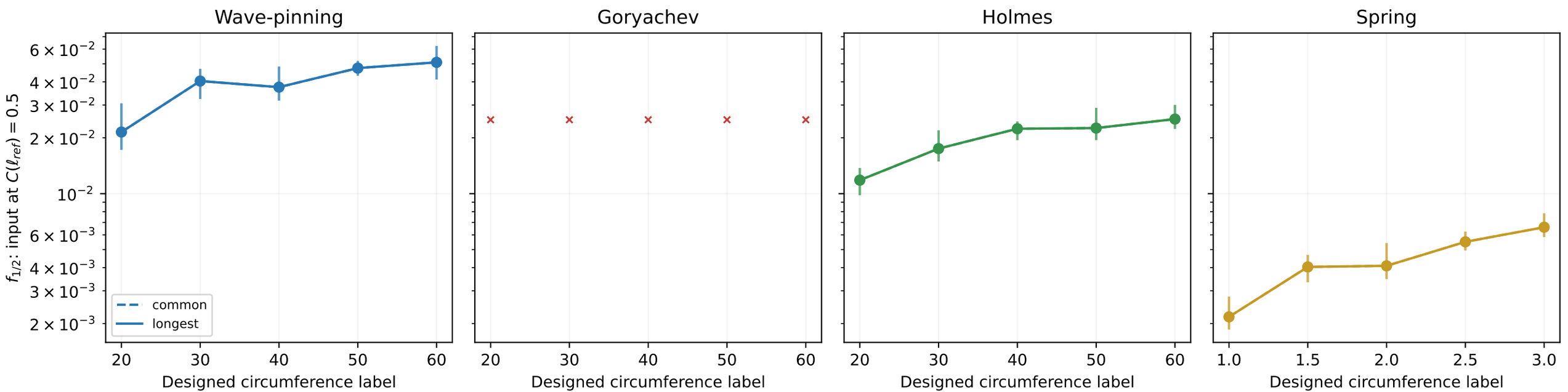

Size advantage relative to single-front geometric expectation — main
Wave-pinning
Goryachev
Holmes
Spring
common
longest
ρ_{geo} relative to single-front geometry
Input fraction f
Input fraction f
Input fraction f
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

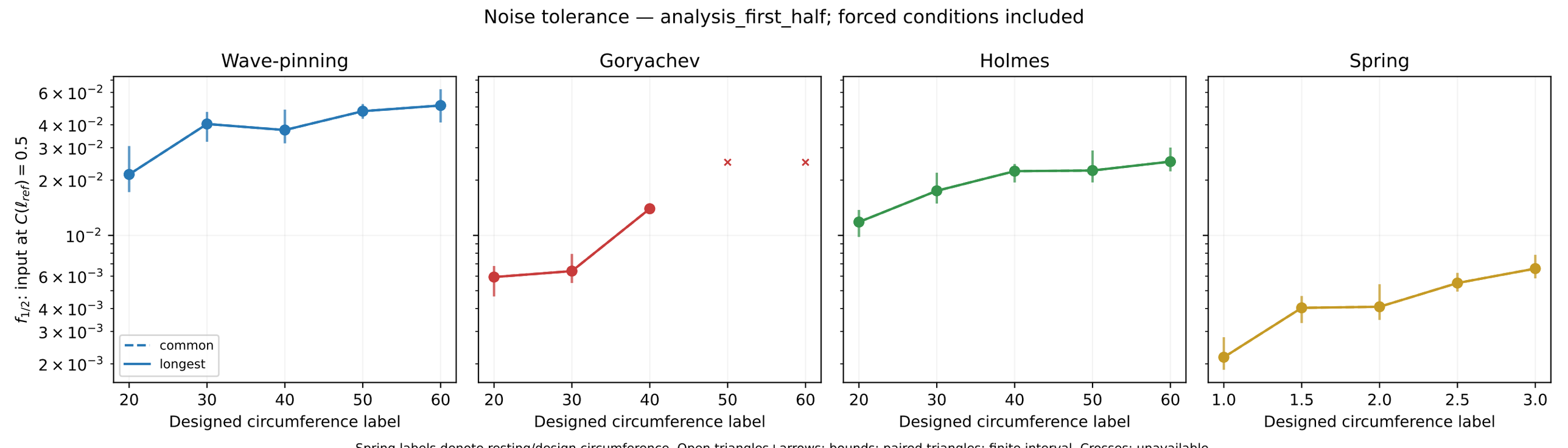

Noise tolerance — main; forced conditions excluded
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles: arrows; bounds; paired triangles: finite interval. Crosses: unavailable

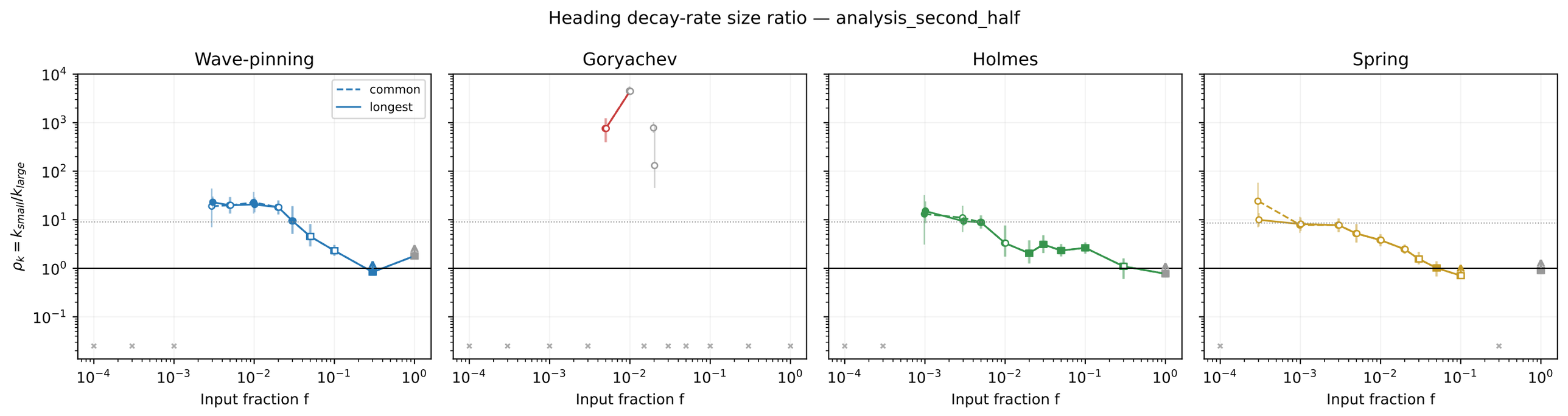

Noise tolerance — main; forced conditions included
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles | arrows: bounds; paired triangles: finite interval. Crosses: unavailable

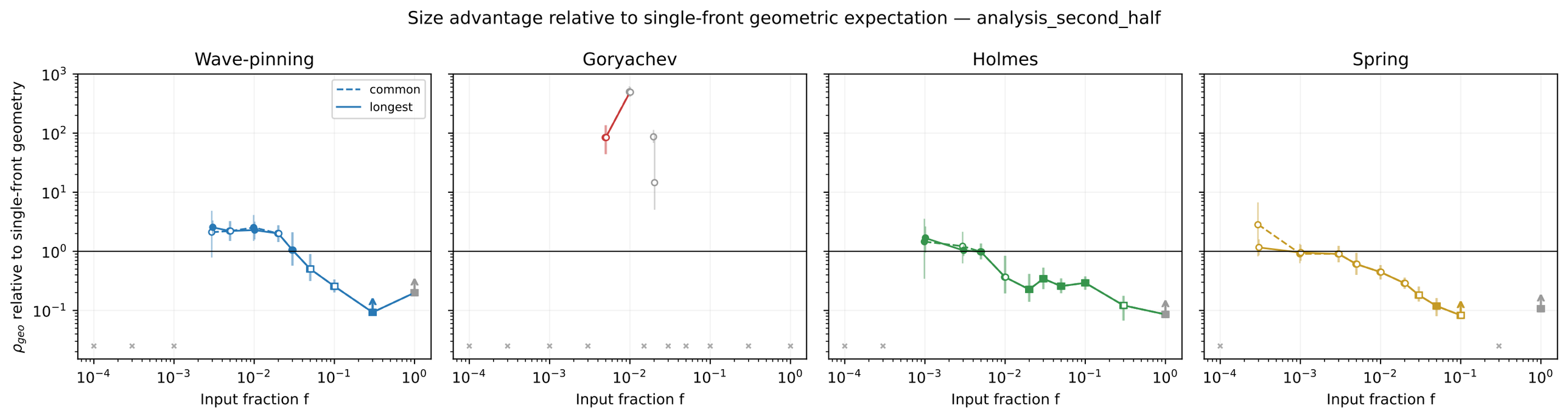

Heading decay-rate size ratio — analysis_first_half
Wave-pinning
Goryachev
Holmes
Spring
common
longest
$\rho_k = k_{small}/k_{large}$
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^0
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

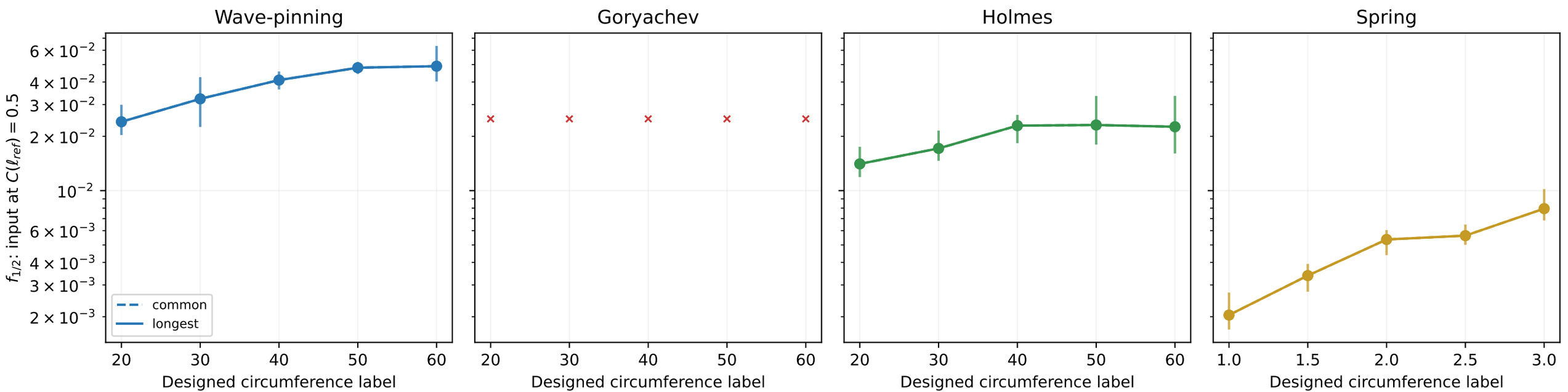

Size advantage relative to single-front geometric expectation — analysis_first_half
Wave-pinning
Goryachev
Holmes
Spring
common
longest
ρ_{geo} relative to single-front geometry
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
Input fraction f
Input fraction f
Input fraction f
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

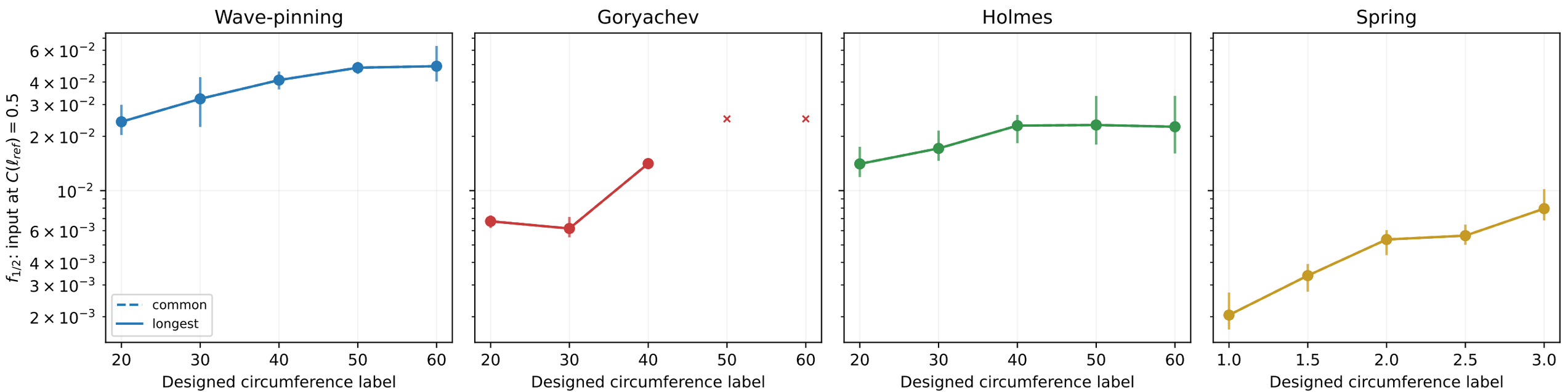

Noise tolerance — analysis_first_half; forced conditions excluded
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles | arrows: bounds; paired triangles: finite interval. Crosses: unavailable

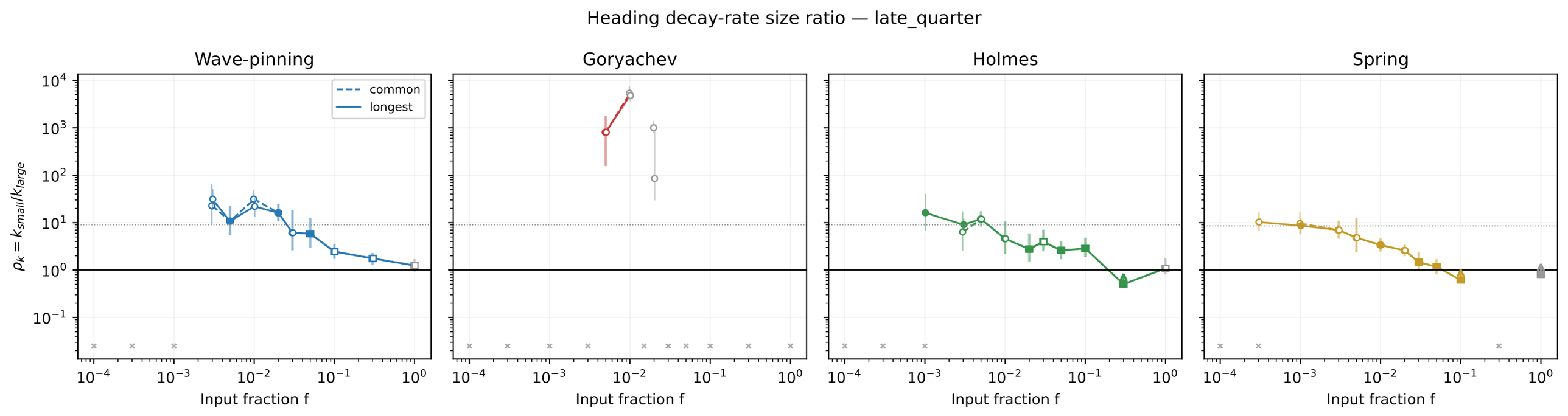

Noise tolerance — analysis_first_half; forced conditions included
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles / arrows: bounds; paired triangles: finite interval. Crosses: unavailable.

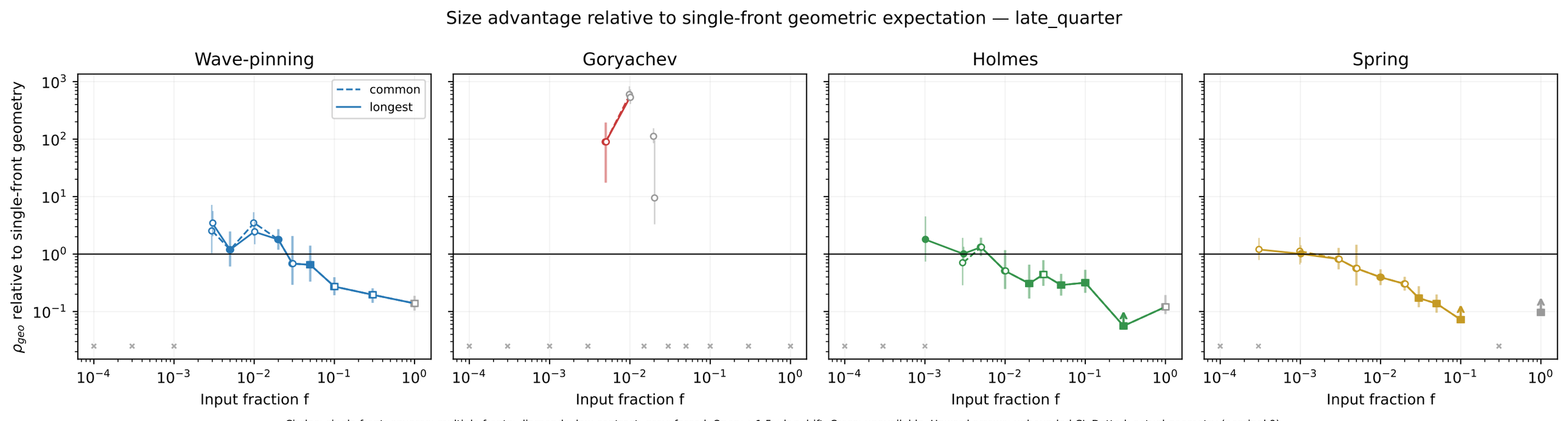

Heading decay-rate size ratio — analysis_second_half
Wave-pinning
Goryachev
Holmes
Spring
common
longest
$\rho_k = k_{small}/k_{large}$
Input fraction f
Input fraction f
Input fraction f
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

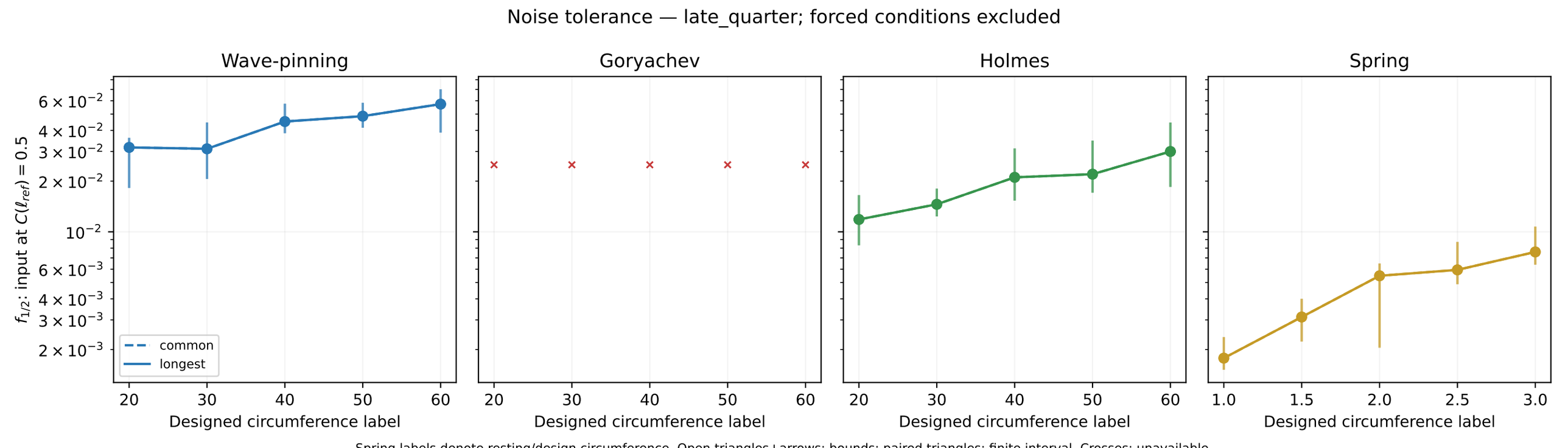

Size advantage relative to single-front geometric expectation — analysis_second_half
Wave-pinning
Goryachev
Holmes
Spring
common
longest
ρ_{geo} relative to single-front geometry
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

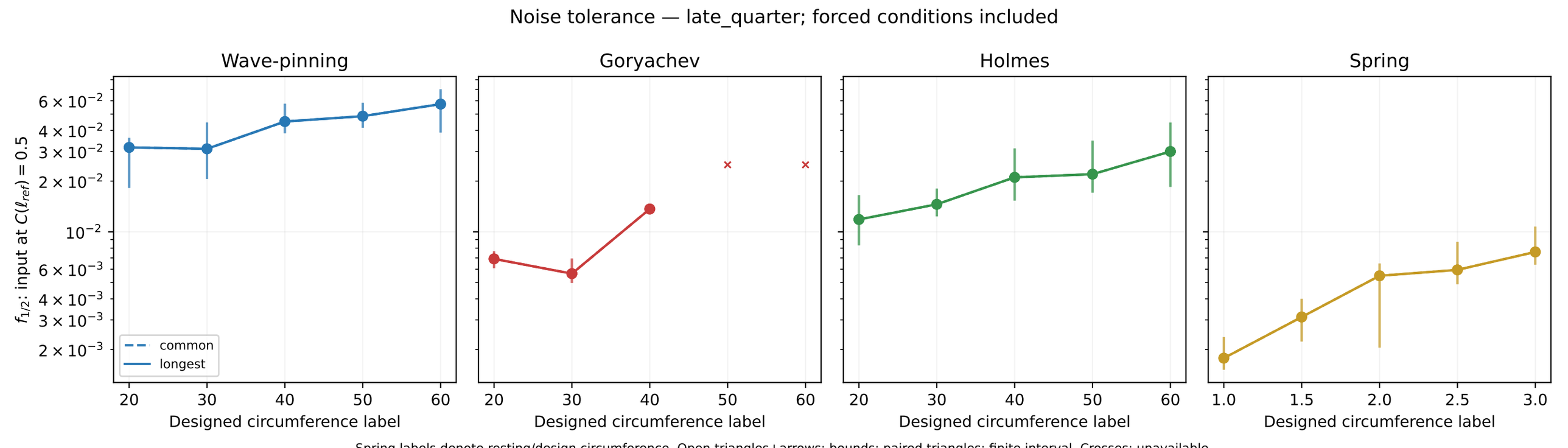

Noise tolerance — analysis_second_half; forced conditions excluded
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles↓ arrows: bounds; paired triangles: finite interval. Crosses: unavailable

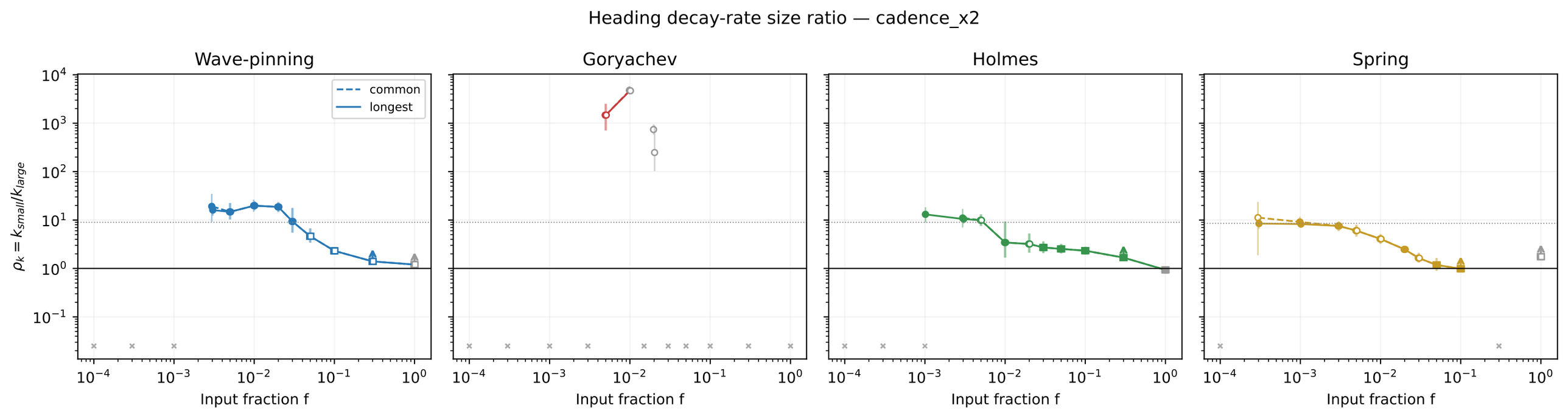

Noise tolerance — analysis_second_half; forced conditions included
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles | arrows: bounds; paired triangles: finite interval. Crosses: unavailable

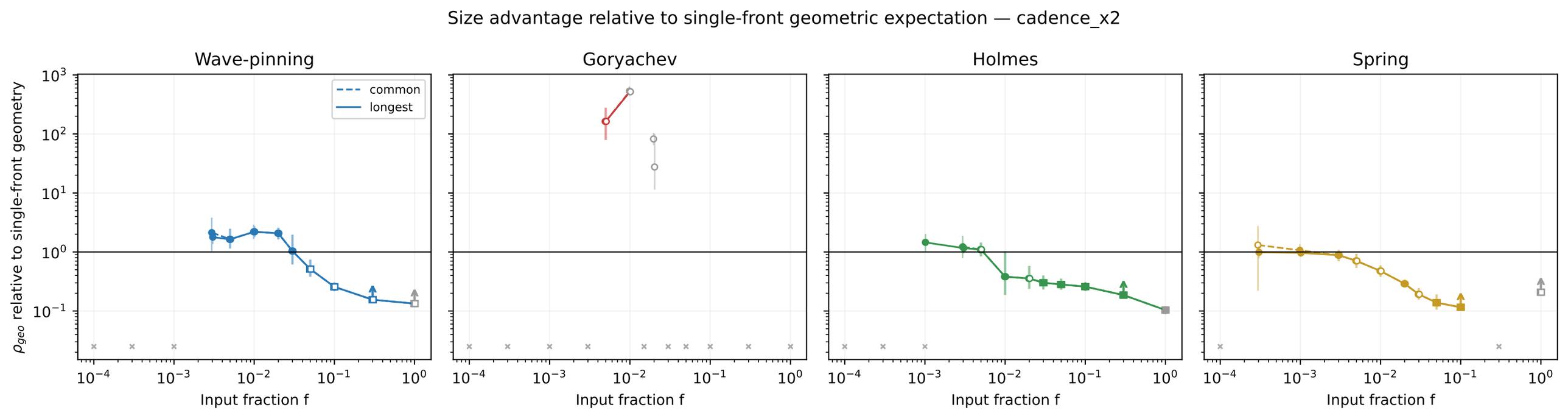

Heading decay-rate size ratio — late_quarter
Wave-pinning
Goryachev
Holmes
Spring
common
longest
$\rho_k = k_{small}/k_{large}$
Input fraction f
Input fraction f
Input fraction f
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

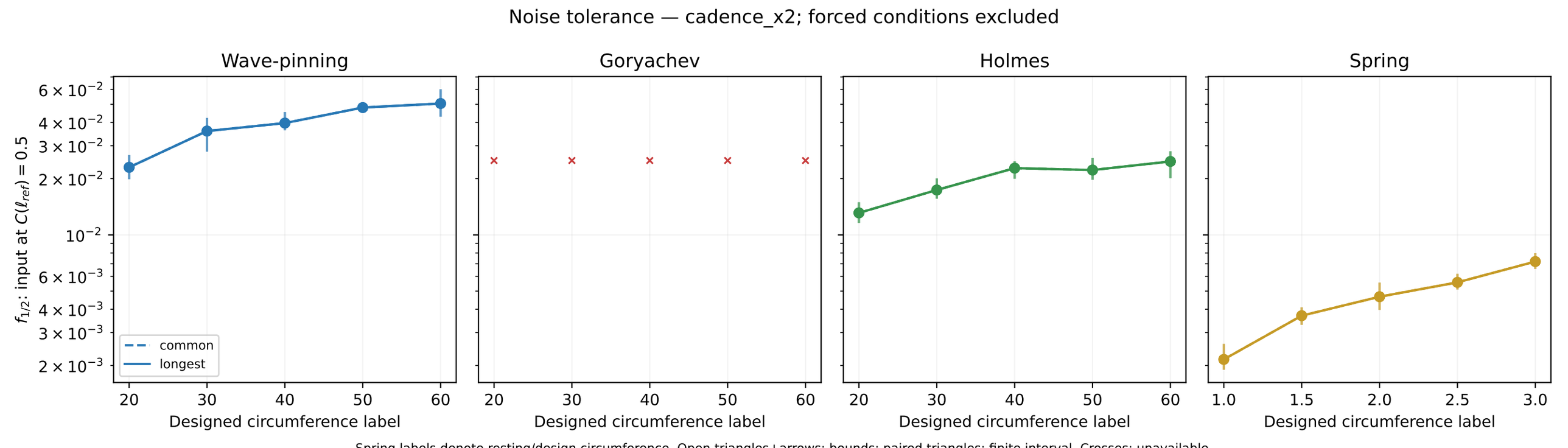

Size advantage relative to single-front geometric expectation — late_quarter
Wave-pinning
Goryachev
Holmes
Spring
common
longest
ρ_{geo} relative to single-front geometry
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

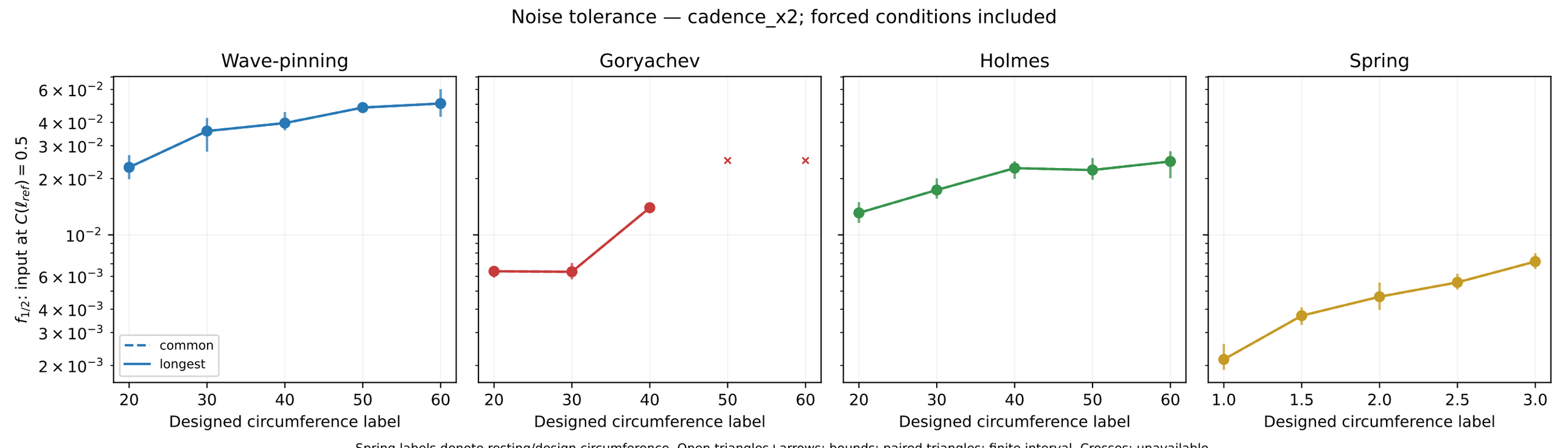

Noise tolerance — late_quarter; forced conditions excluded
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles / arrows: bounds; paired triangles: finite interval. Crosses: unavailable.

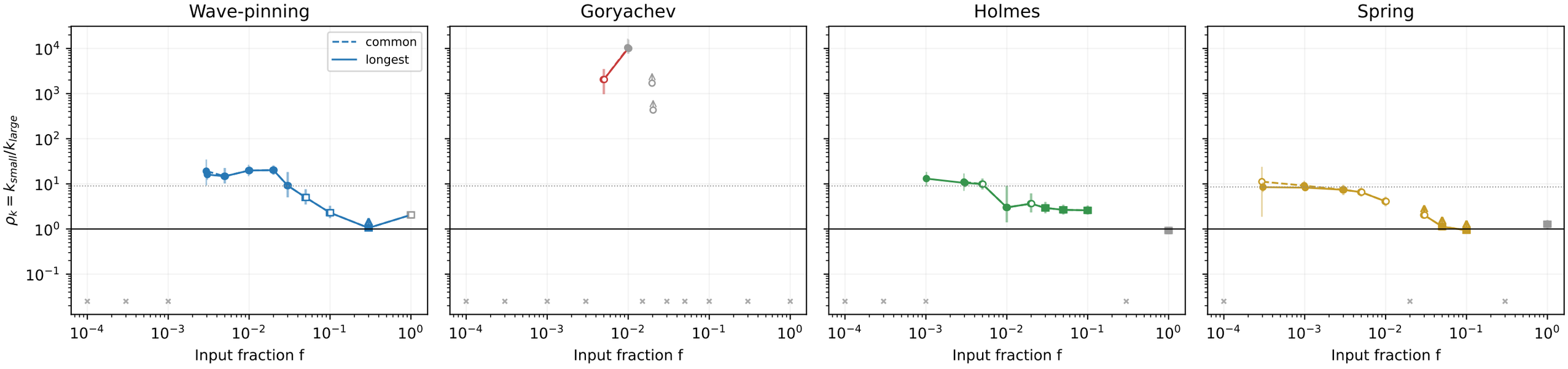

Noise tolerance — late_quarter; forced conditions included
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles / arrows: bounds; paired triangles: finite interval. Crosses: unavailable.

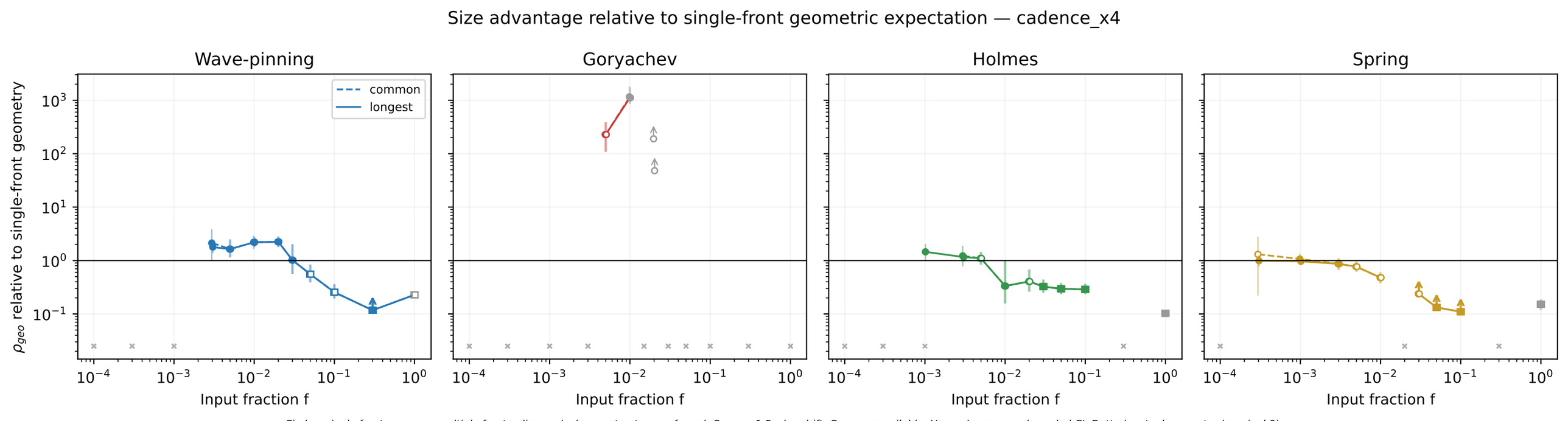

Heading decay-rate size ratio — cadence_x2
Wave-pinning
Goryachev
Holmes
Spring
common
longest
$\rho_k = k_{small}/k_{large}$
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^0
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

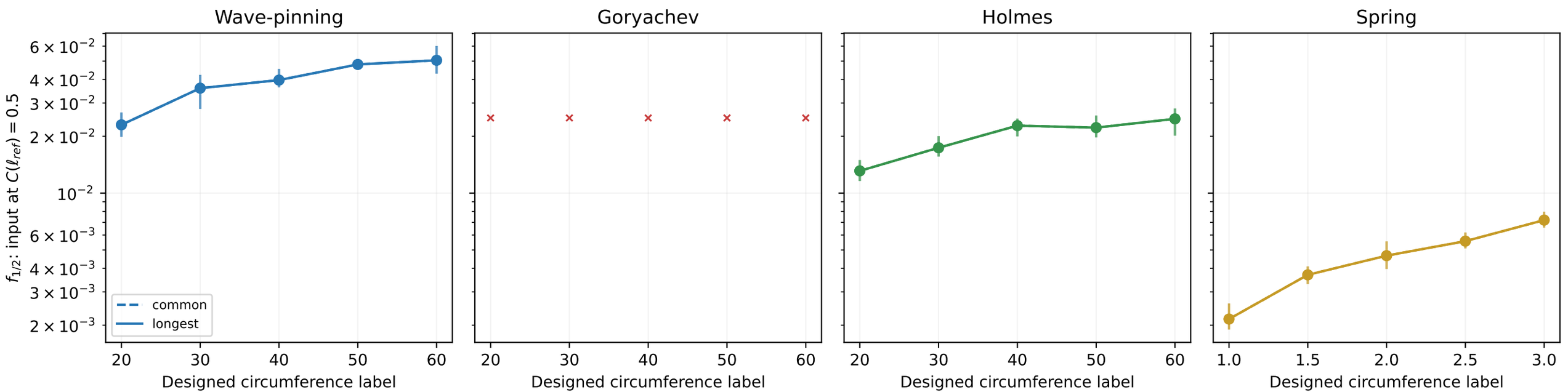

Size advantage relative to single-front geometric expectation — cadence_x2
Wave-pinning
Goryachev
Holmes
Spring
common
longest
ρ_{geo} relative to single-front geometry
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
Input fraction f
Input fraction f
Input fraction f
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

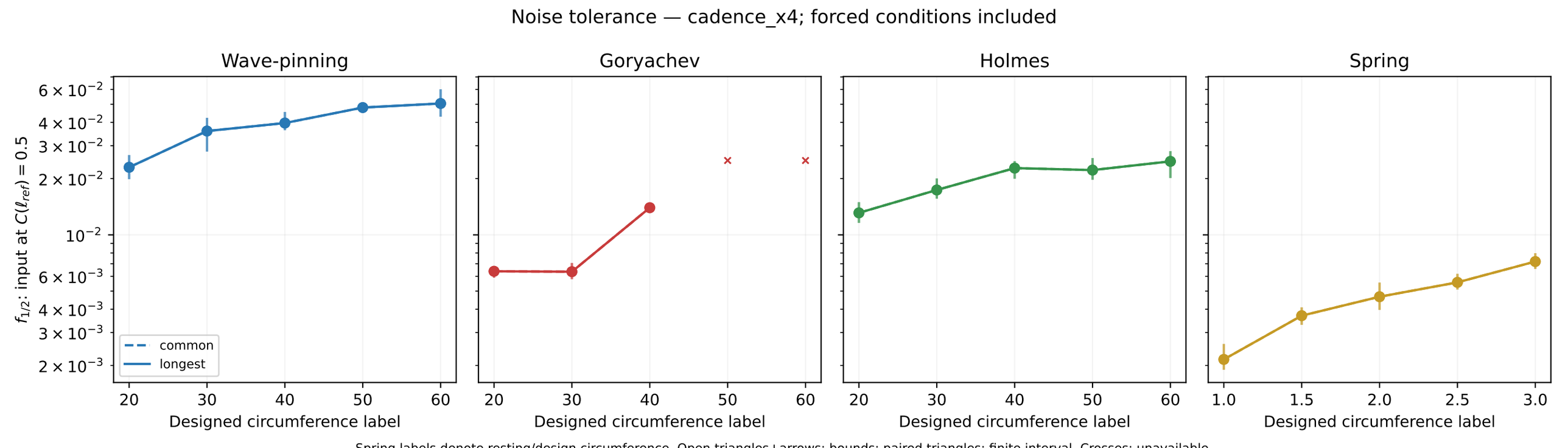

Noise tolerance — cadence_x2; forced conditions excluded
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles↓ arrows: bounds; paired triangles: finite interval. Crosses: unavailable

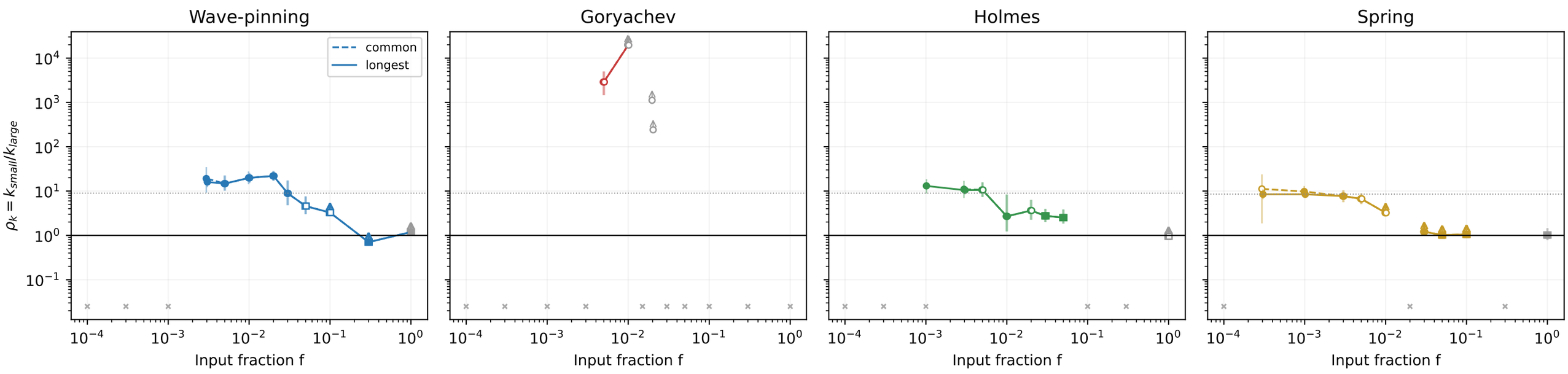

Noise tolerance — cadence_x2; forced conditions included
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles | arrows: bounds; paired triangles: finite interval. Crosses: unavailable

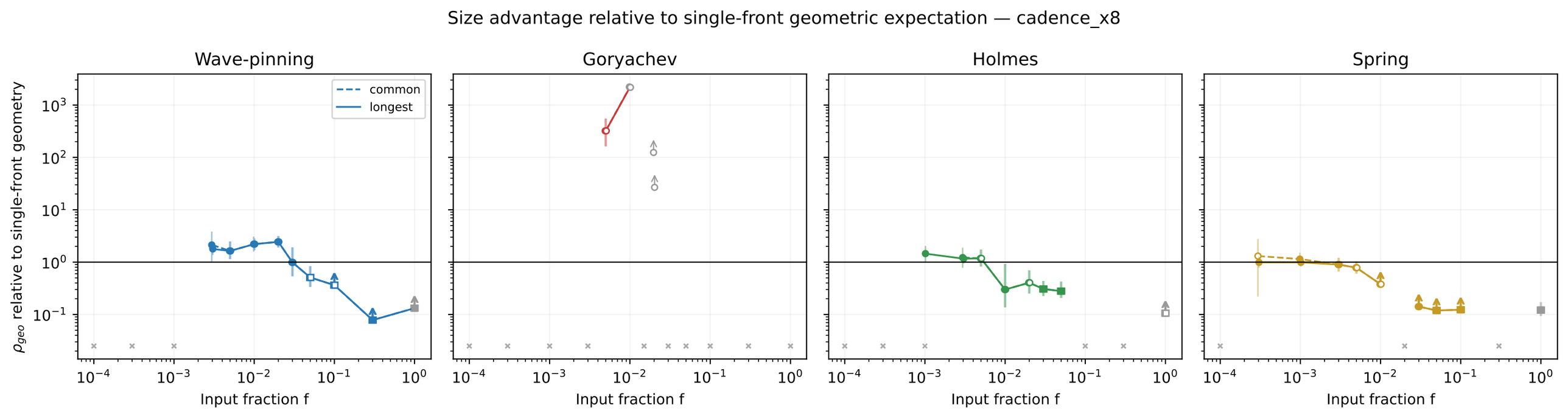

Heading decay-rate size ratio — cadence_x4
Wave-pinning
Goryachev
Holmes
Spring
common
longest
$\rho_k = k_{small}/k_{large}$
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^0
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

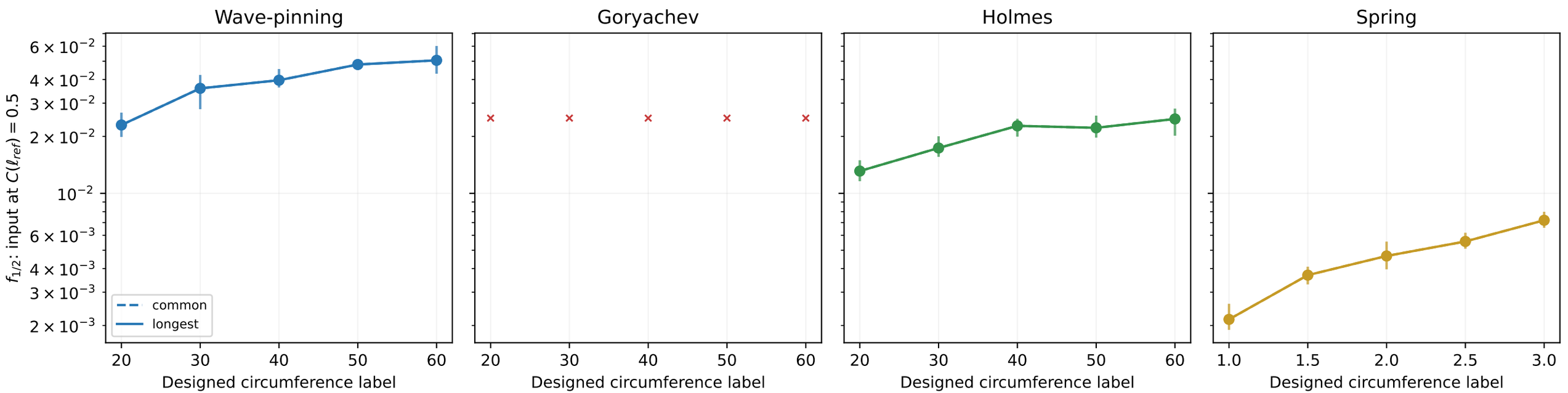

Size advantage relative to single-front geometric expectation — cadence_x4
Wave-pinning
Goryachev
Holmes
Spring
common
longest
ρ_{geo} relative to single-front geometry
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

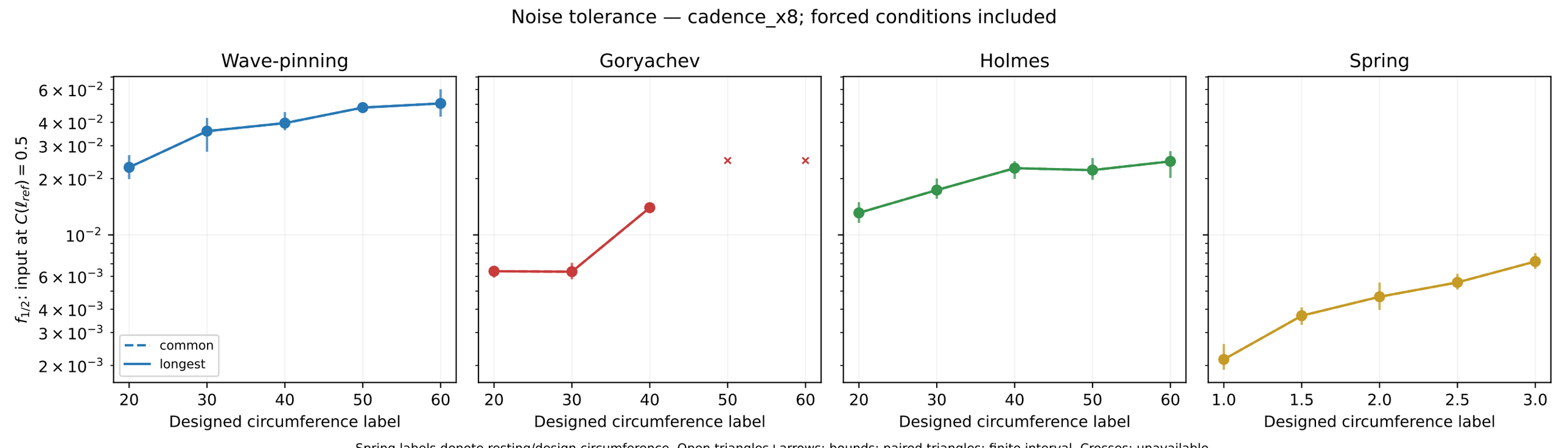

Noise tolerance — cadence_x4; forced conditions excluded
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles↓ arrows: bounds; paired triangles: finite interval. Crosses: unavailable

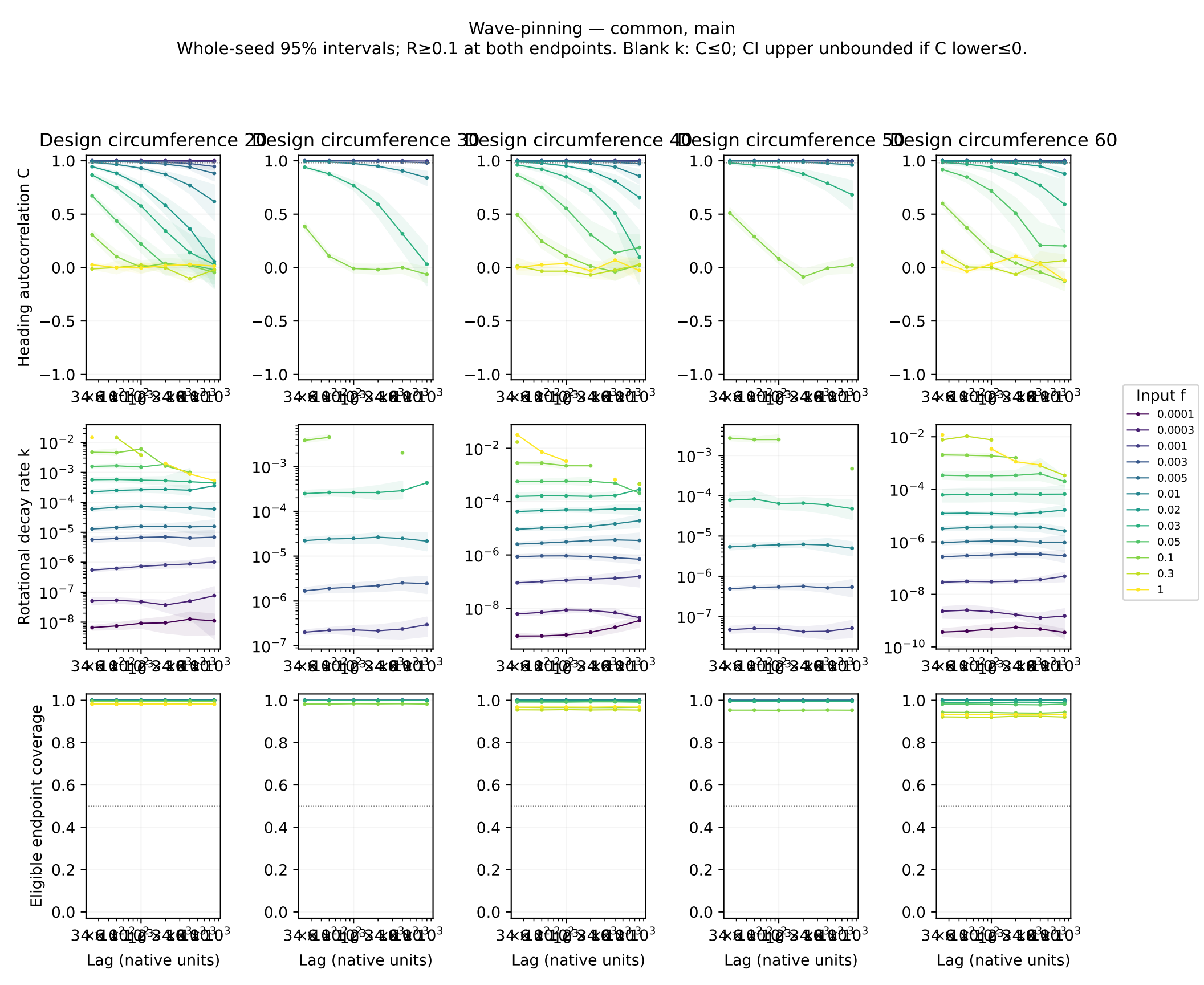

Noise tolerance — cadence_x4; forced conditions included
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles: arrows; bounds; paired triangles: finite interval. Crosses: unavailable

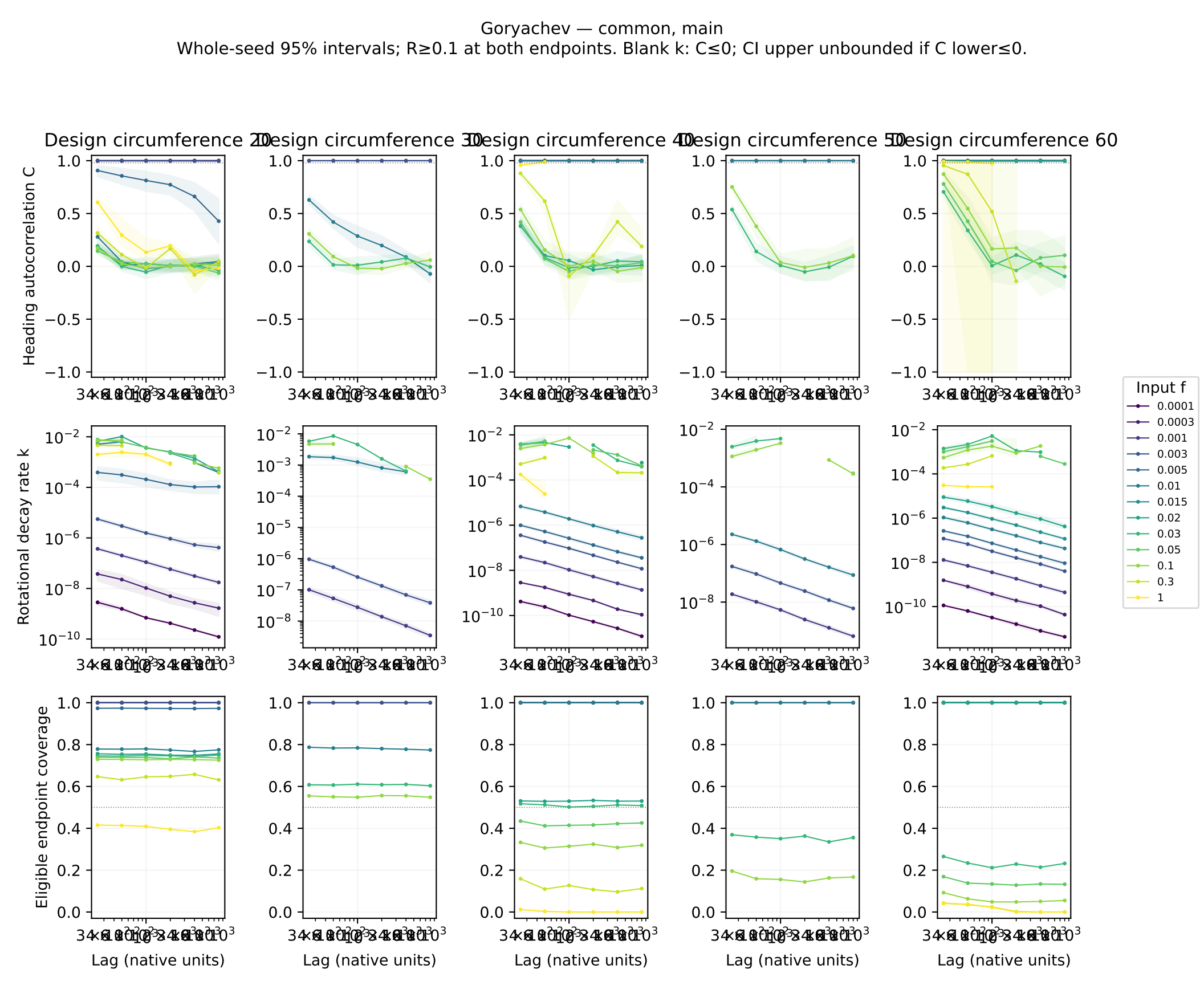

Heading decay-rate size ratio — cadence_x8
Wave-pinning
Goryachev
Holmes
Spring
common
longest
$\rho_k = k_{small}/k_{large}$
10^4
10^3
10^2
10^1
10^0
10^{-1}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^0
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

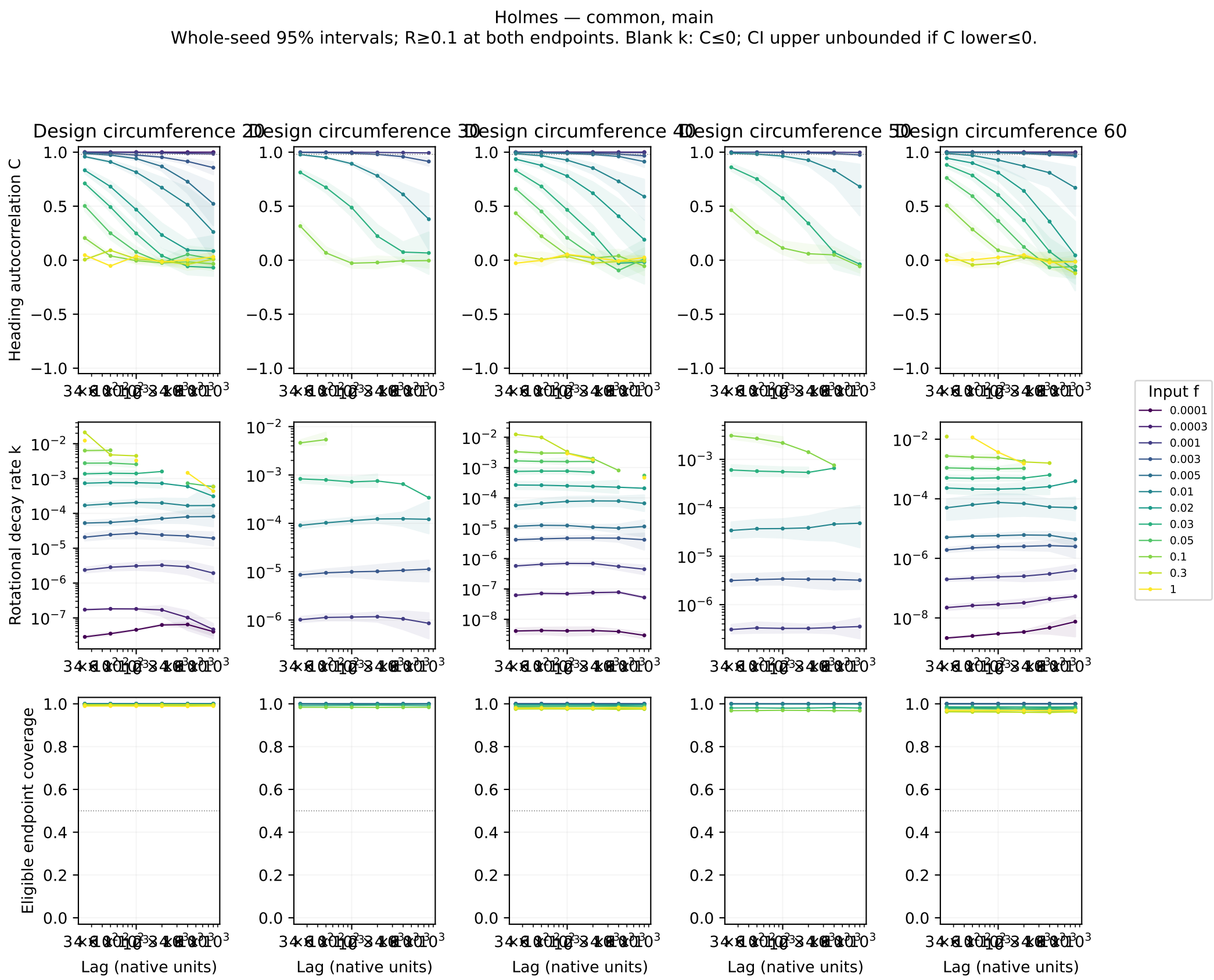

Size advantage relative to single-front geometric expectation — cadence_x8
Wave-pinning
Goryachev
Holmes
Spring
common
longest
ρ_{geo} relative to single-front geometry
10^{3}
10^{2}
10^{1}
10^{0}
10^{-1}
10^{-4}
10^{-3}
10^{-2}
10^{-1}
10^{0}
Input fraction f
Circles: single front; squares: multiple fronts; diamonds: low contrast; grey: forced. Open: >1.5× lag drift. Cross: unavailable. Upward arrows: unbounded CI. Dotted: actual geometry (nominal 9).

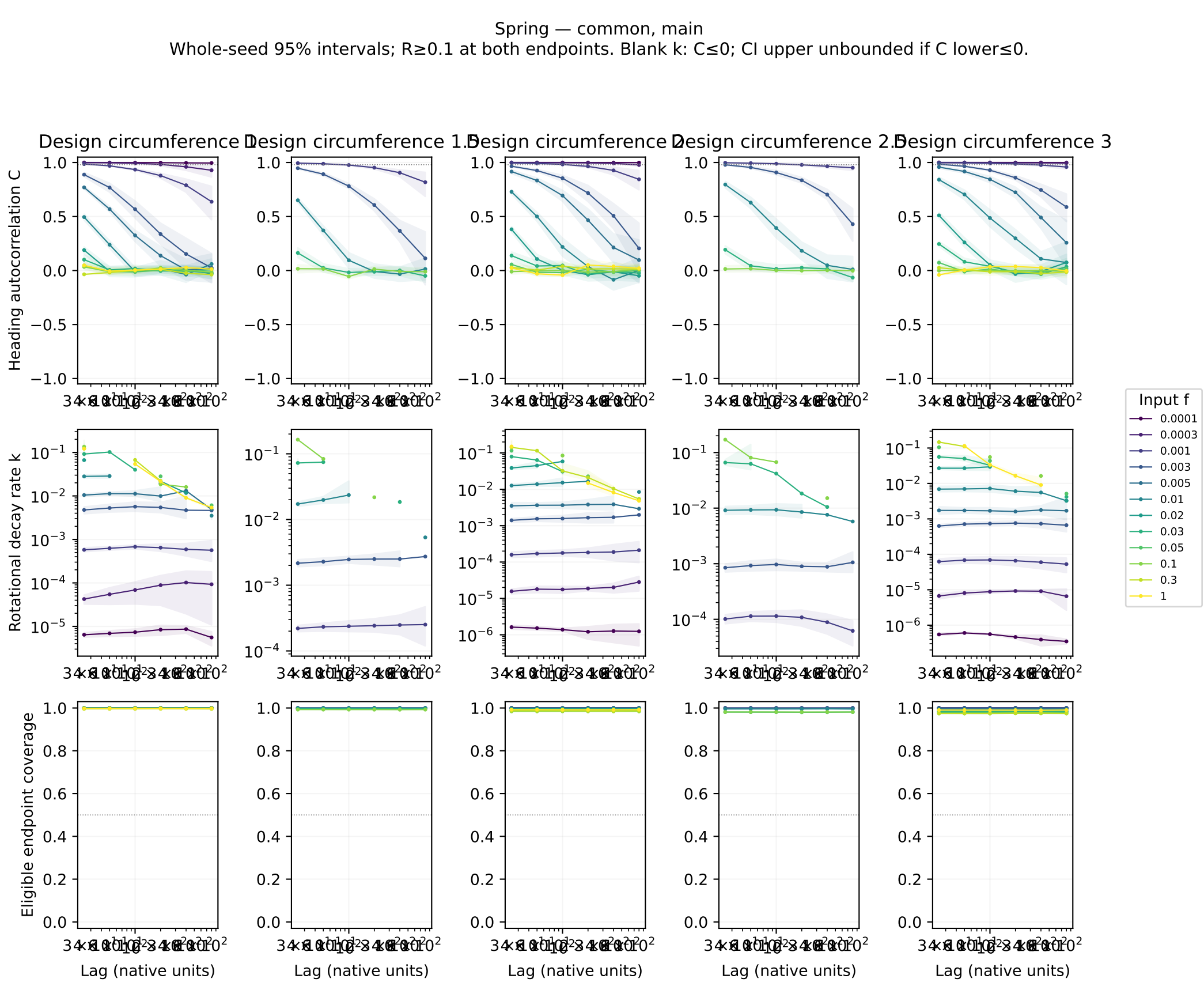

Noise tolerance — cadence_x8; forced conditions excluded
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles↓ arrows: bounds; paired triangles: finite interval. Crosses: unavailable

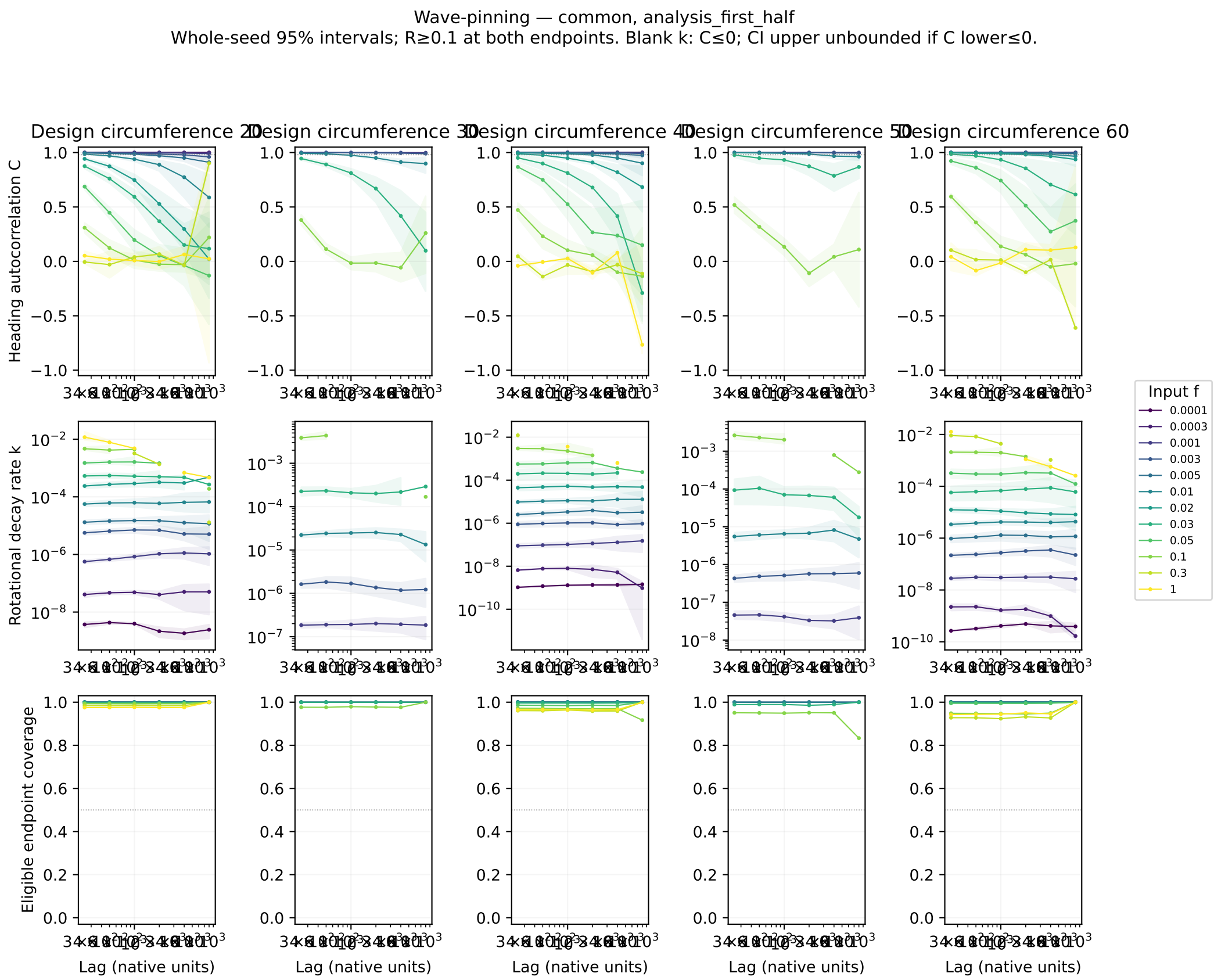

Noise tolerance — cadence_x8; forced conditions included
Wave-pinning
Goryachev
Holmes
Spring
$f_{1/2}$: input at $C(\ell_{ref}) = 0.5$
6×10^{-2}
4×10^{-2}
3×10^{-2}
2×10^{-2}
10^{-2}
6×10^{-3}
4×10^{-3}
3×10^{-3}
2×10^{-3}
common
longest
20
30
40
50
60
1.0
1.5
2.0
2.5
3.0
Designed circumference label
Spring labels denote resting/design circumference. Open triangles / arrows: bounds; paired triangles: finite interval. Crosses: unavailable.

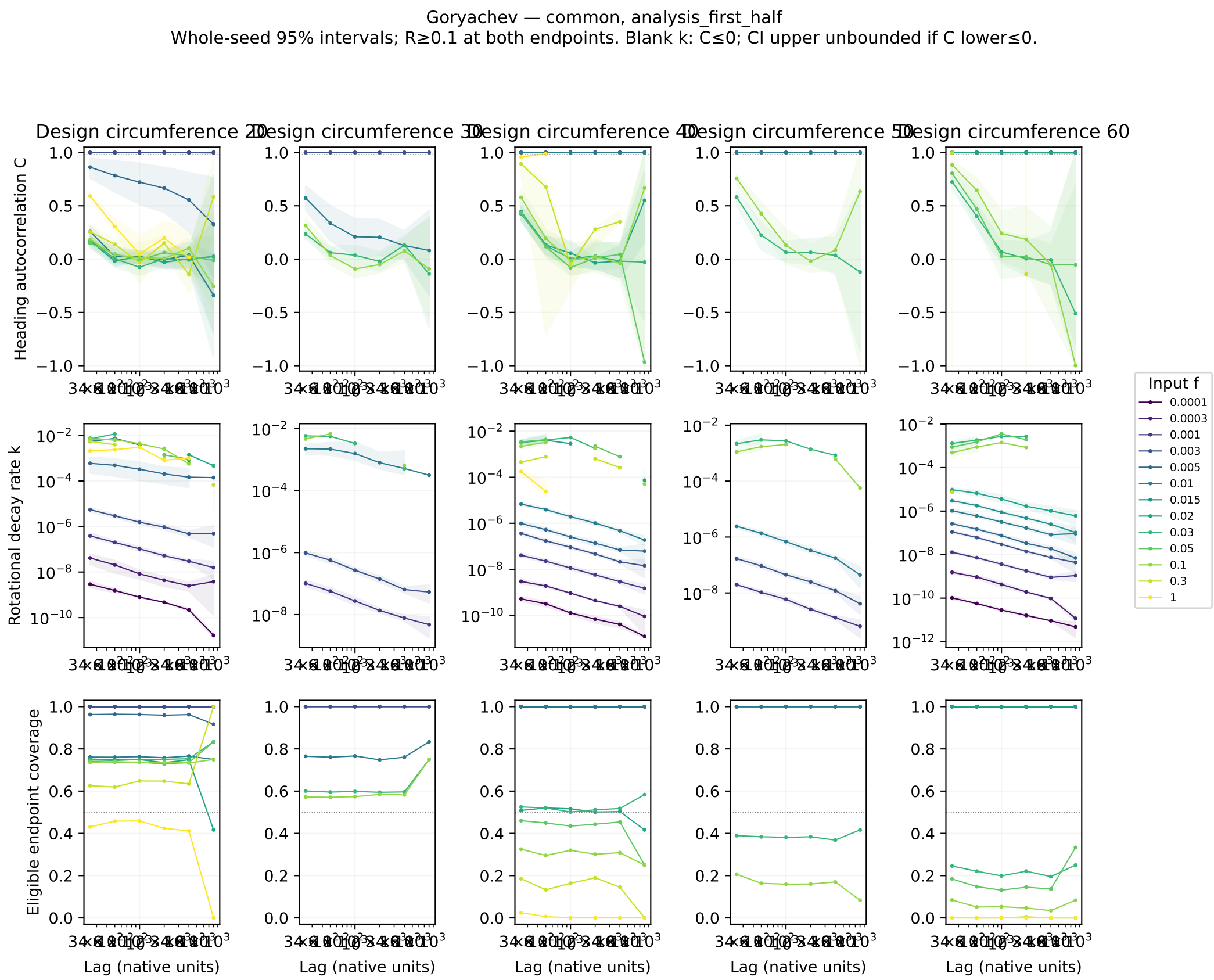

Wave-pinning — common, main
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

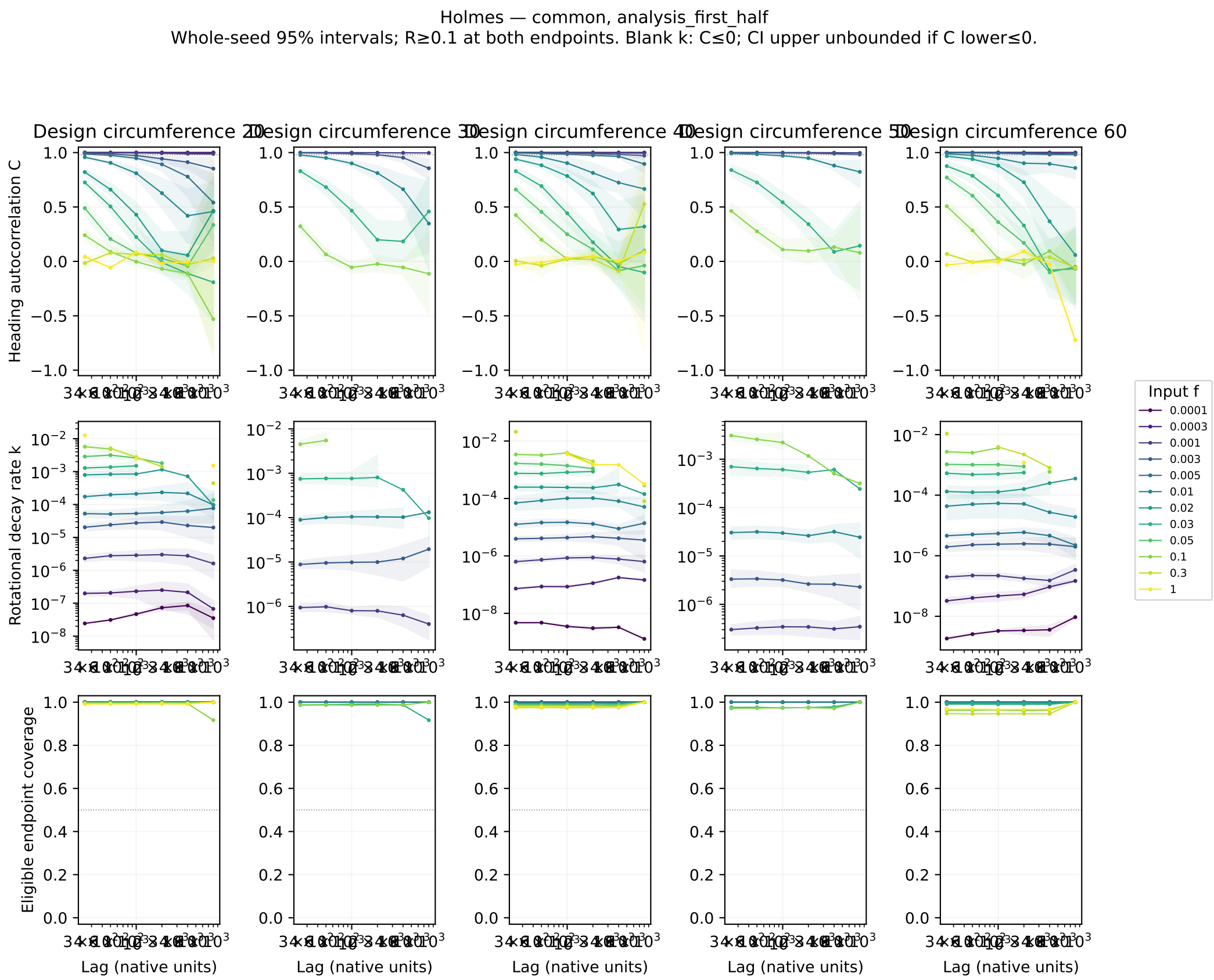

Goryachev — common, main
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1

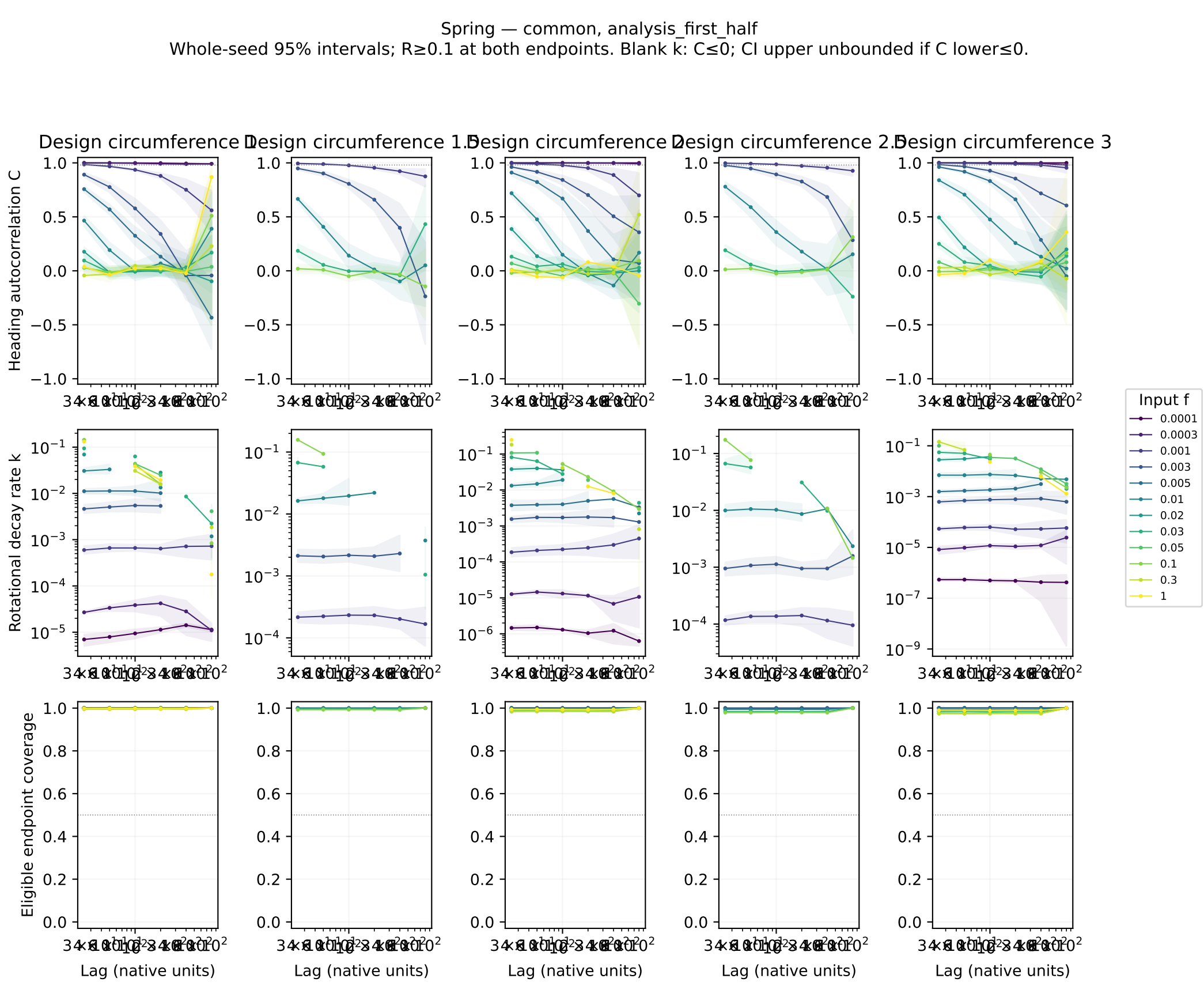

Holmes — common, main
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

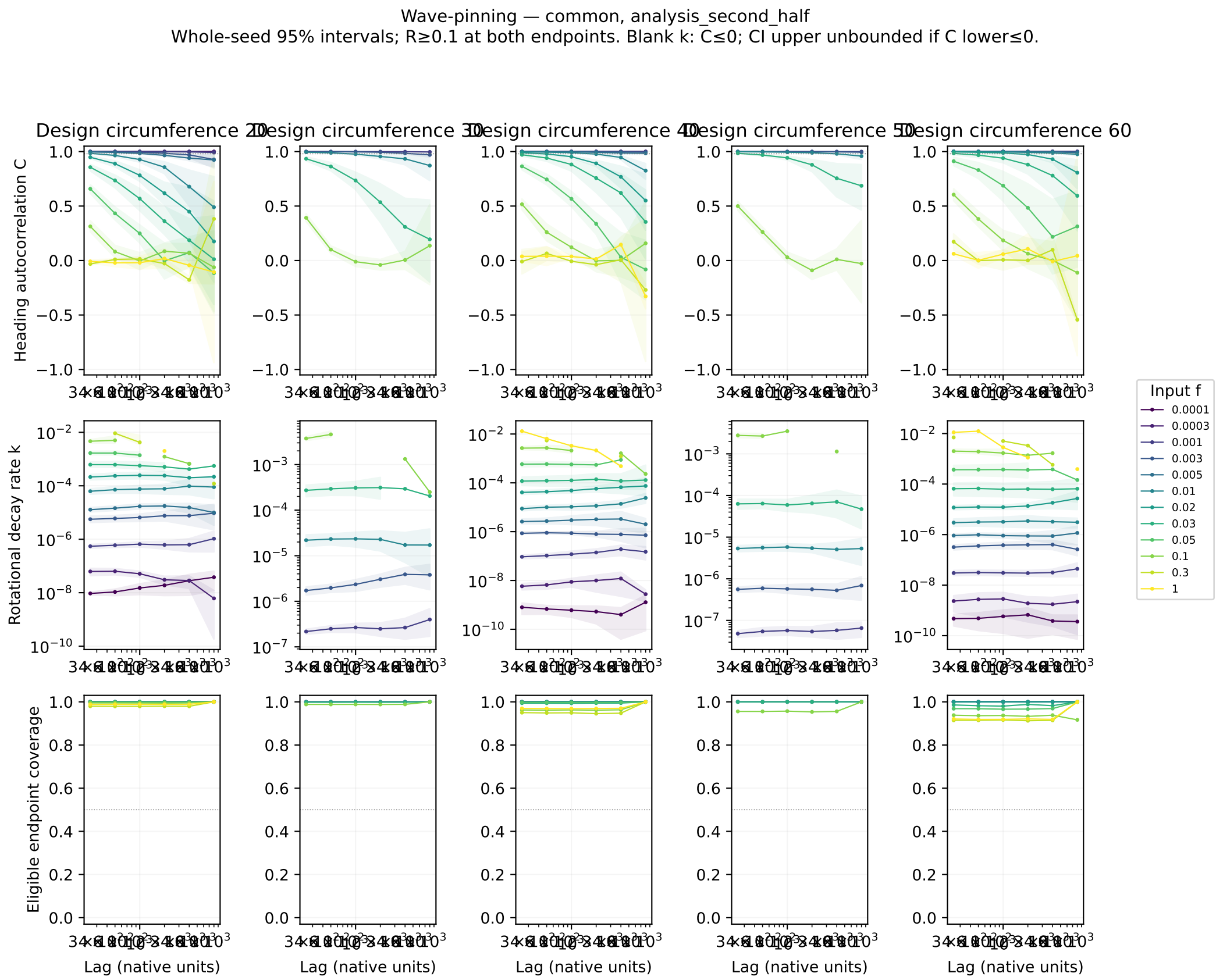

Spring — common, main
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

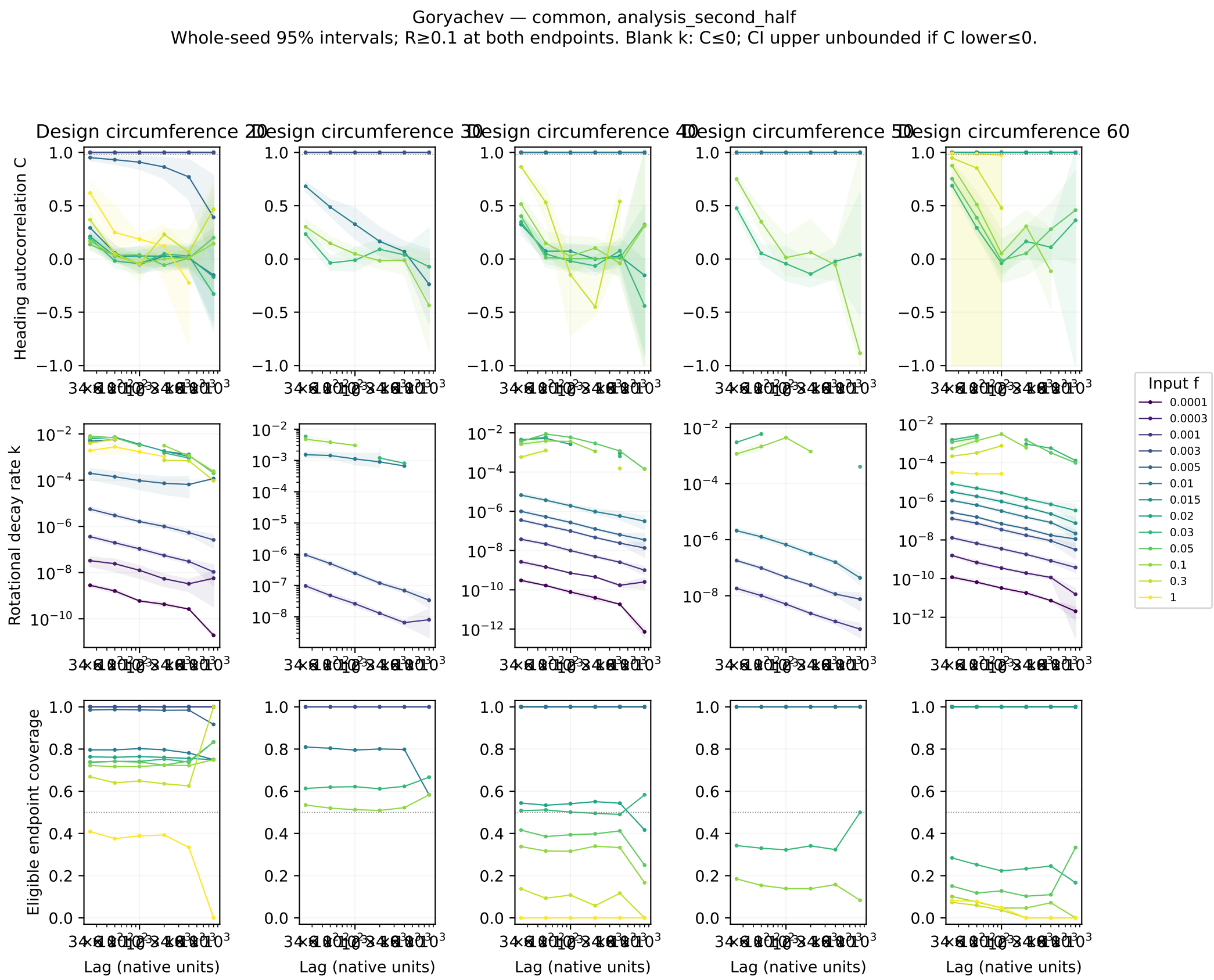

Wave-pinning — common, analysis_first_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

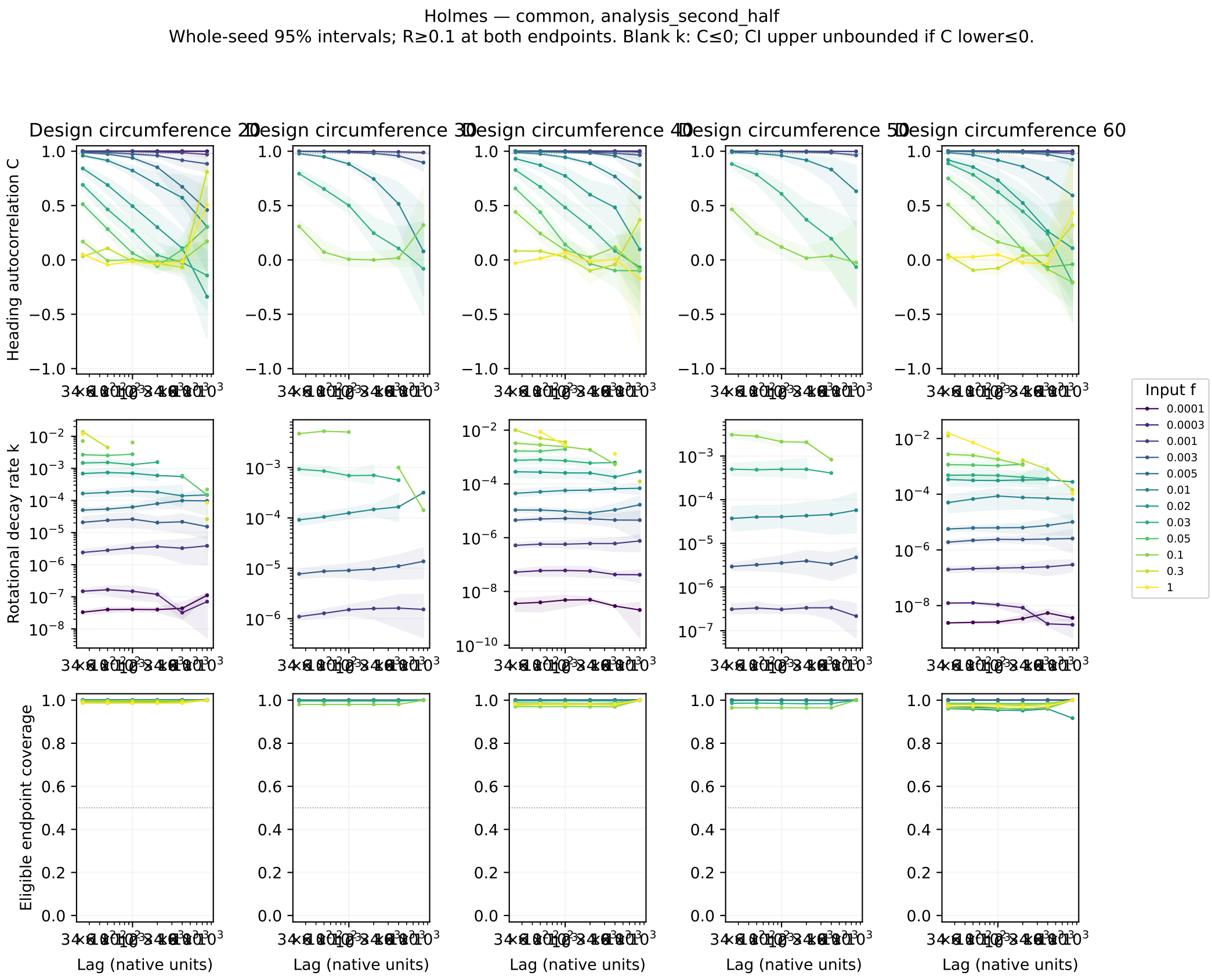

Goryachev — common, analysis_first_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1

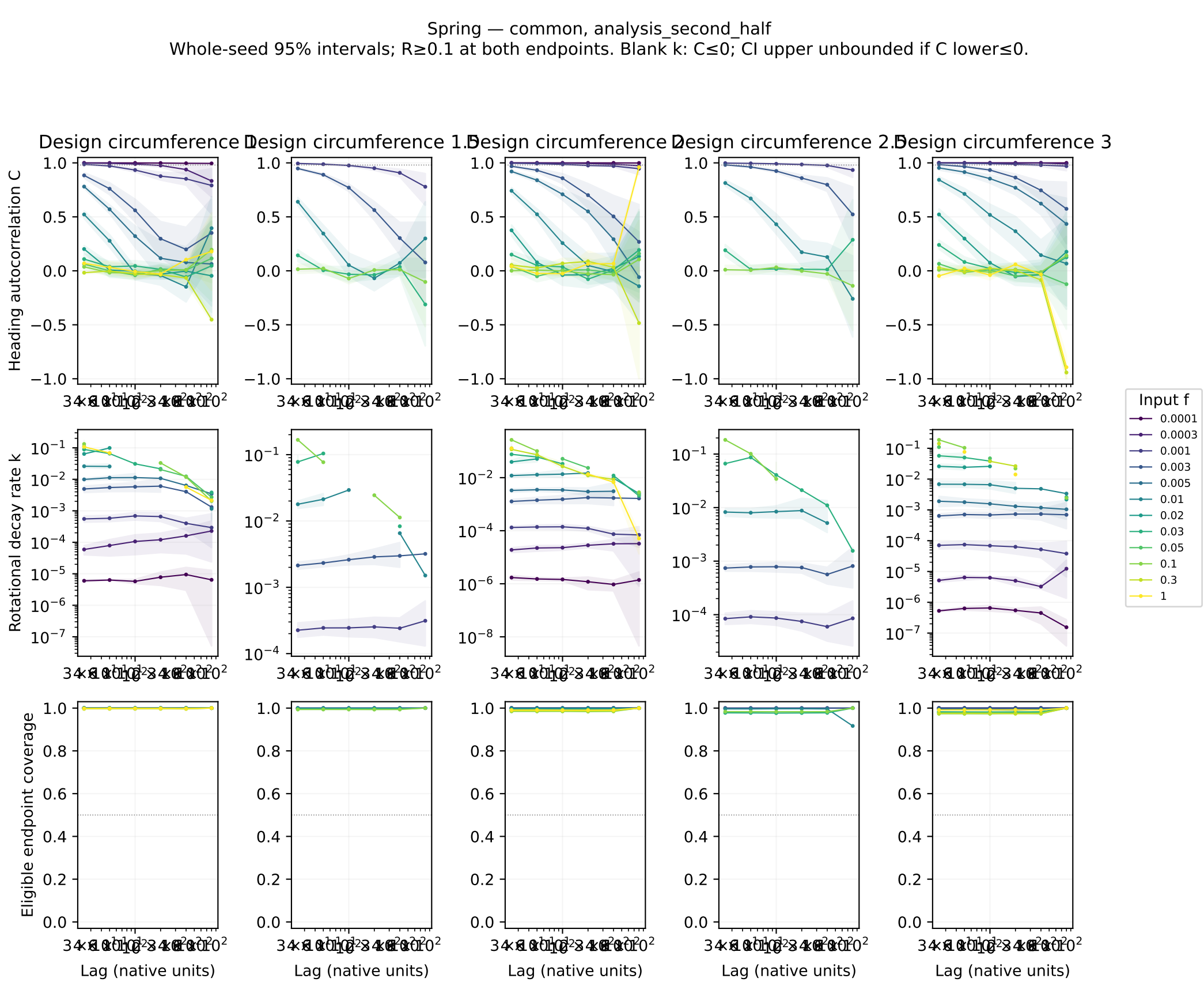

Holmes — common, analysis_first_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

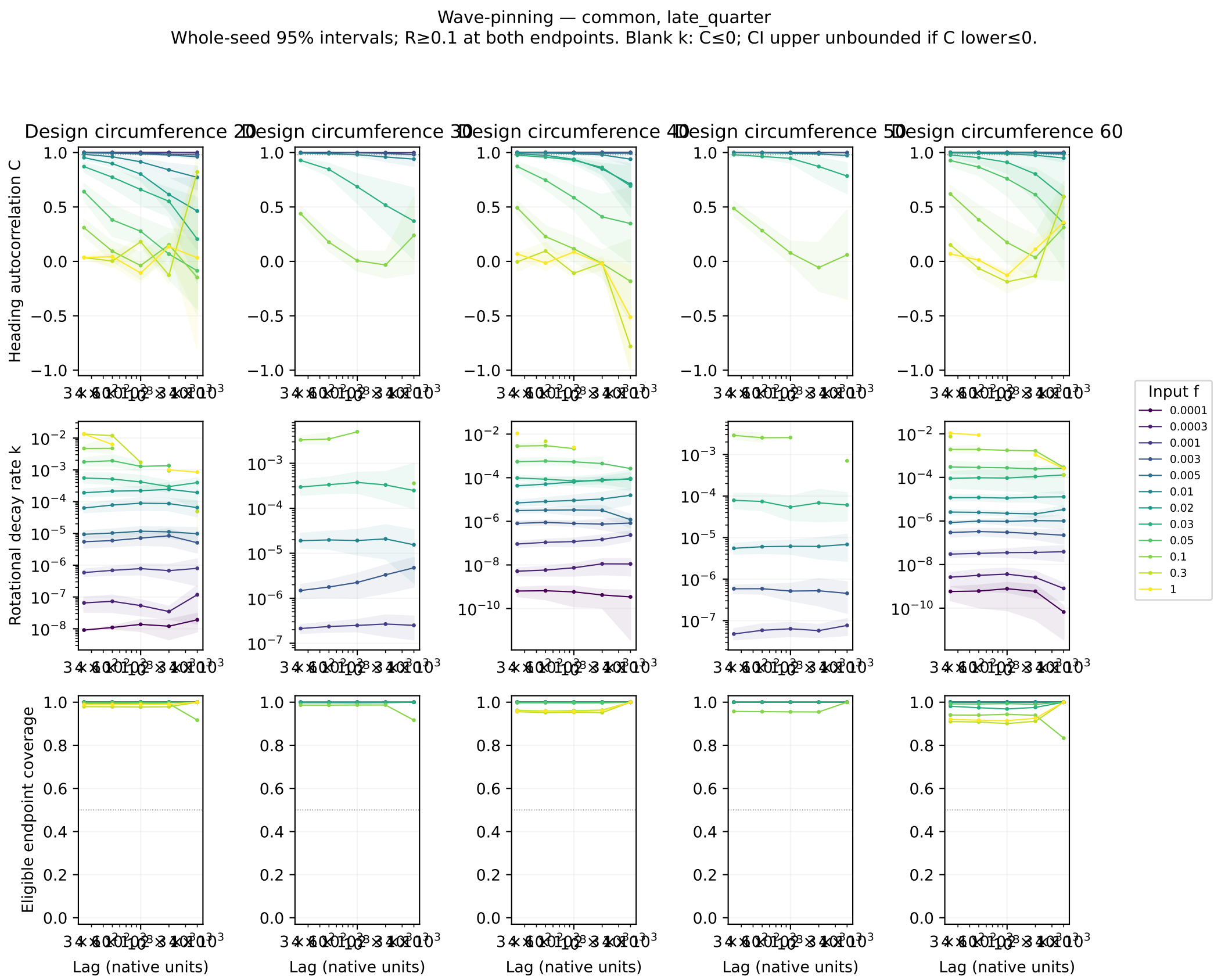

Spring — common, analysis_first_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

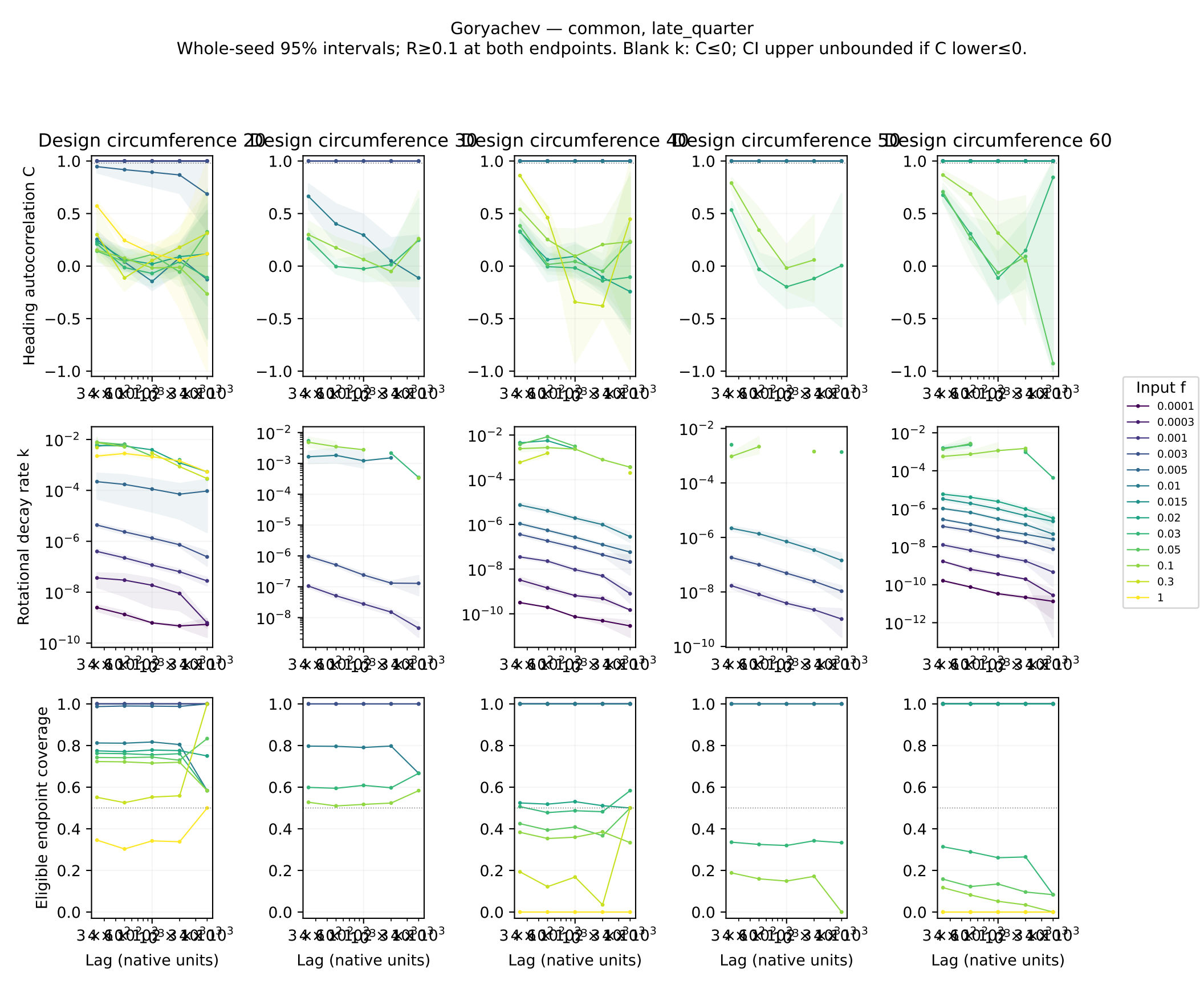

Wave-pinning — common, analysis_second_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

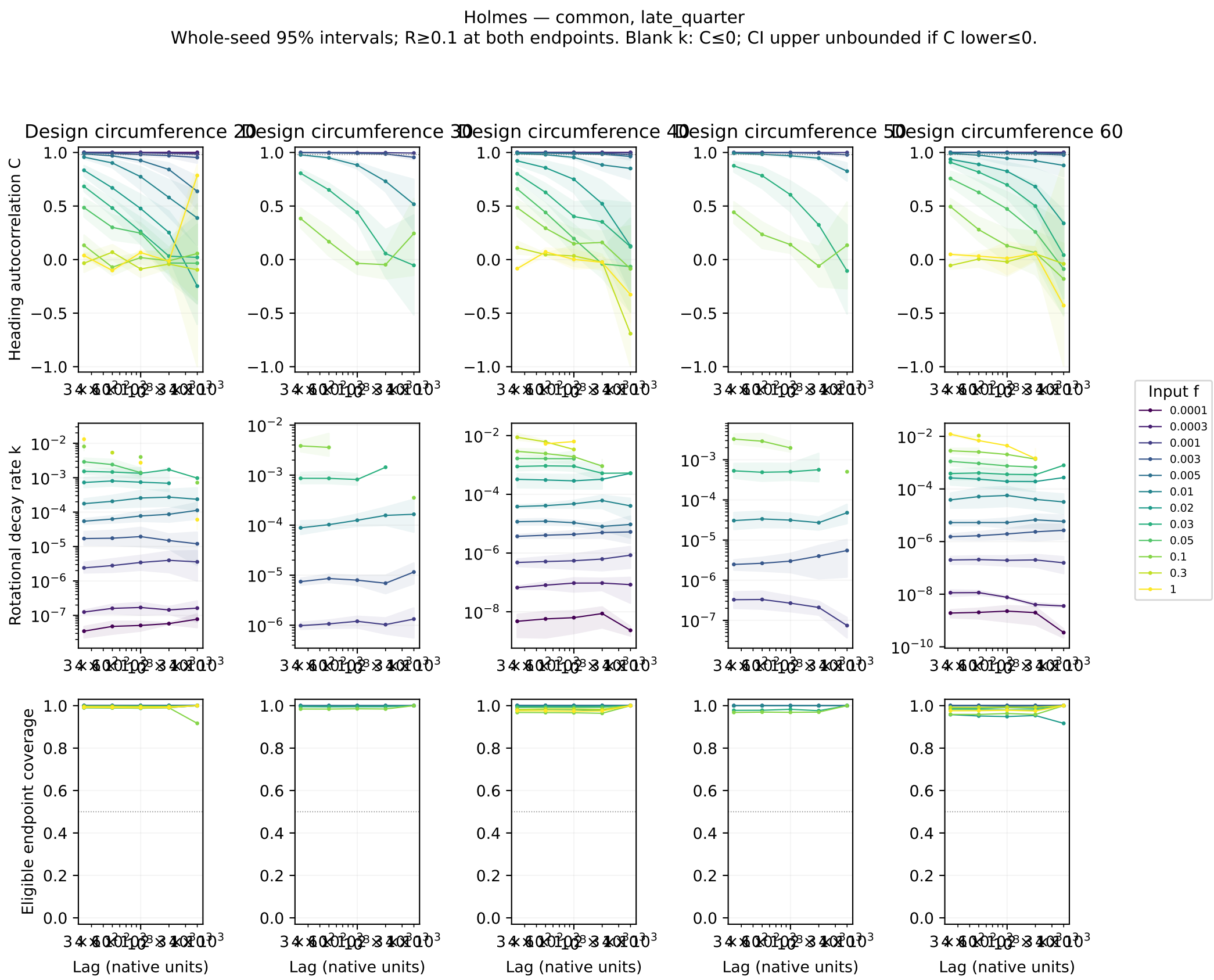

Goryachev — common, analysis_second_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1

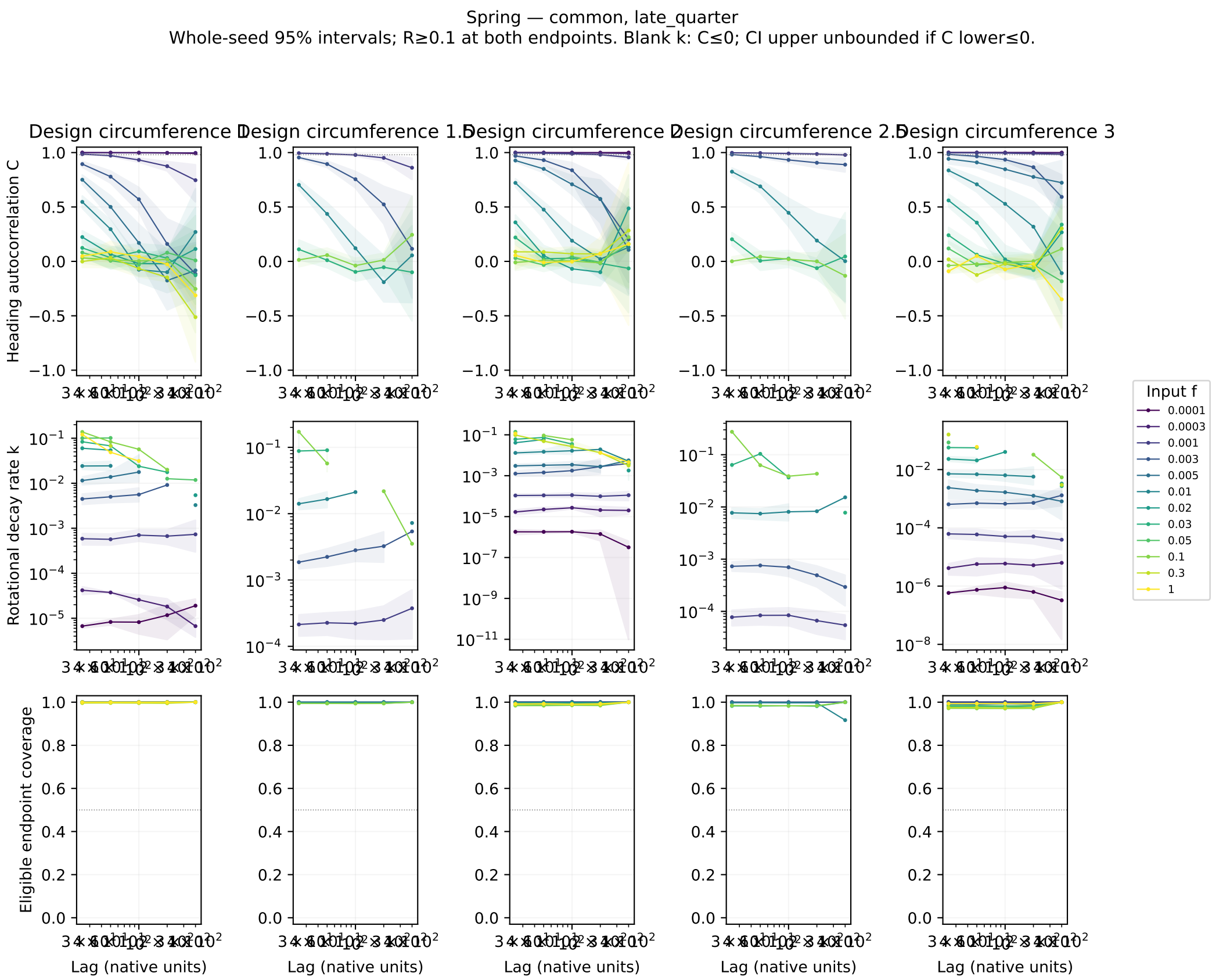

Holmes — common, analysis_second_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

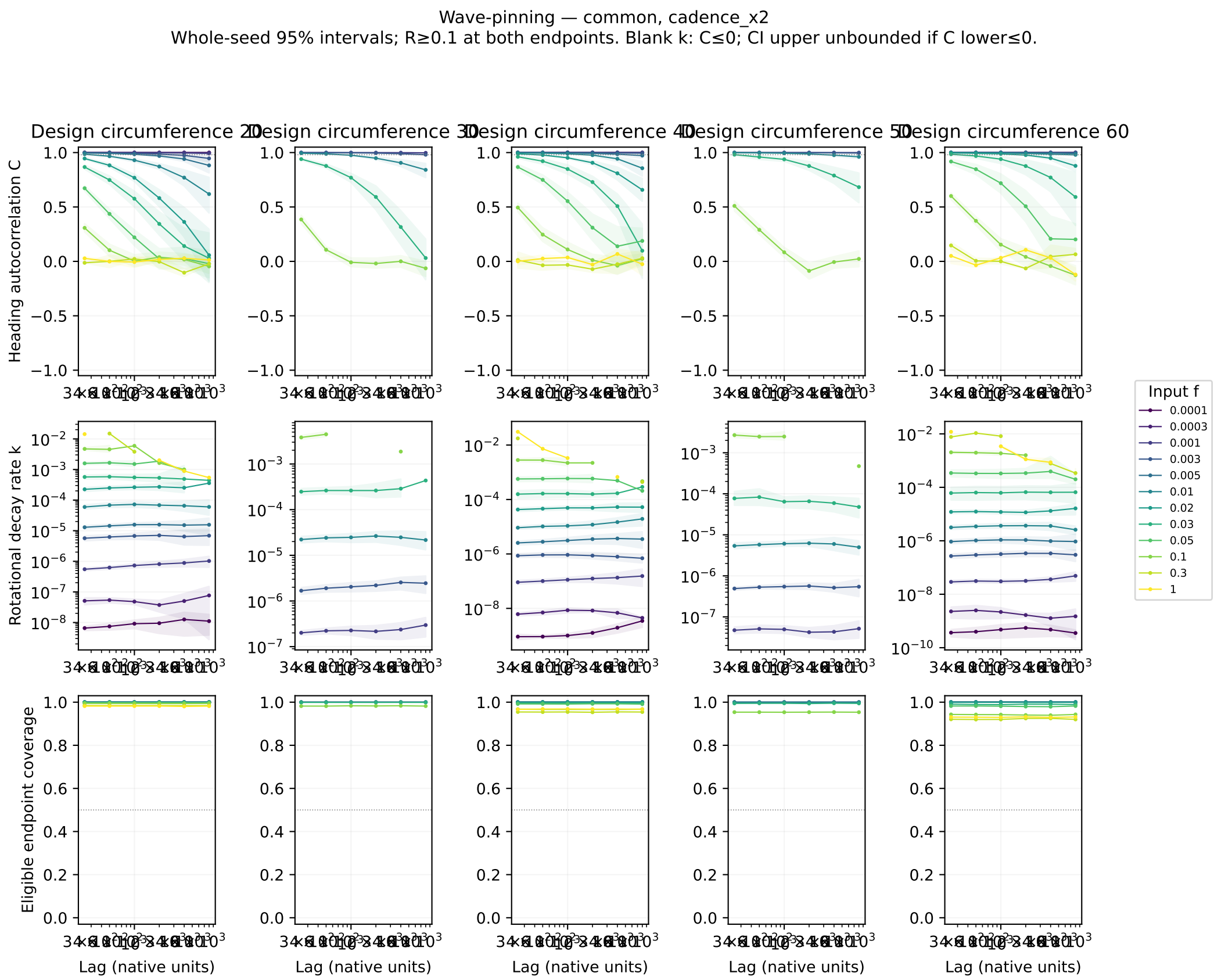

Spring — common, analysis_second_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

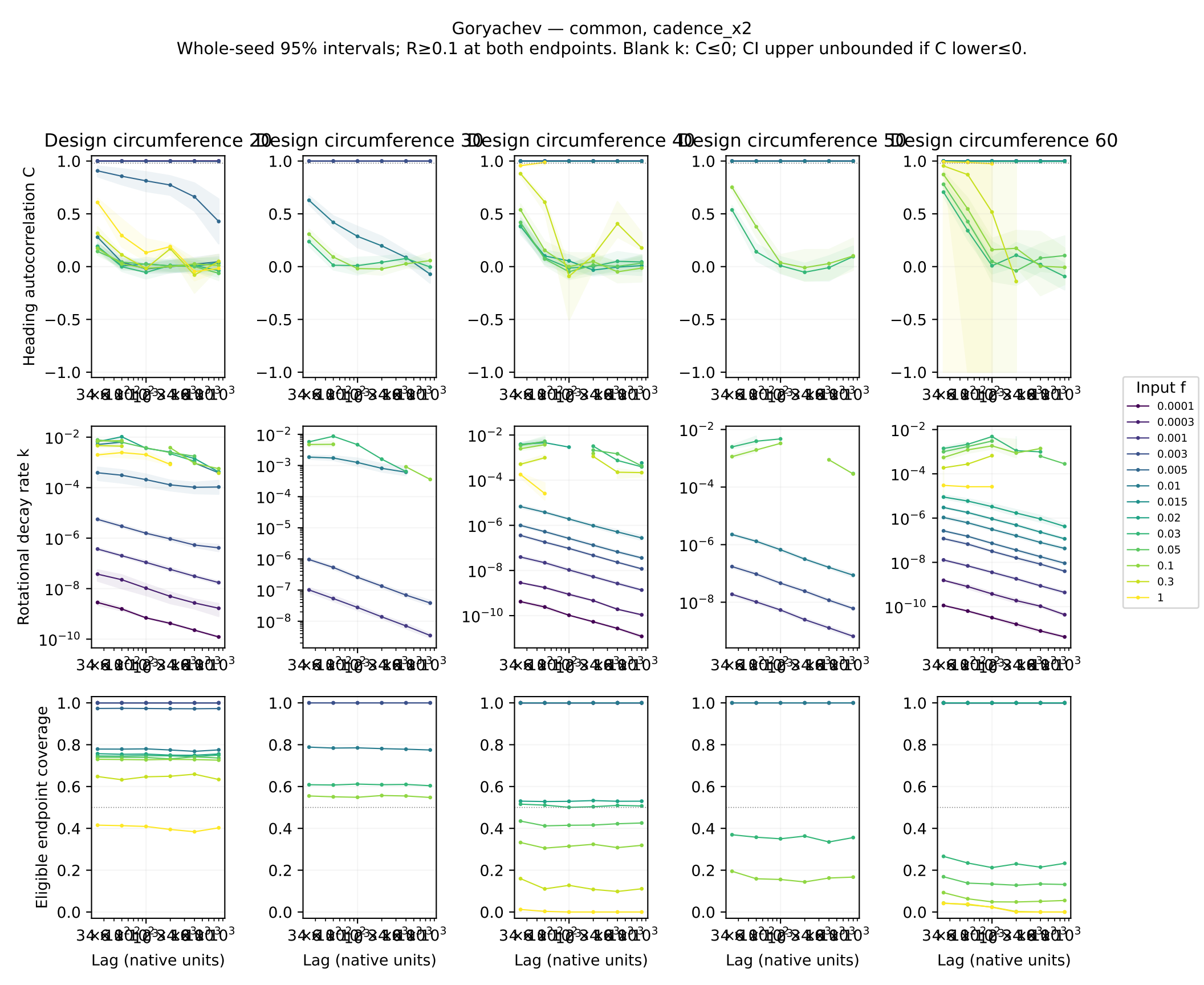

Wave-pinning — common, late_quarter
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

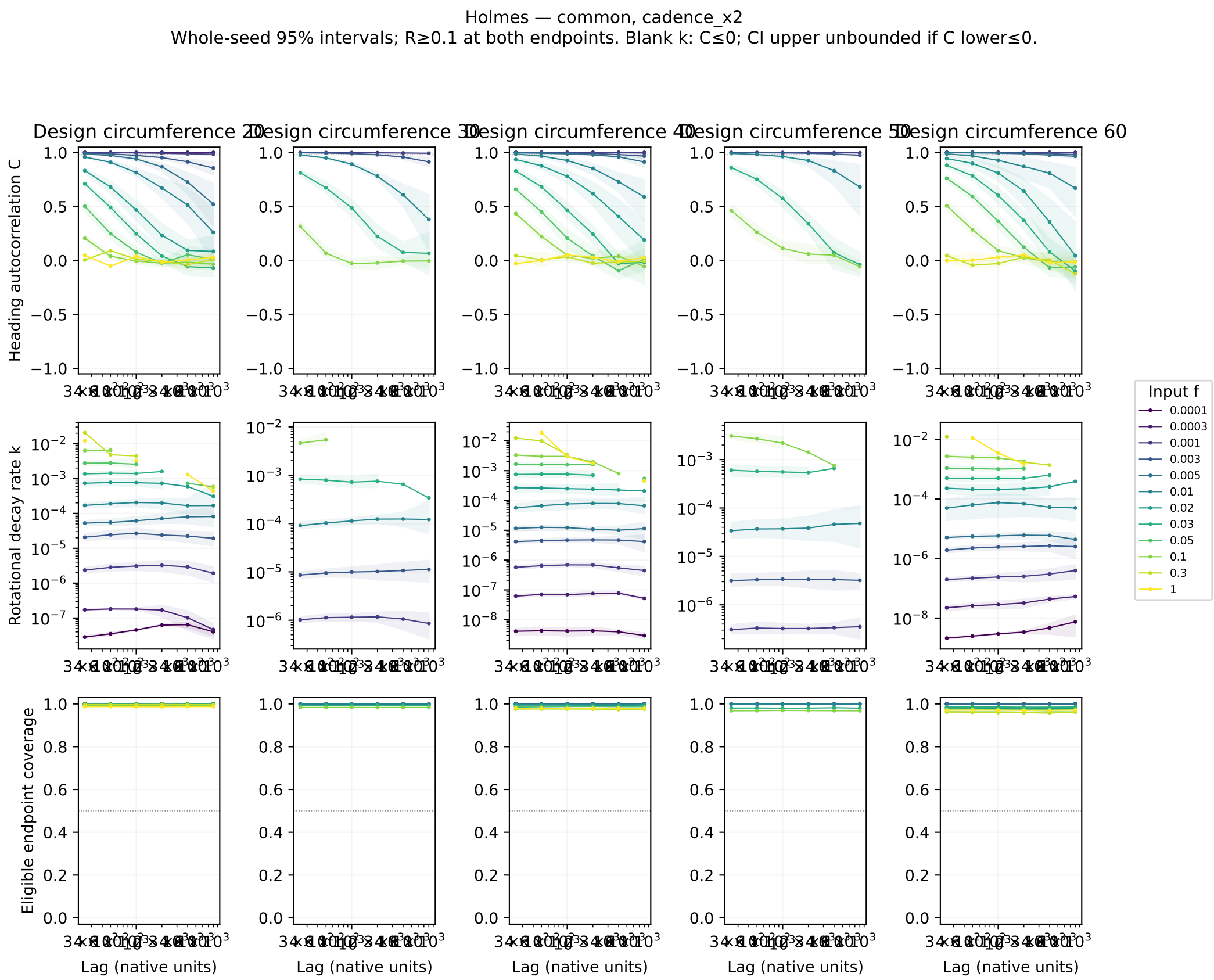

Goryachev — common, late_quarter
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1

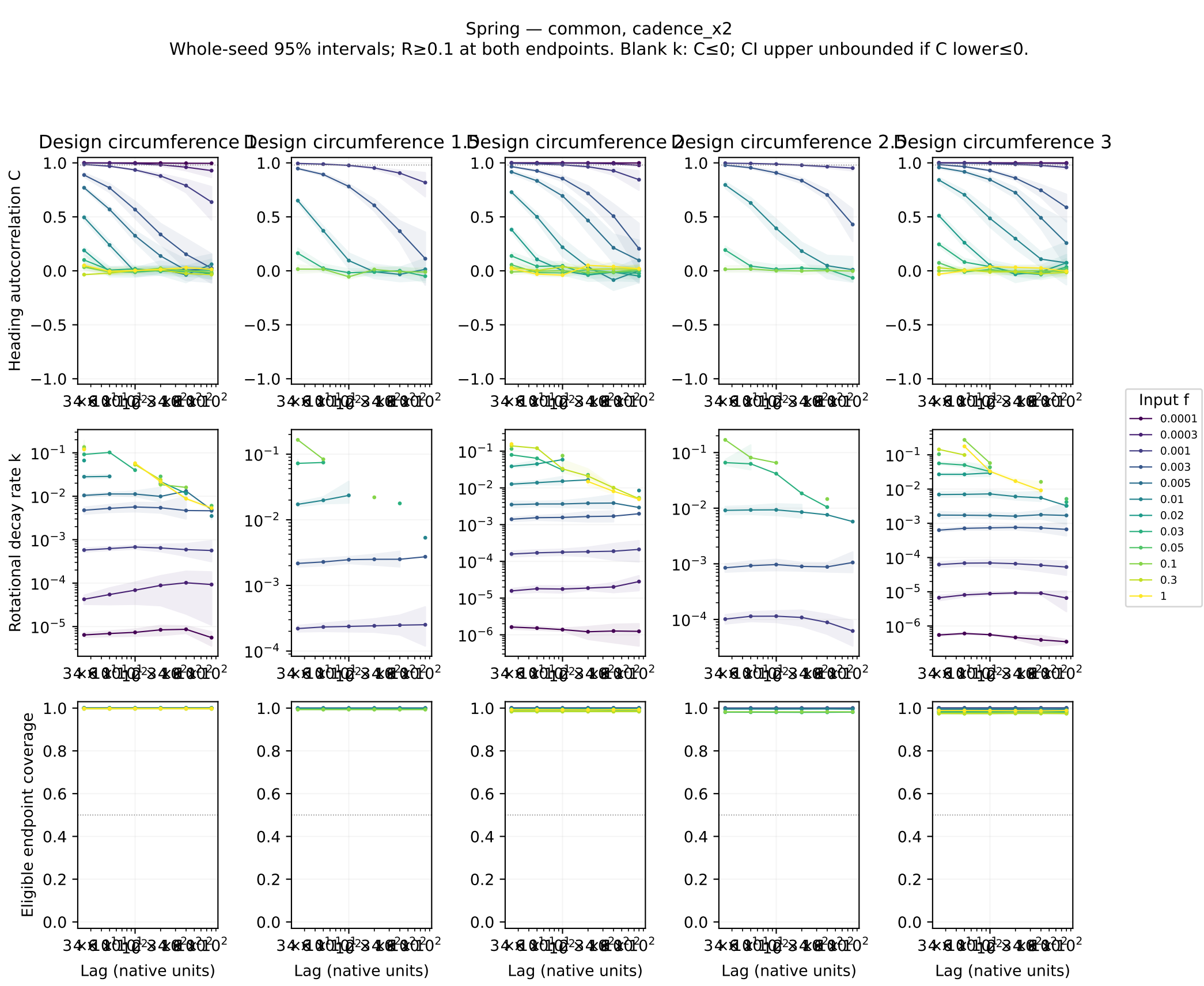

Holmes — common, late_quarter
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

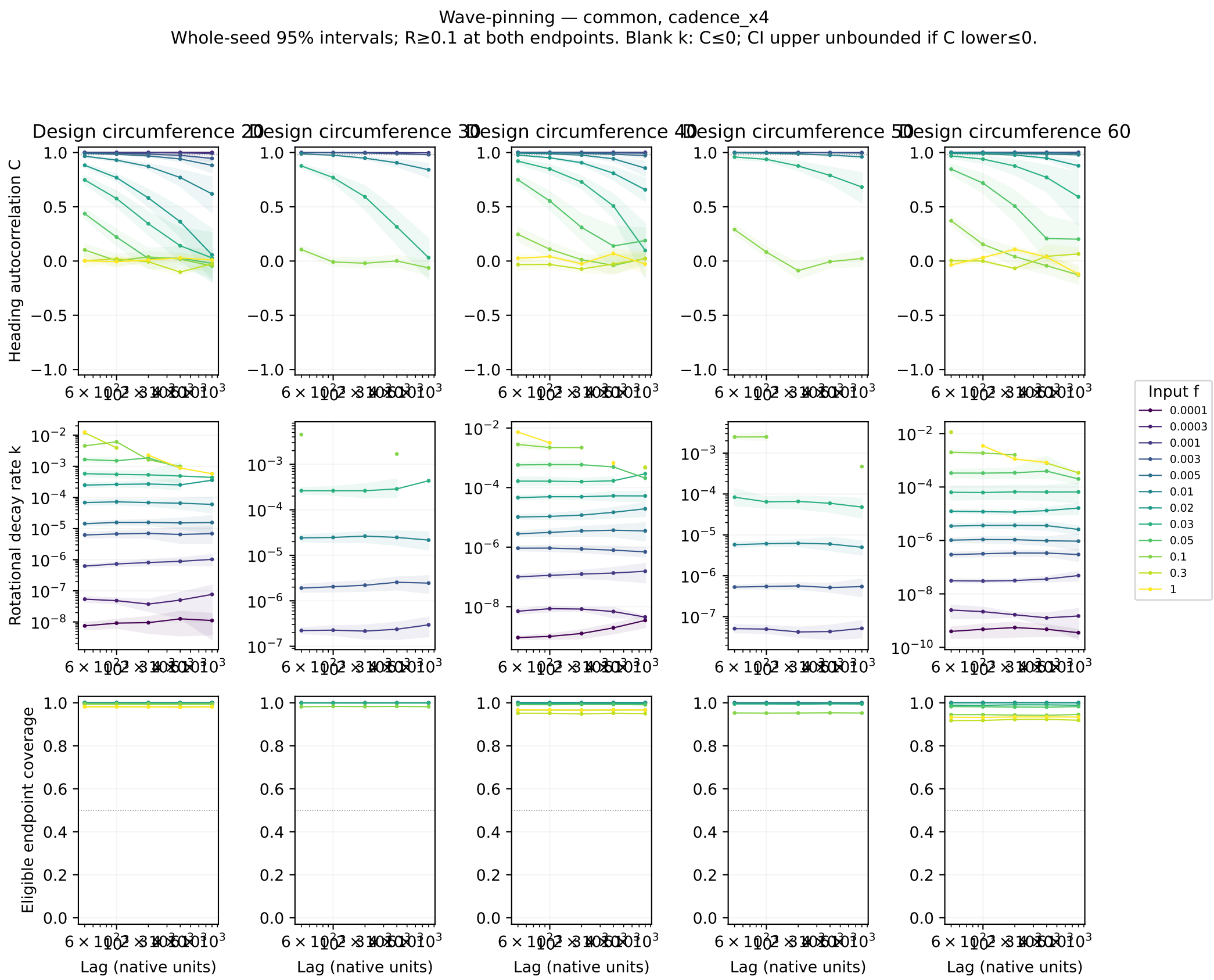

Spring — common, late_quarter
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

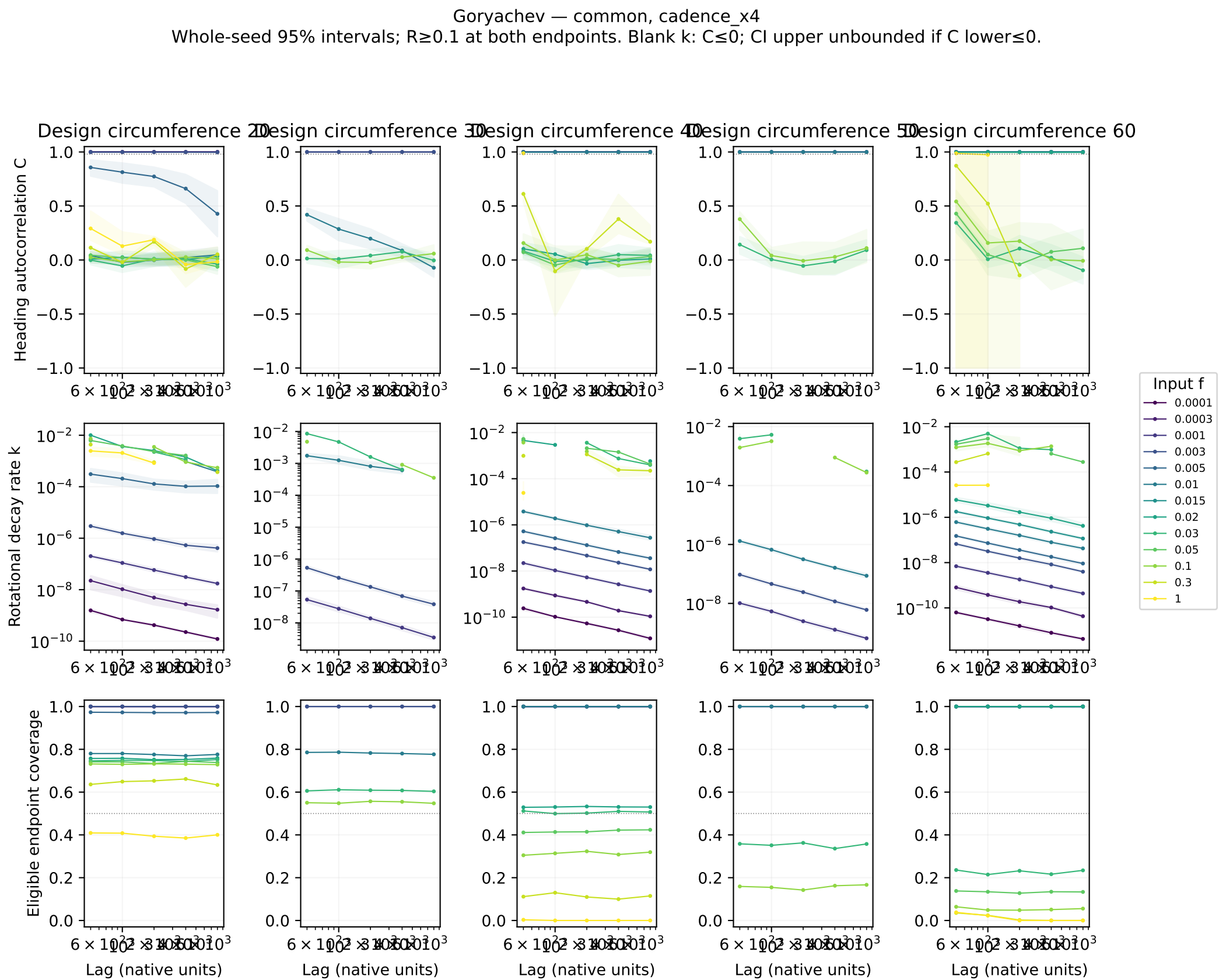

Wave-pinning — common, cadence_x2
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

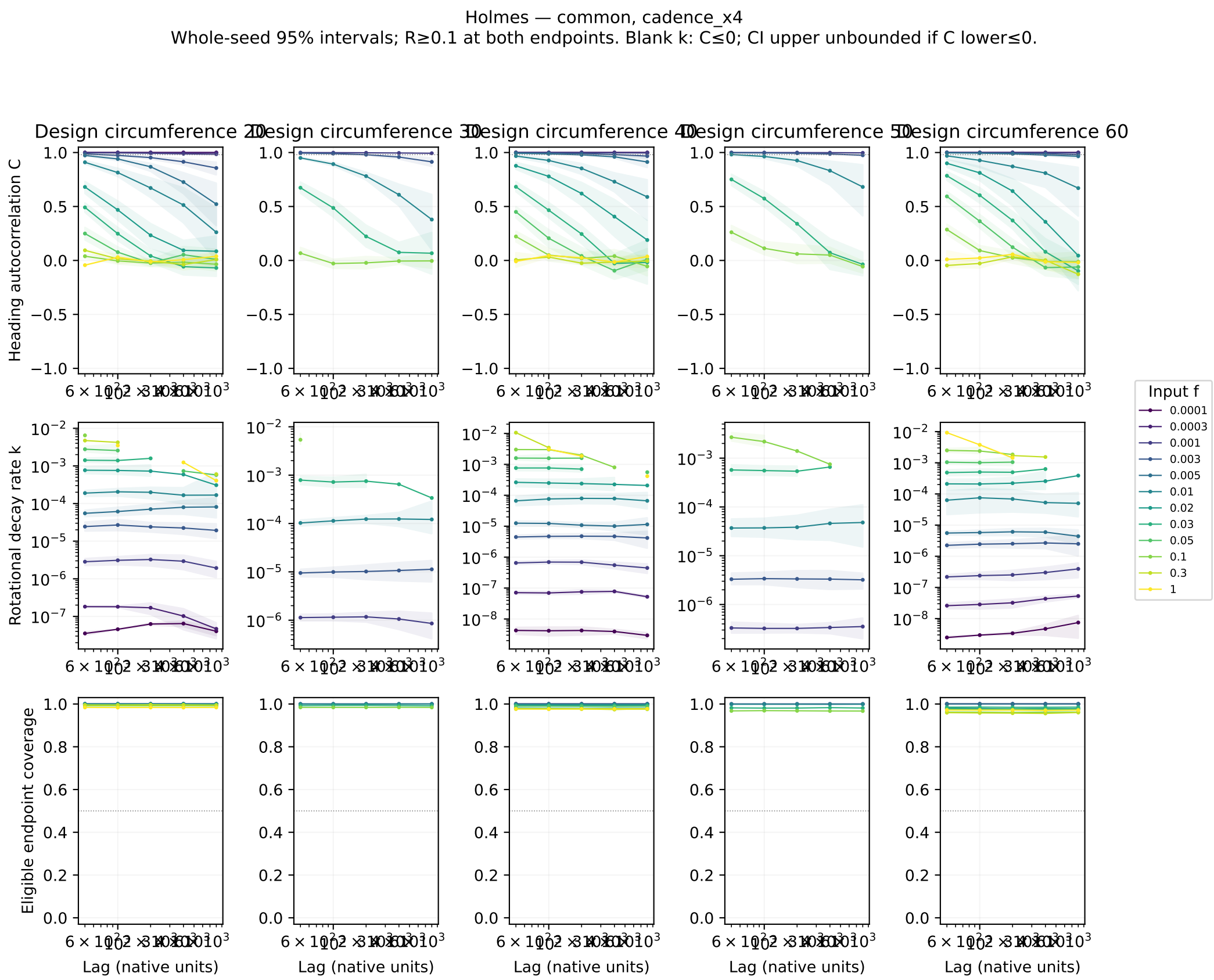

Goryachev — common, cadence_x2
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1

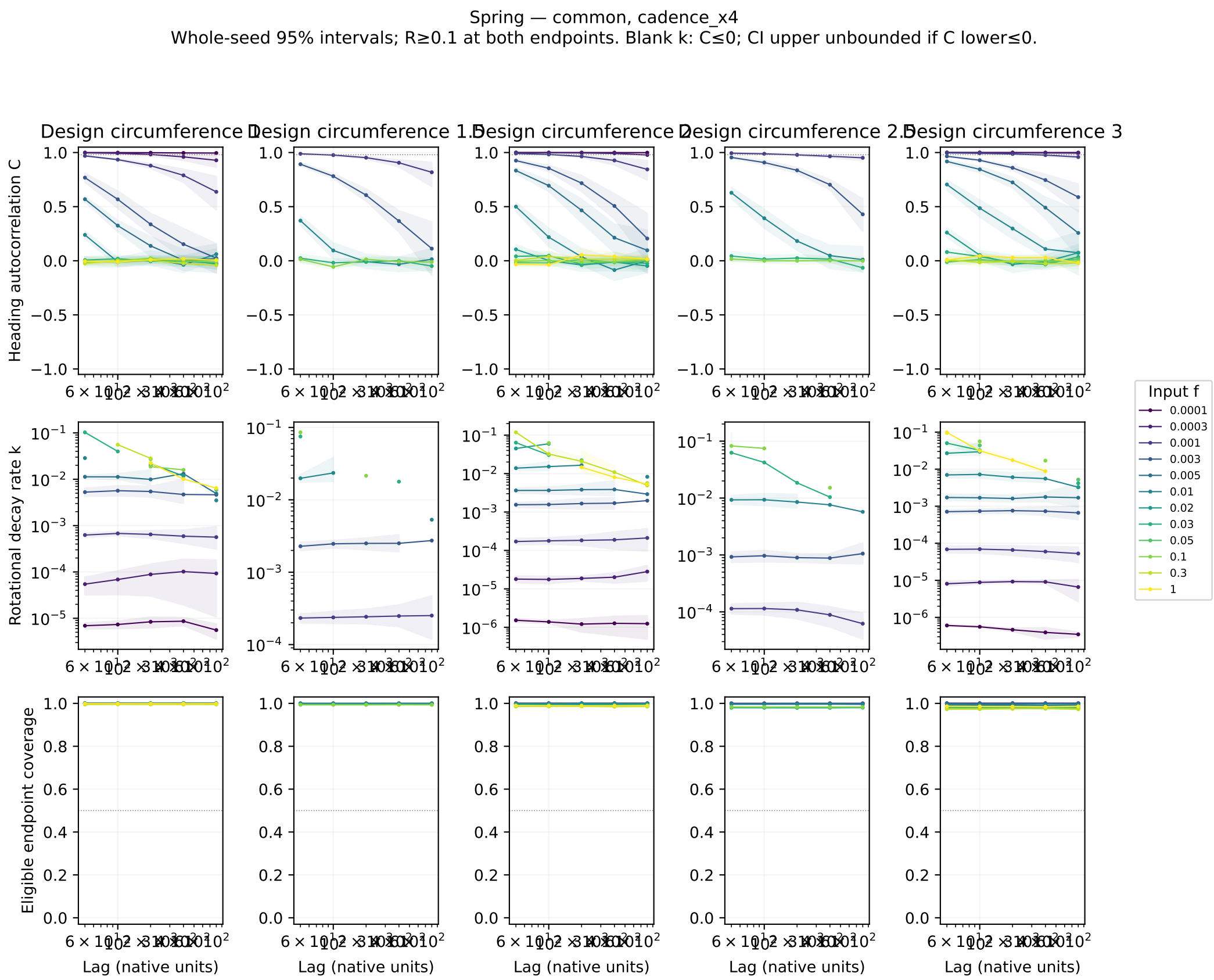

Holmes — common, cadence_x2
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

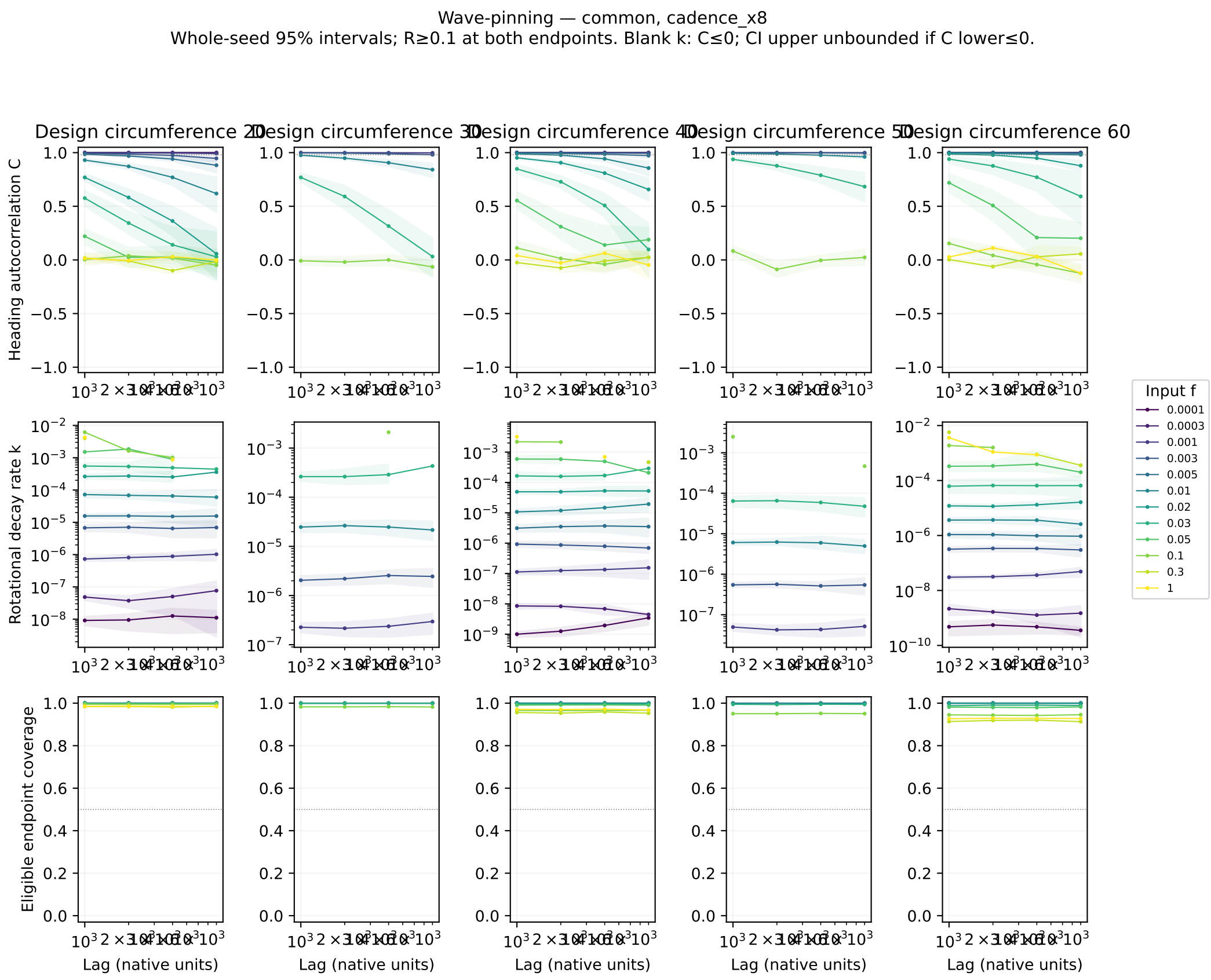

Spring — common, cadence_x2
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

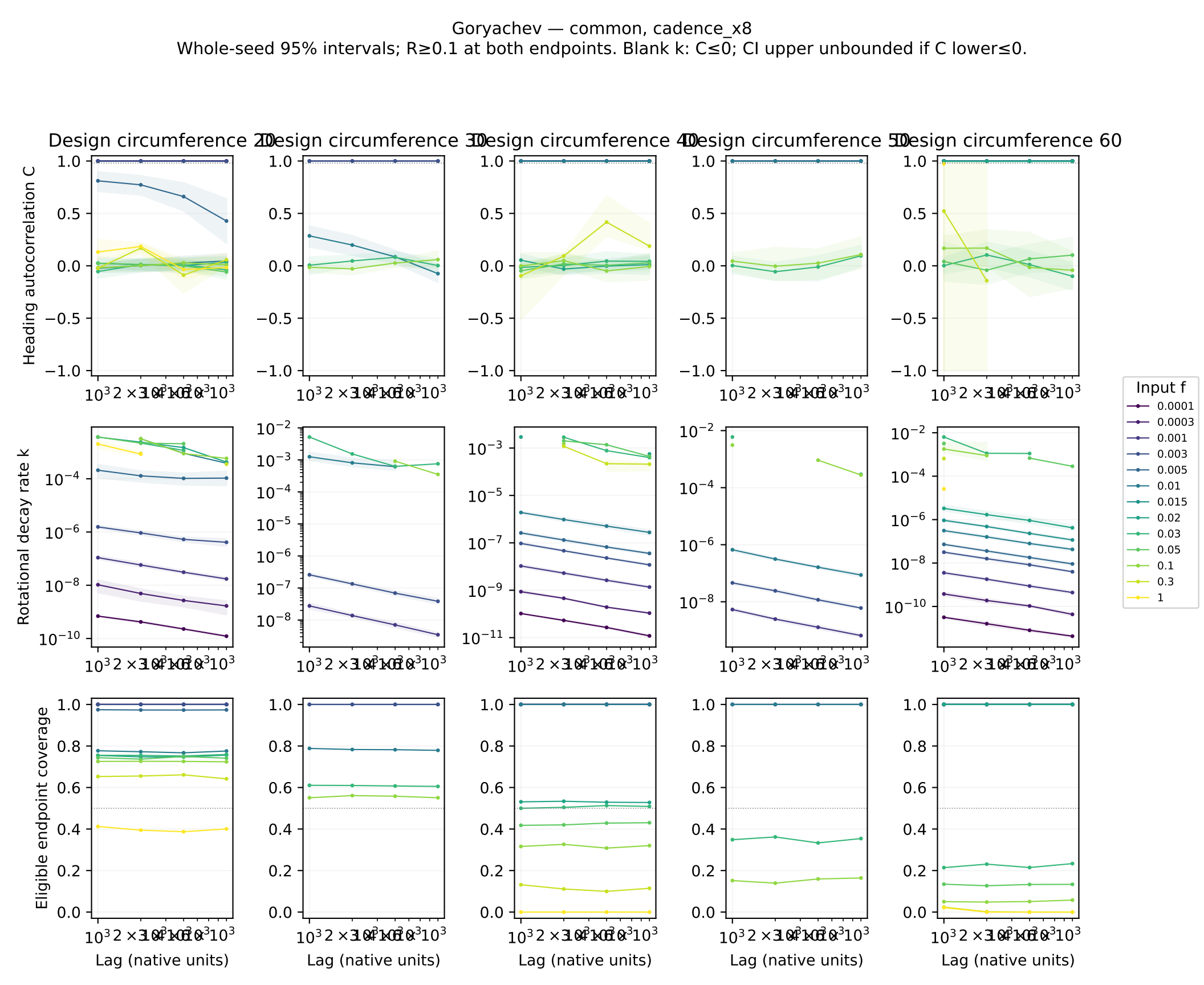

Wave-pinning — common, cadence_x4
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

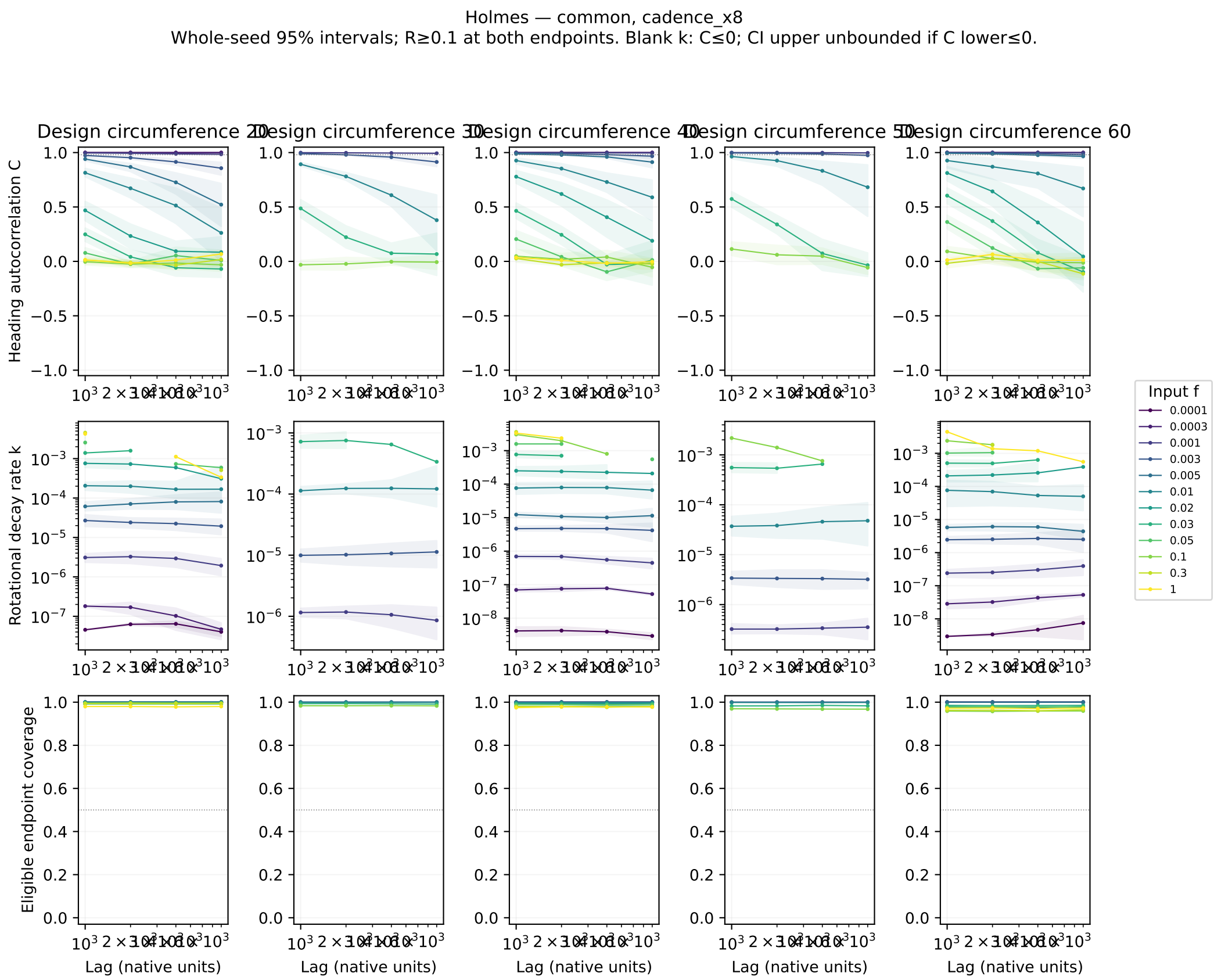

Goryachev — common, cadence_x4
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1

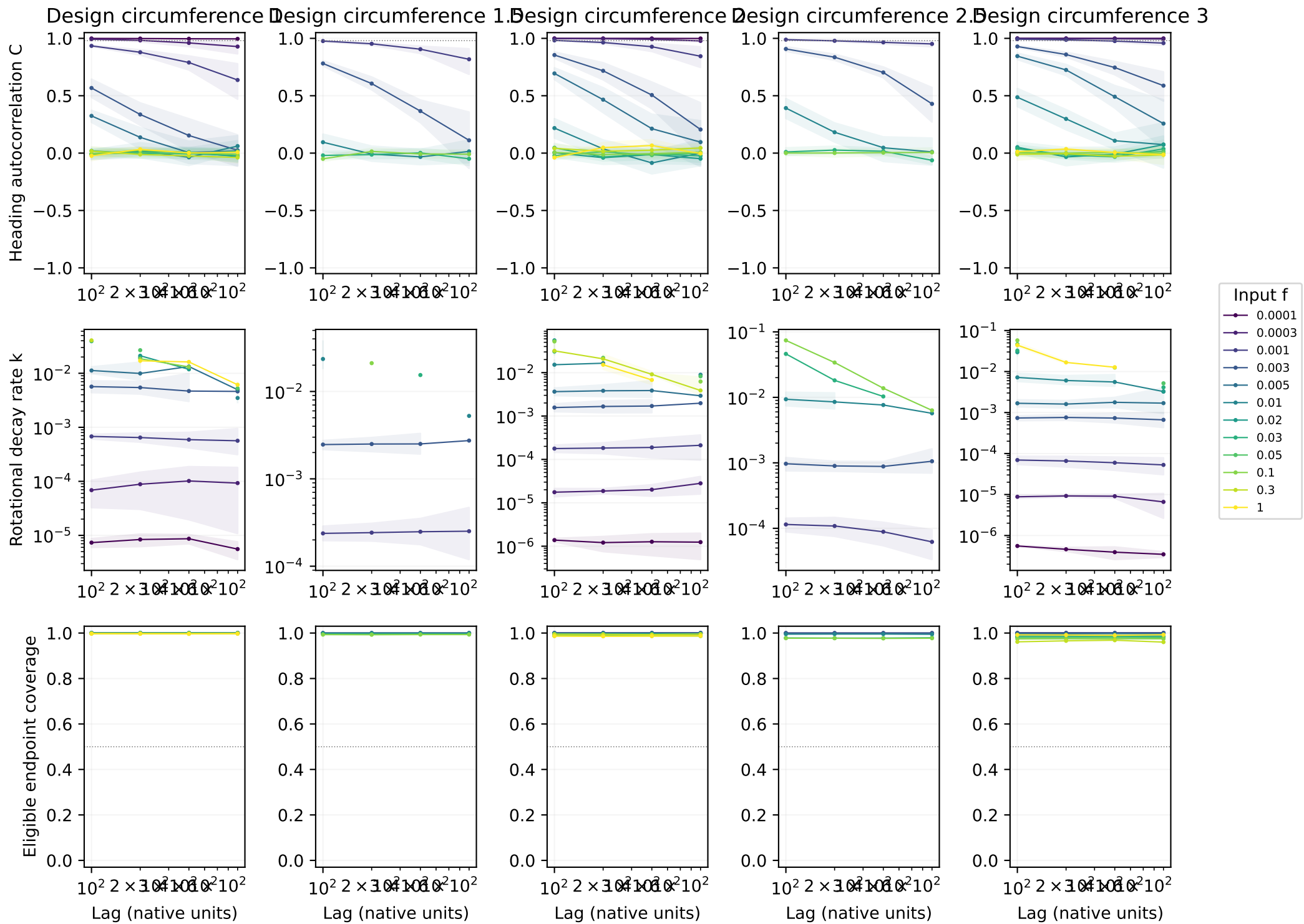

Holmes — common, cadence_x4
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

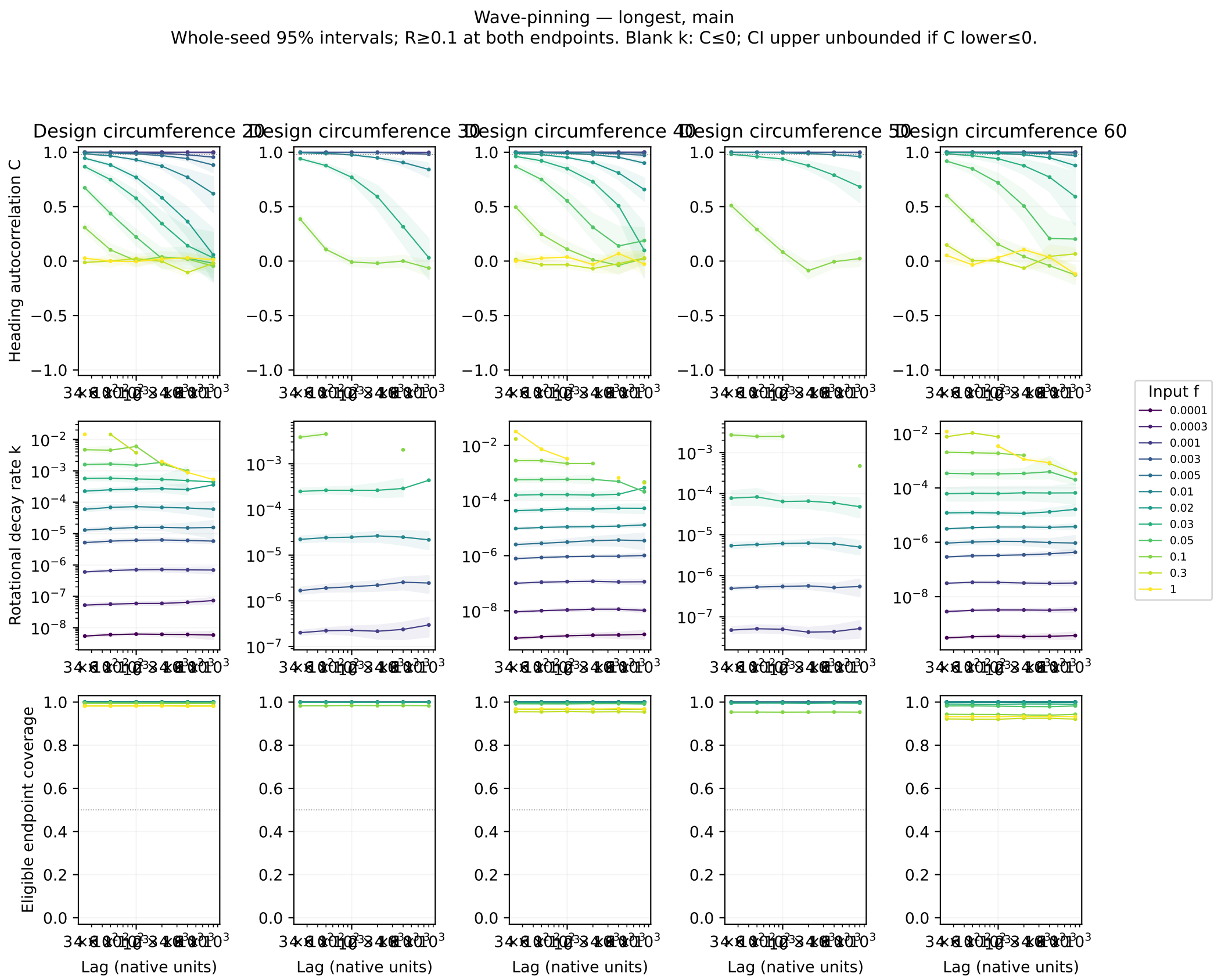

Spring — common, cadence_x4
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

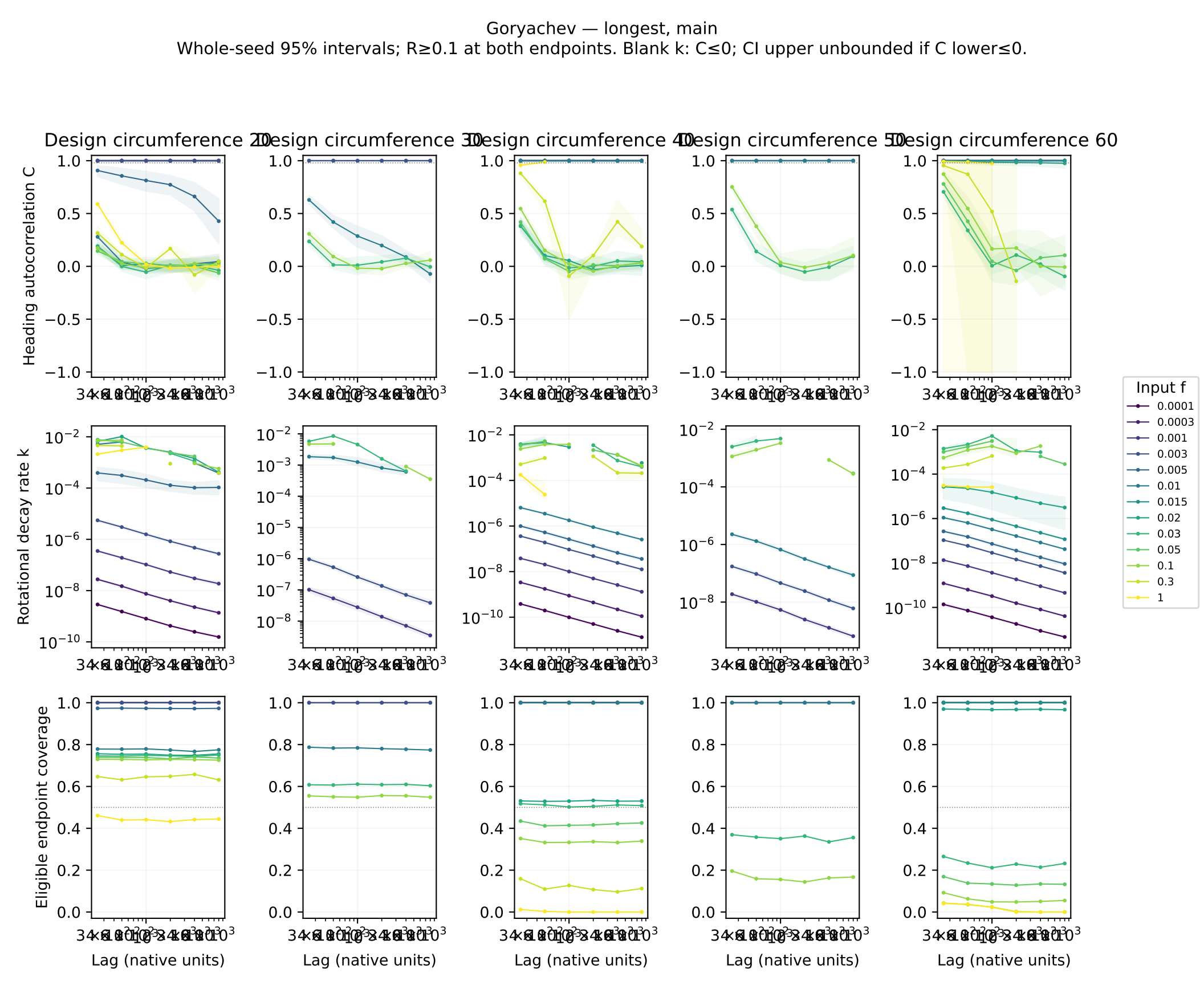

Wave-pinning — common, cadence_x8
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

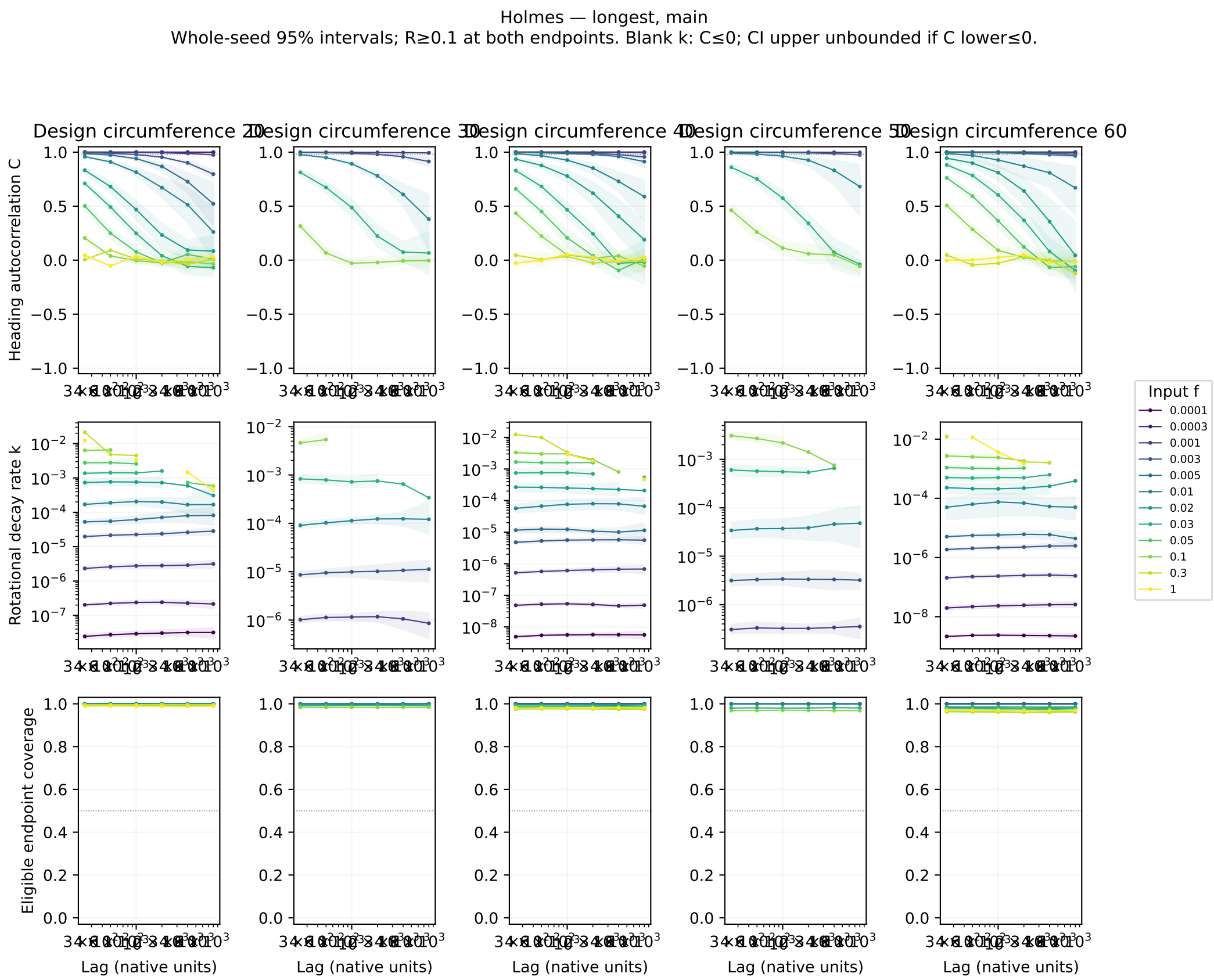

Goryachev — common, cadence_x8
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1

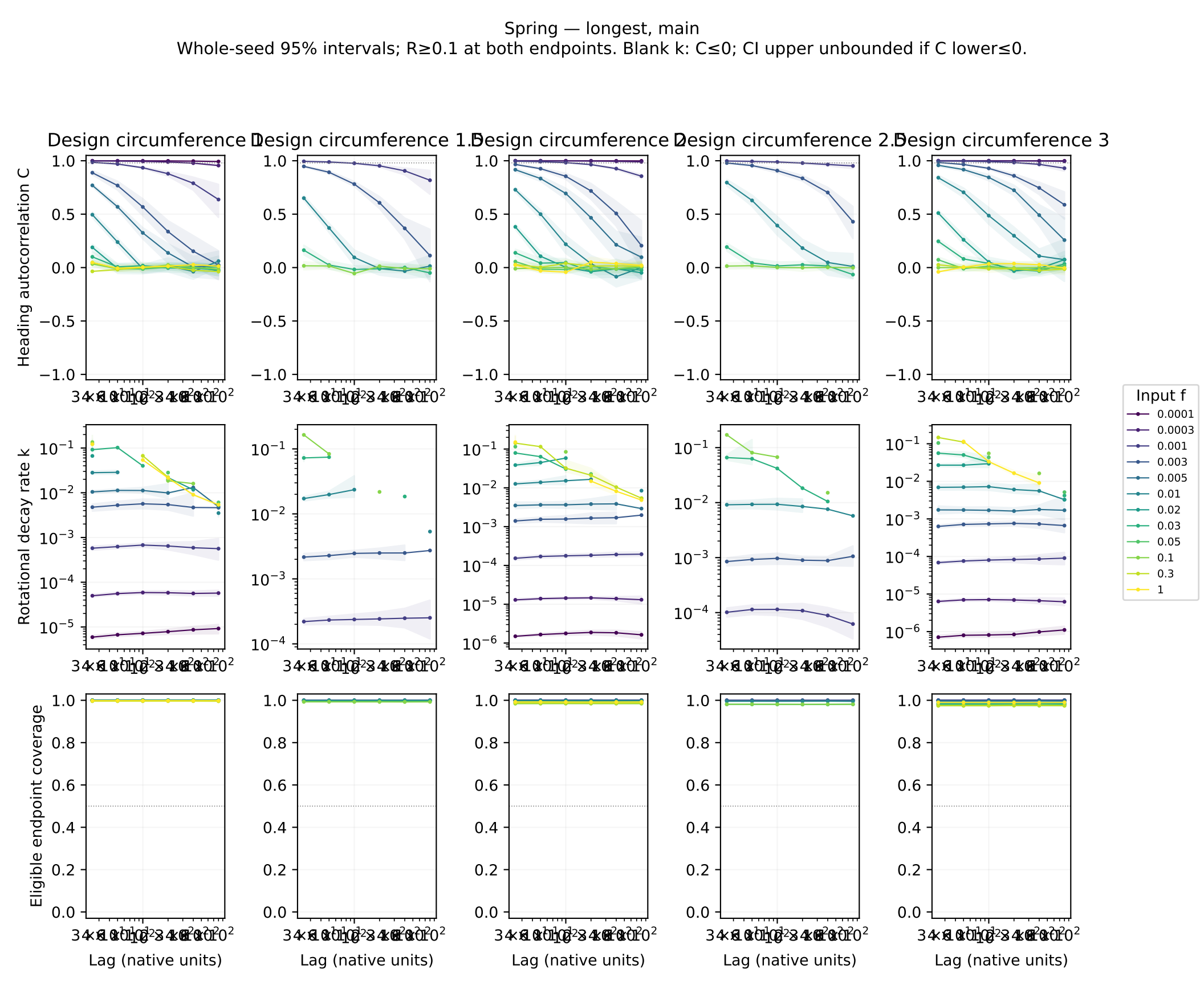

Holmes — common, cadence_x8
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

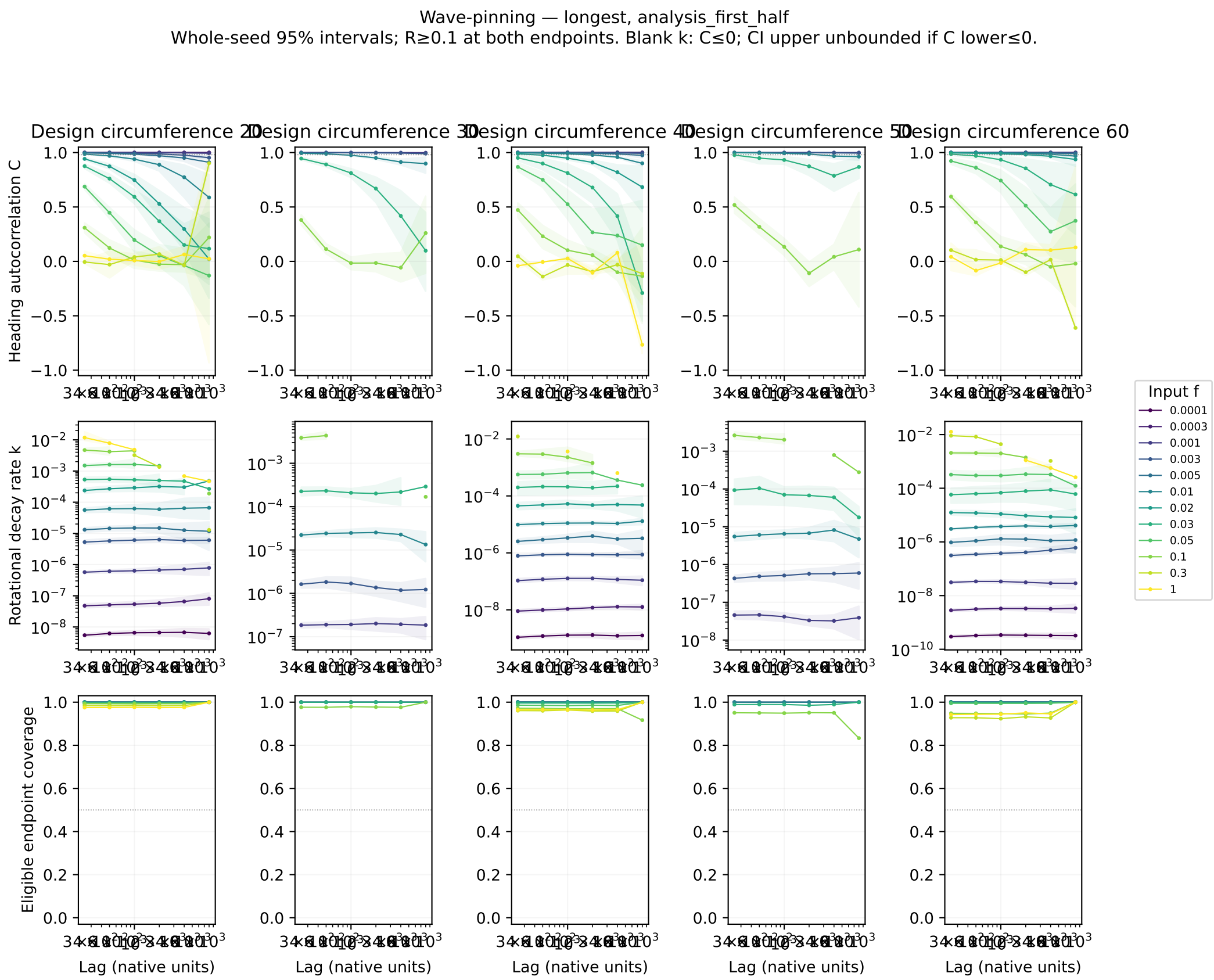

Spring — common, cadence_x8
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

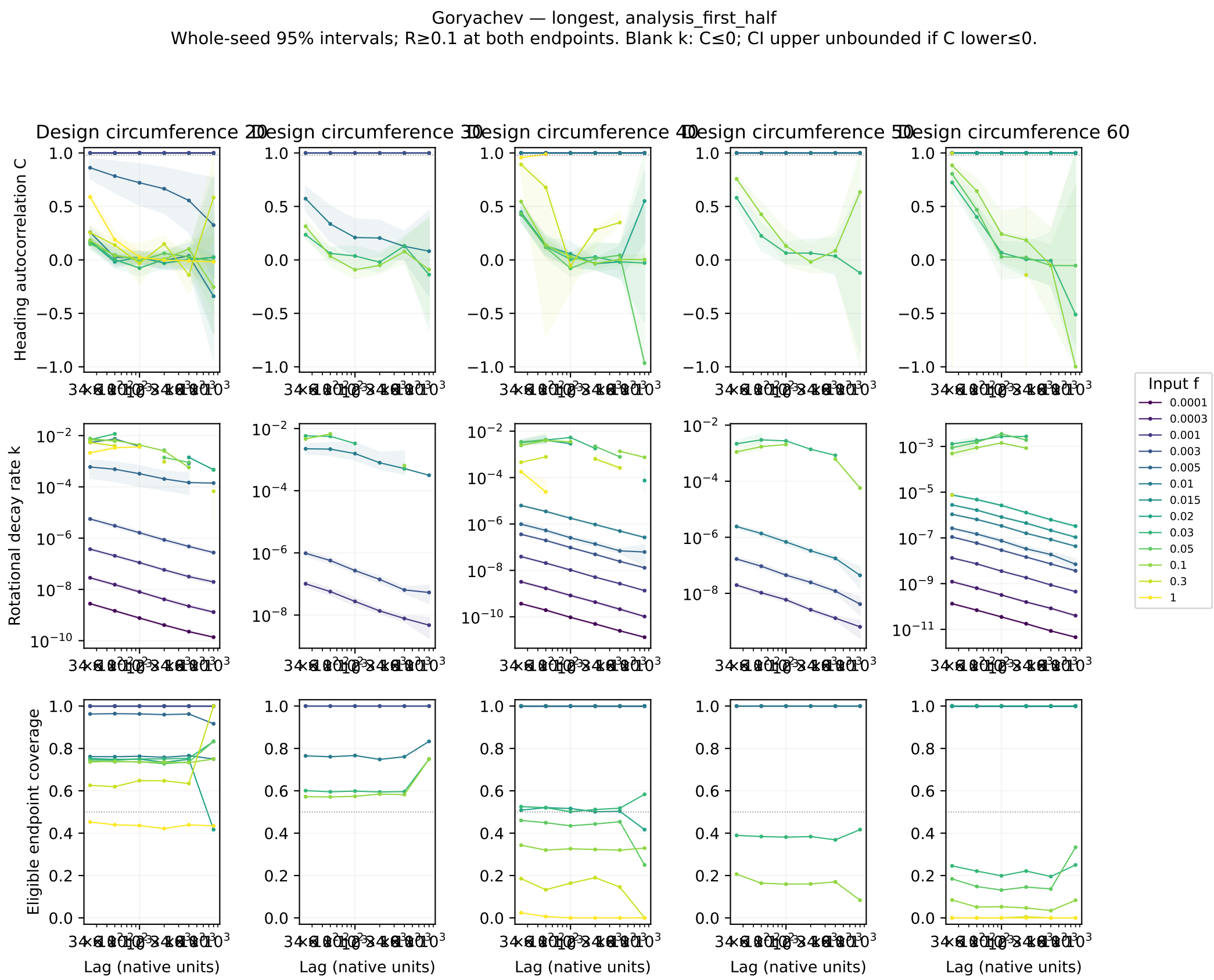

Wave-pinning — longest, main
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

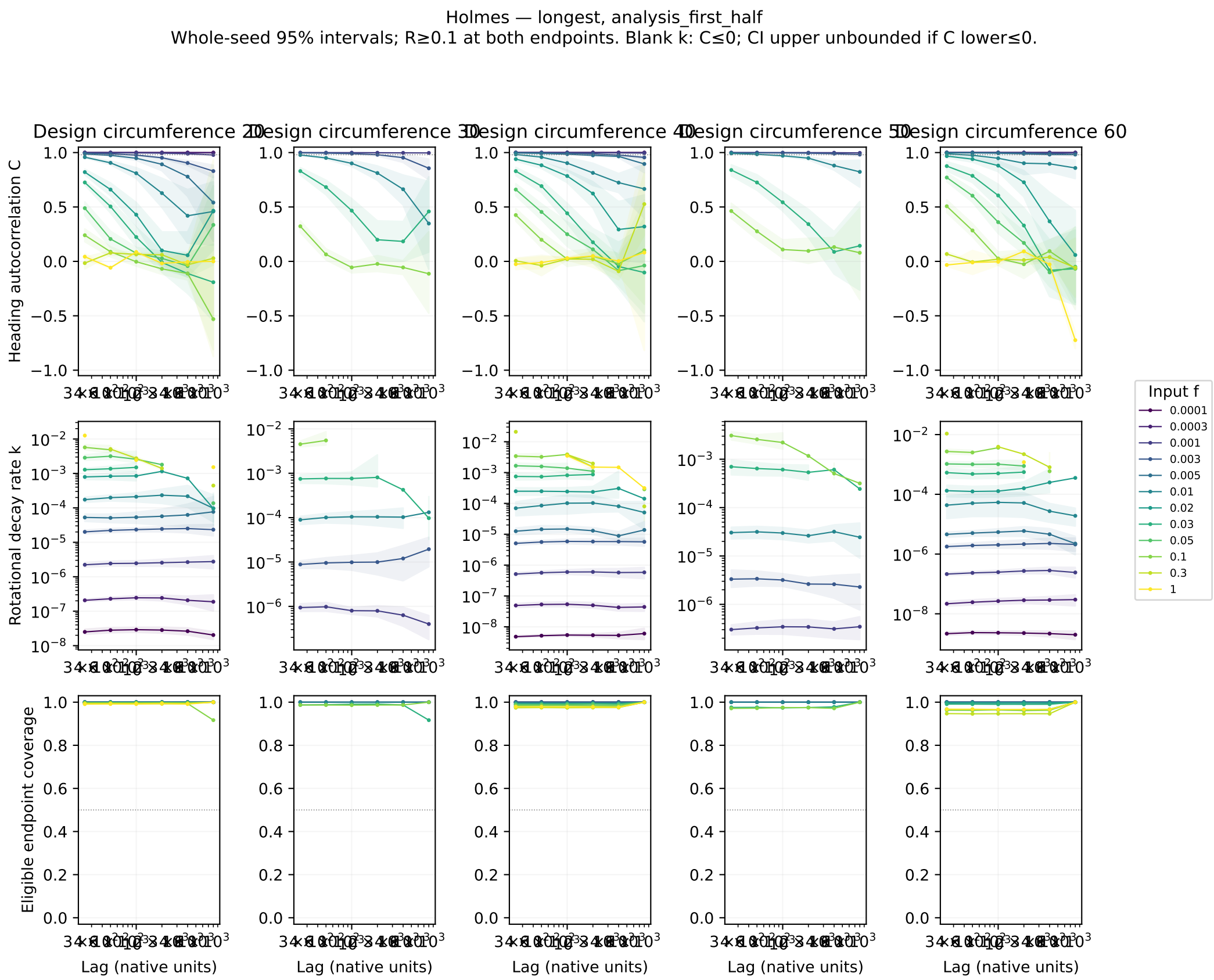

Goryachev — longest, main
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1

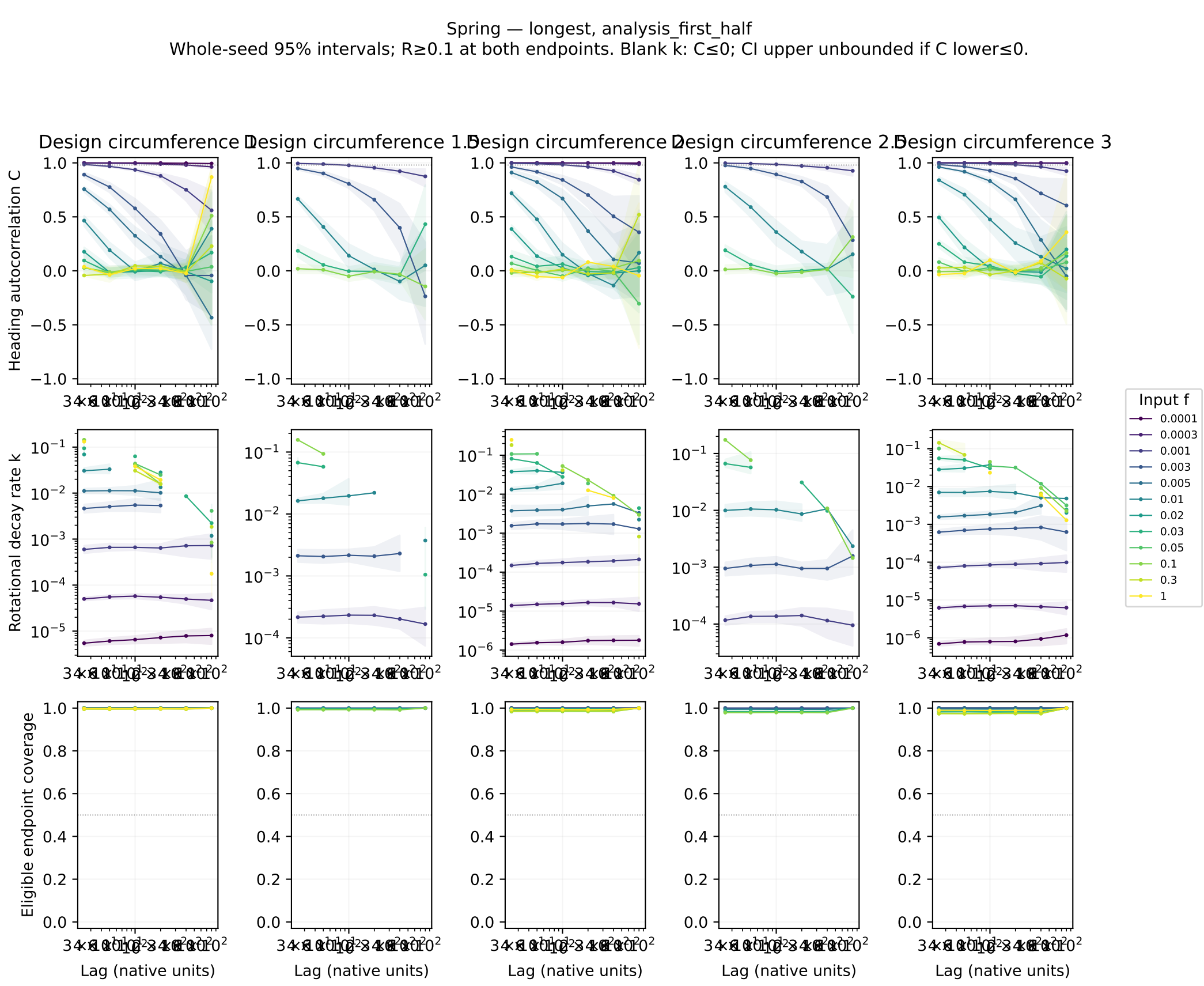

Holmes — longest, main
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

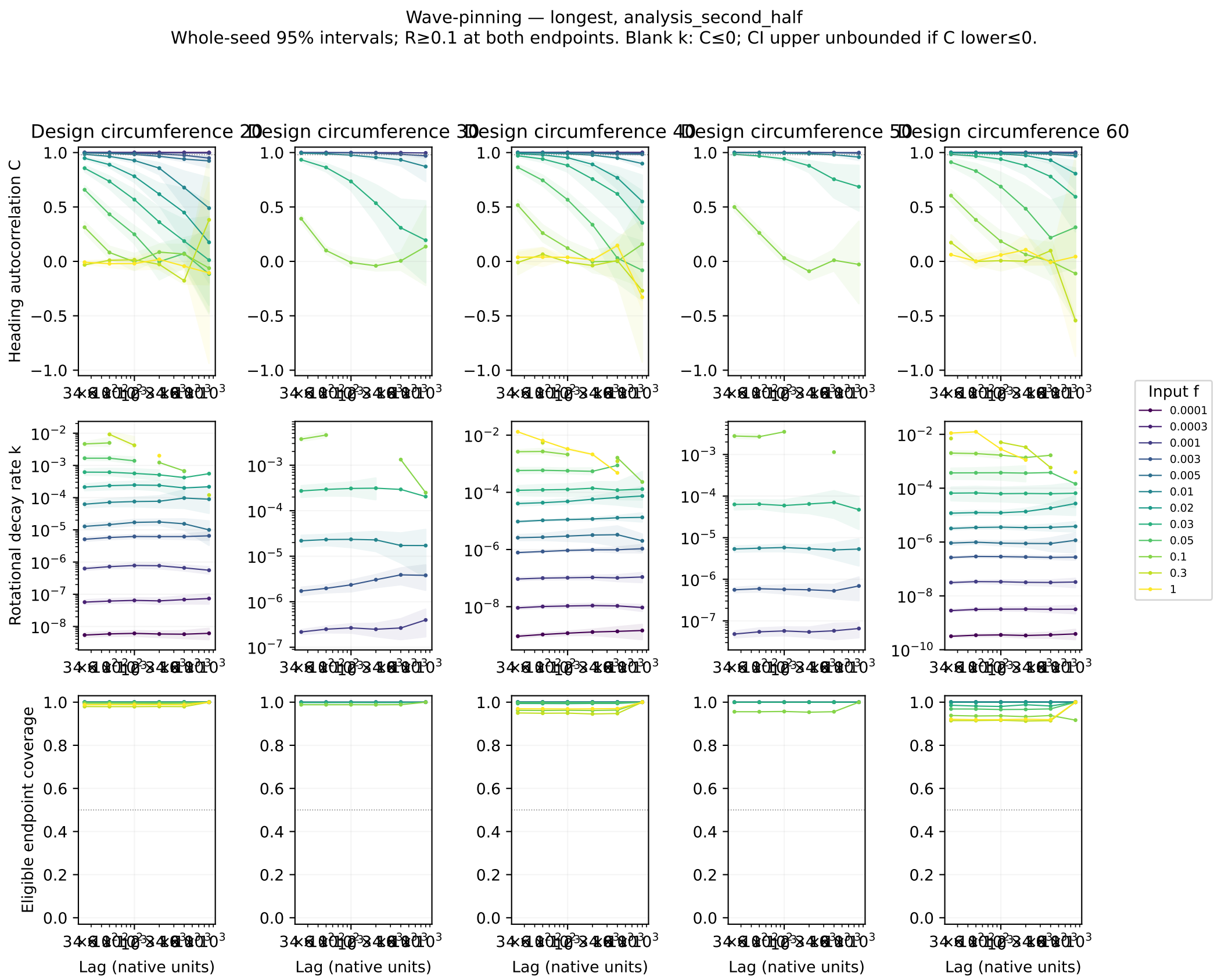

Spring — longest, main
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

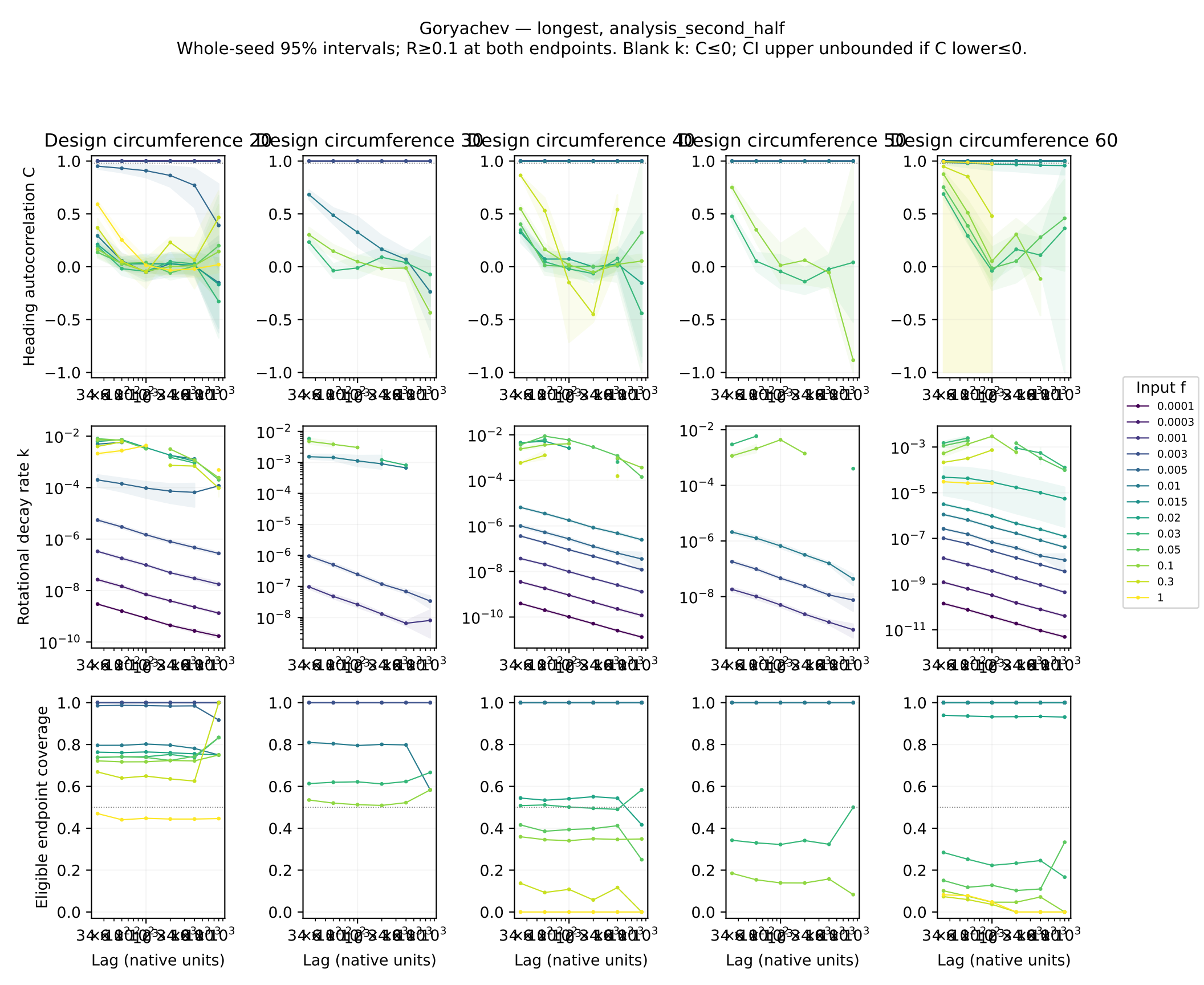

Wave-pinning — longest, analysis_first_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

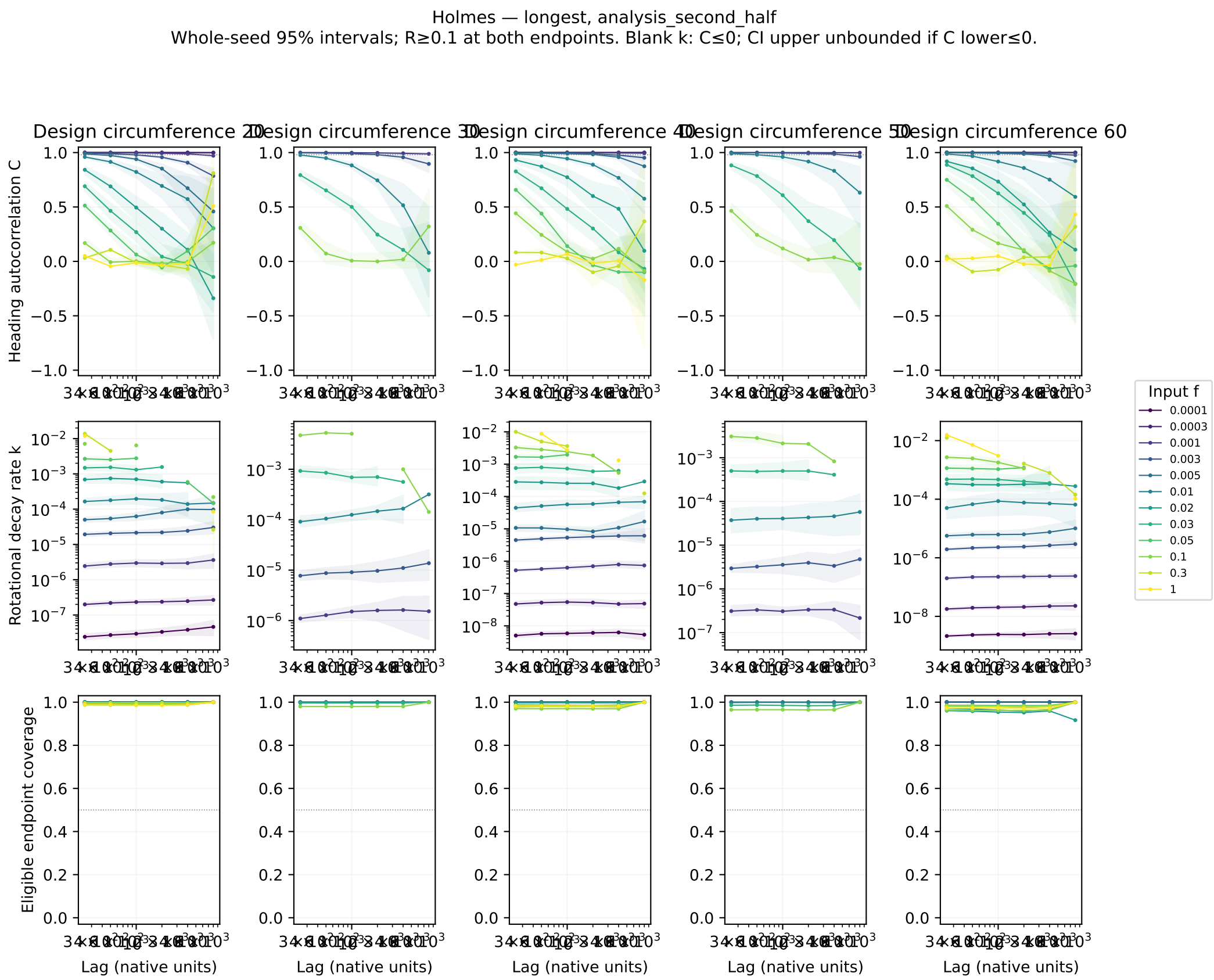

Goryachev — longest, analysis_first_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1

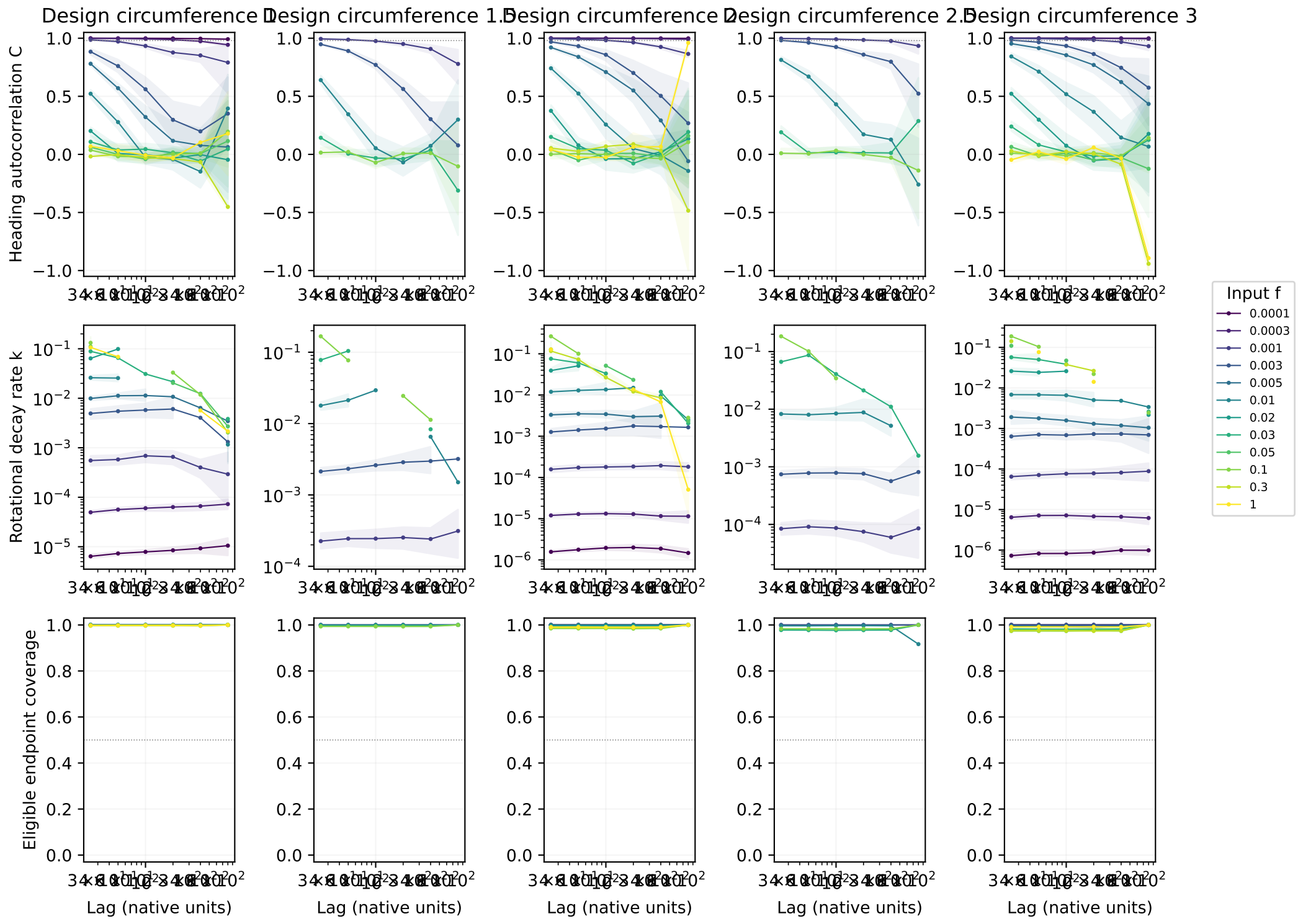

Holmes — longest, analysis_first_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

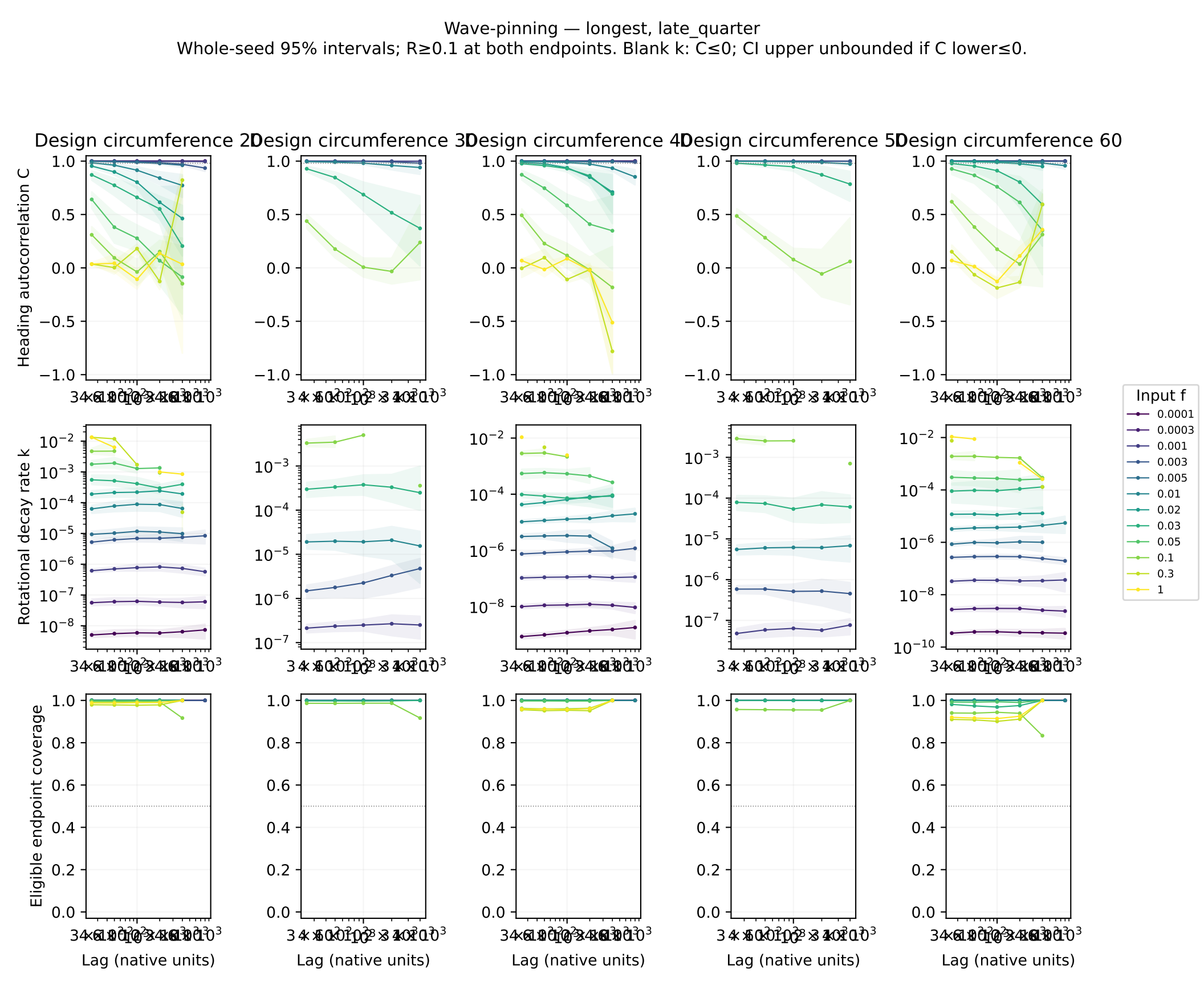

Spring — longest, analysis_first_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

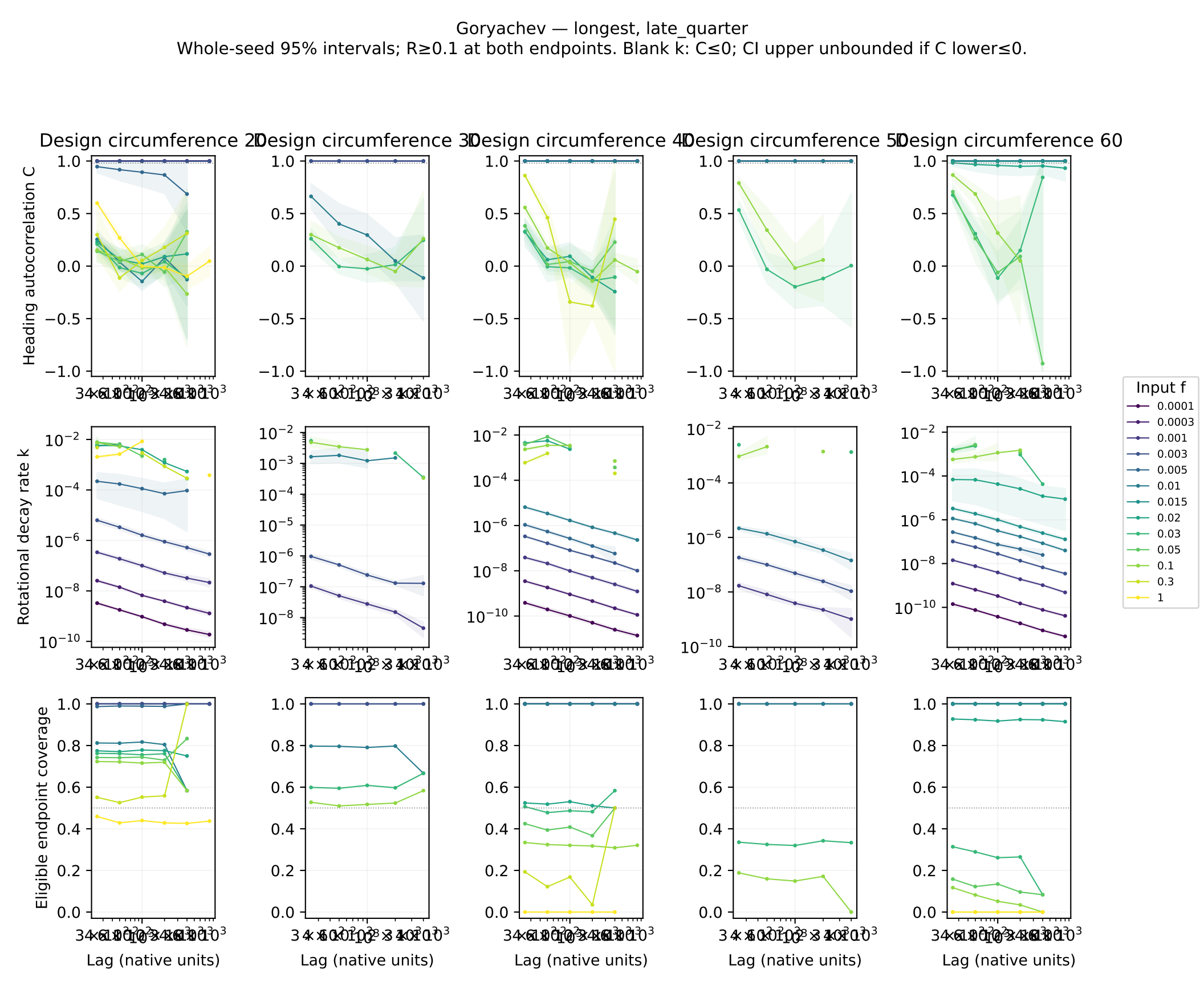

Wave-pinning — longest, analysis_second_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

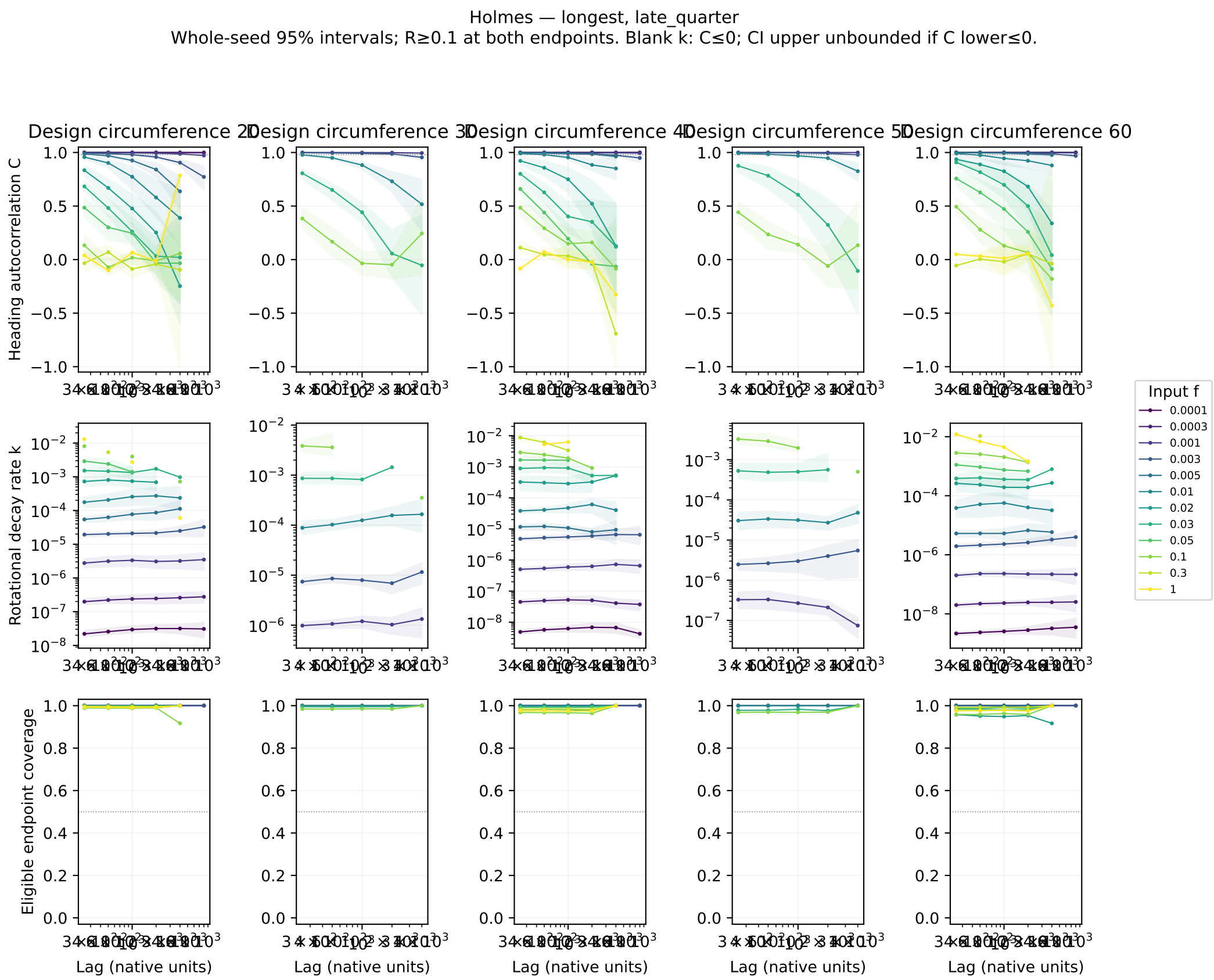

Goryachev — longest, analysis_second_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1

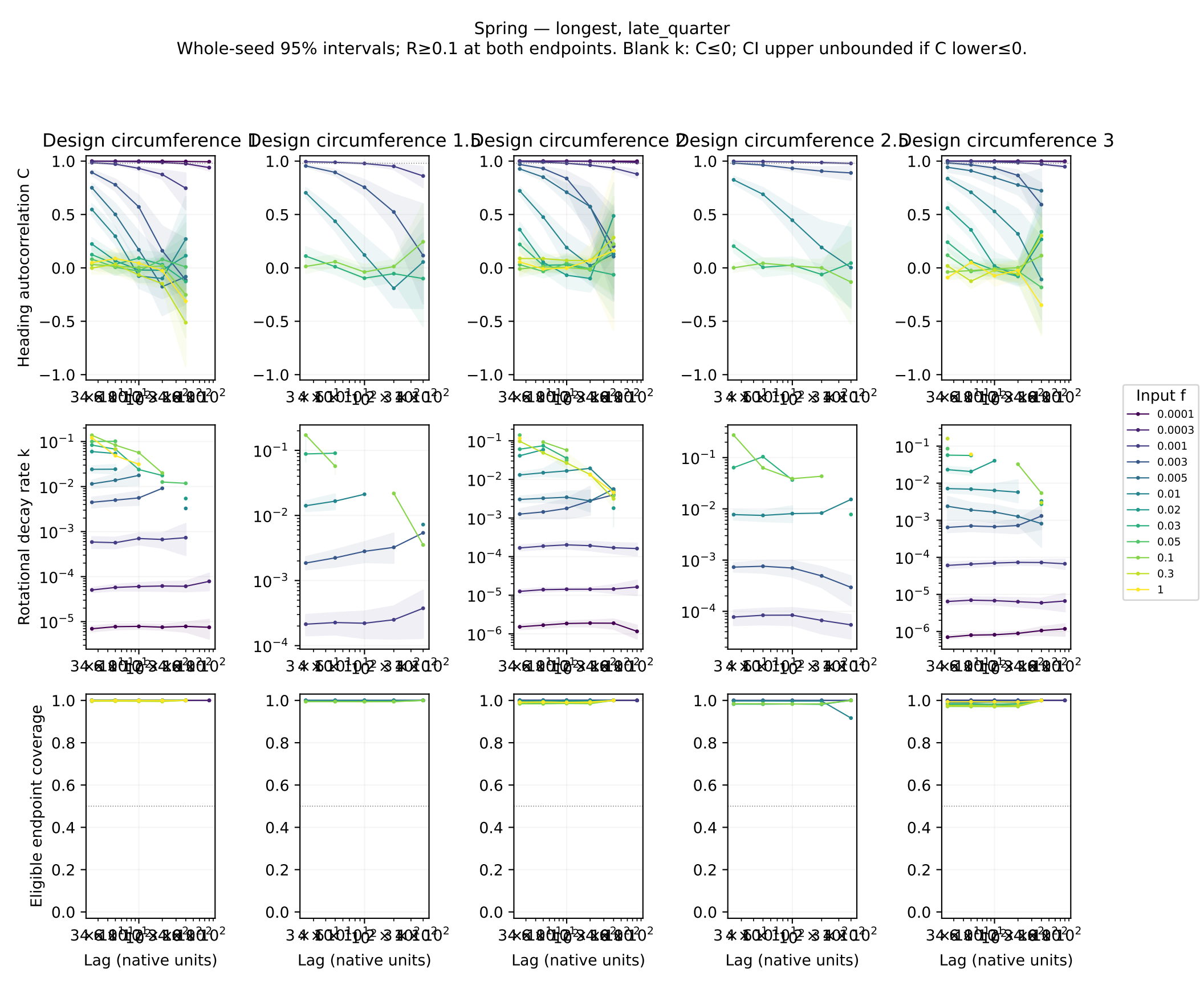

Holmes — longest, analysis_second_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

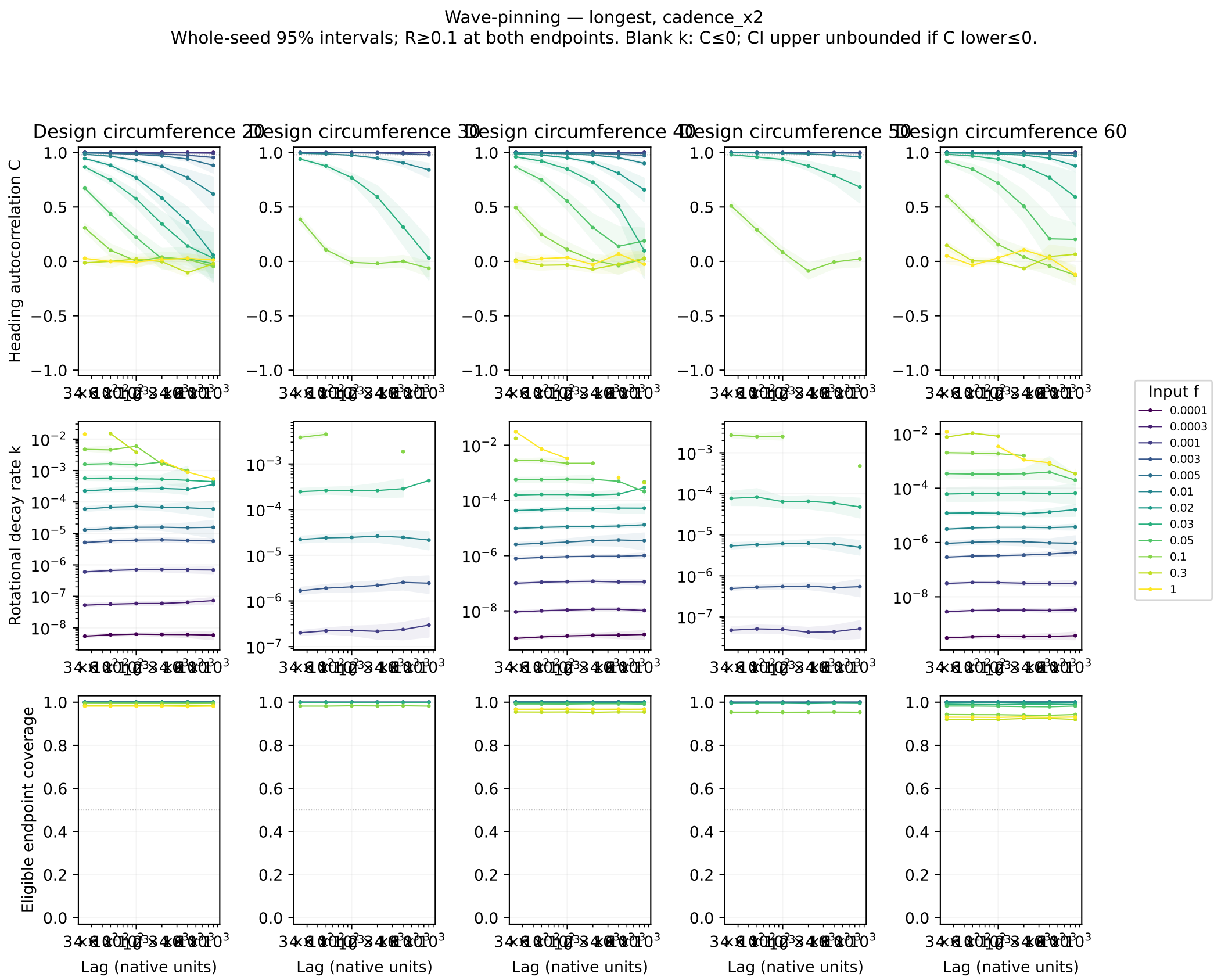

Spring — longest, analysis_second_half
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

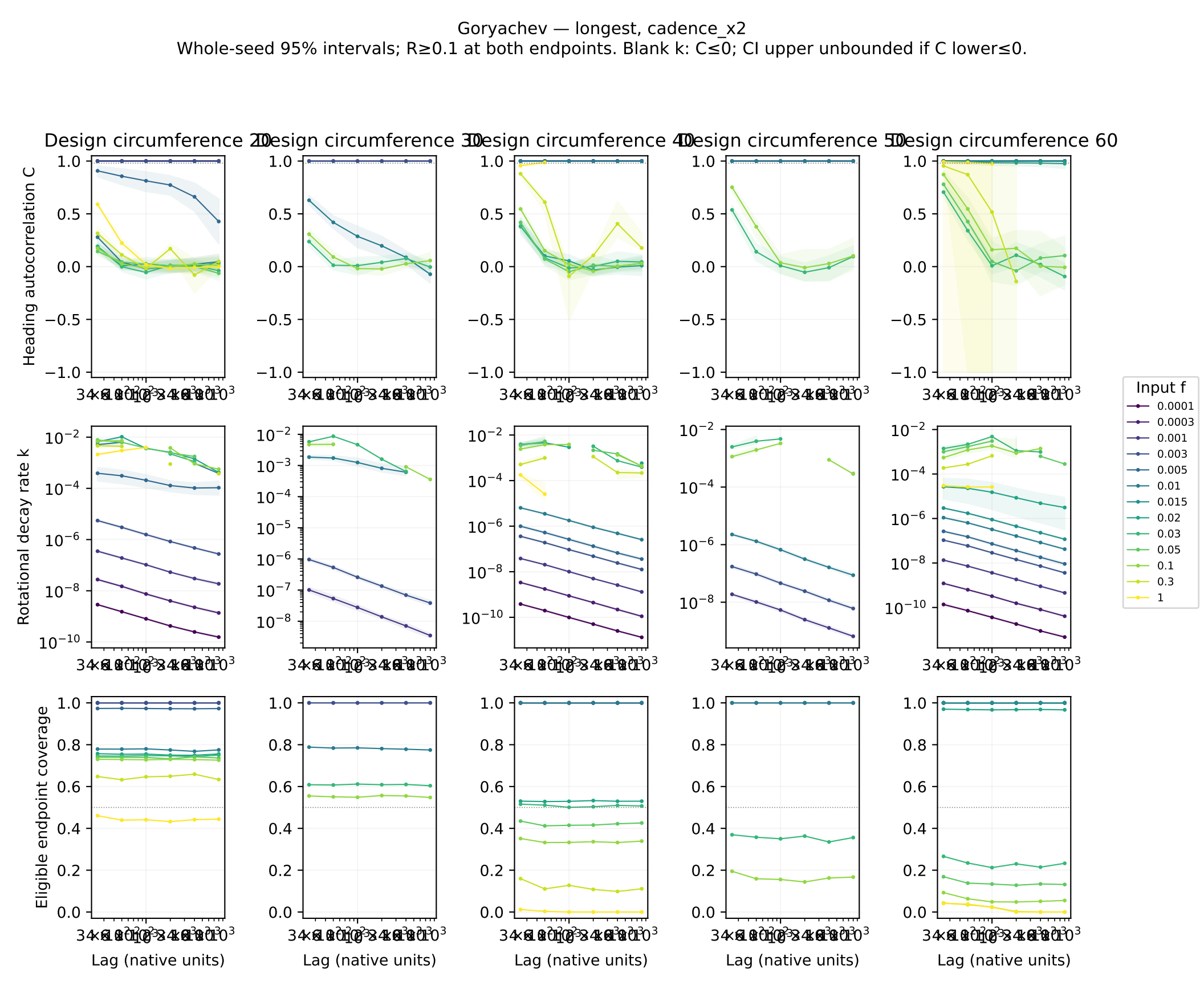

Wave-pinning — longest, late_quarter
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

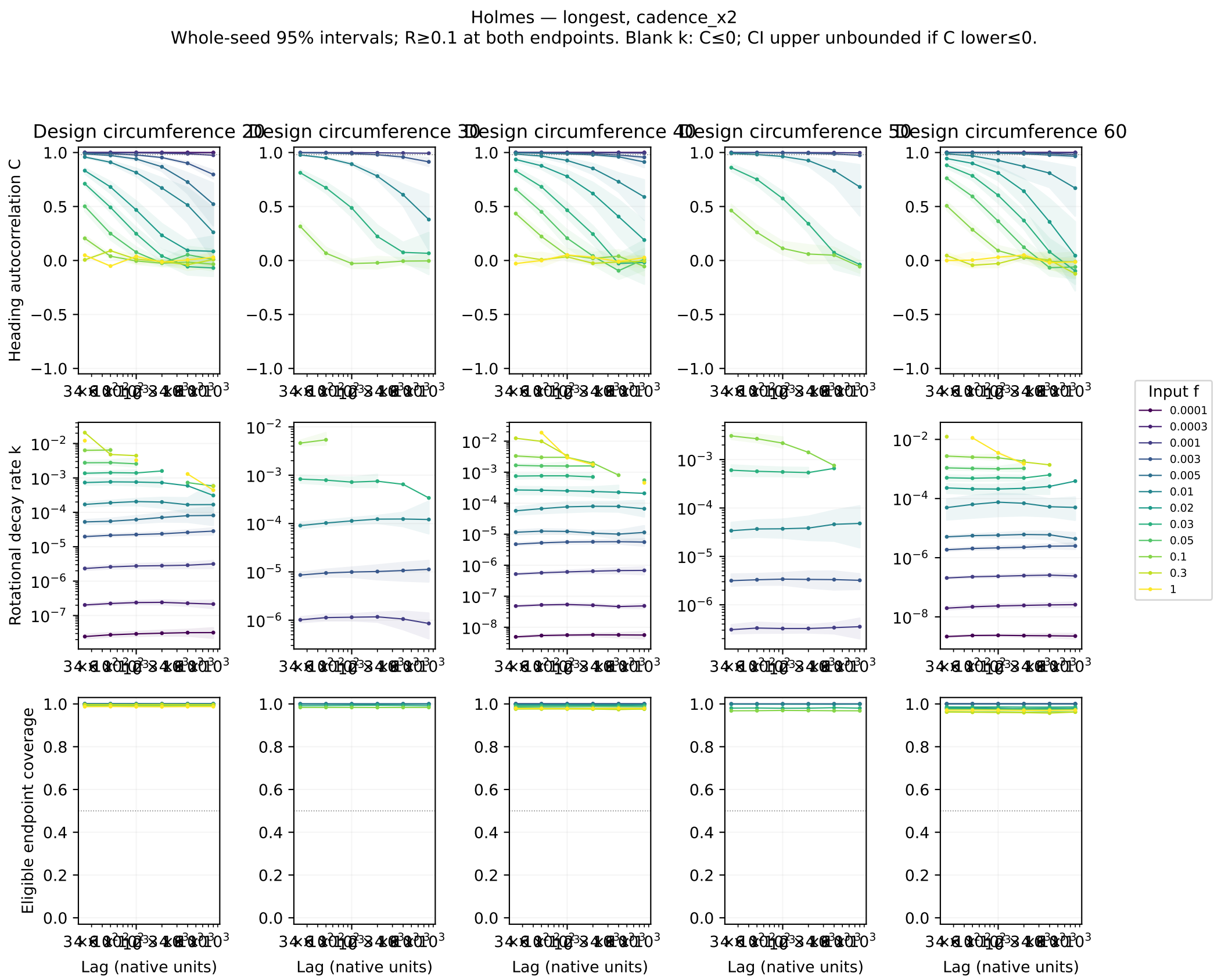

Goryachev — longest, late_quarter
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1

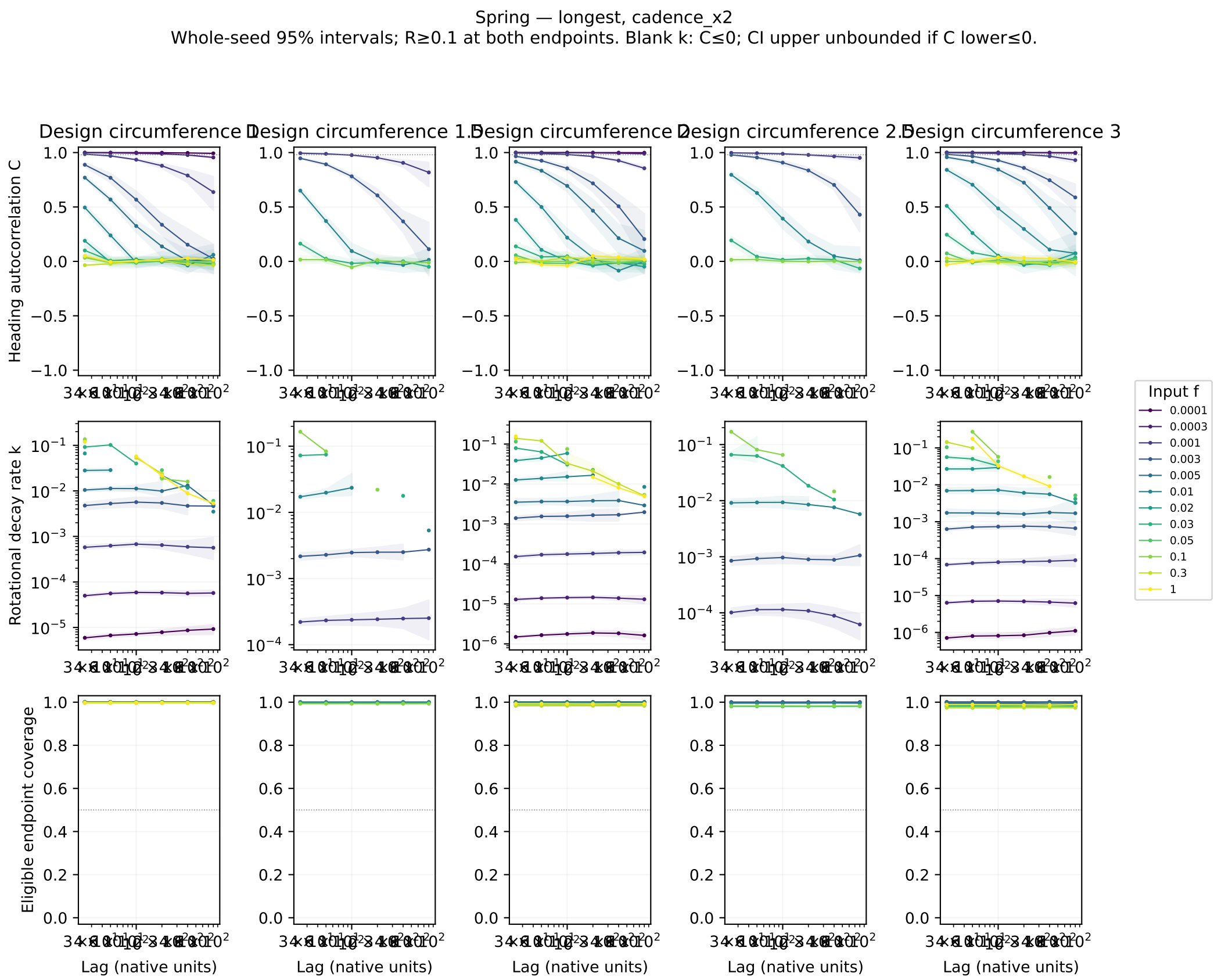

Holmes — longest, late_quarter
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

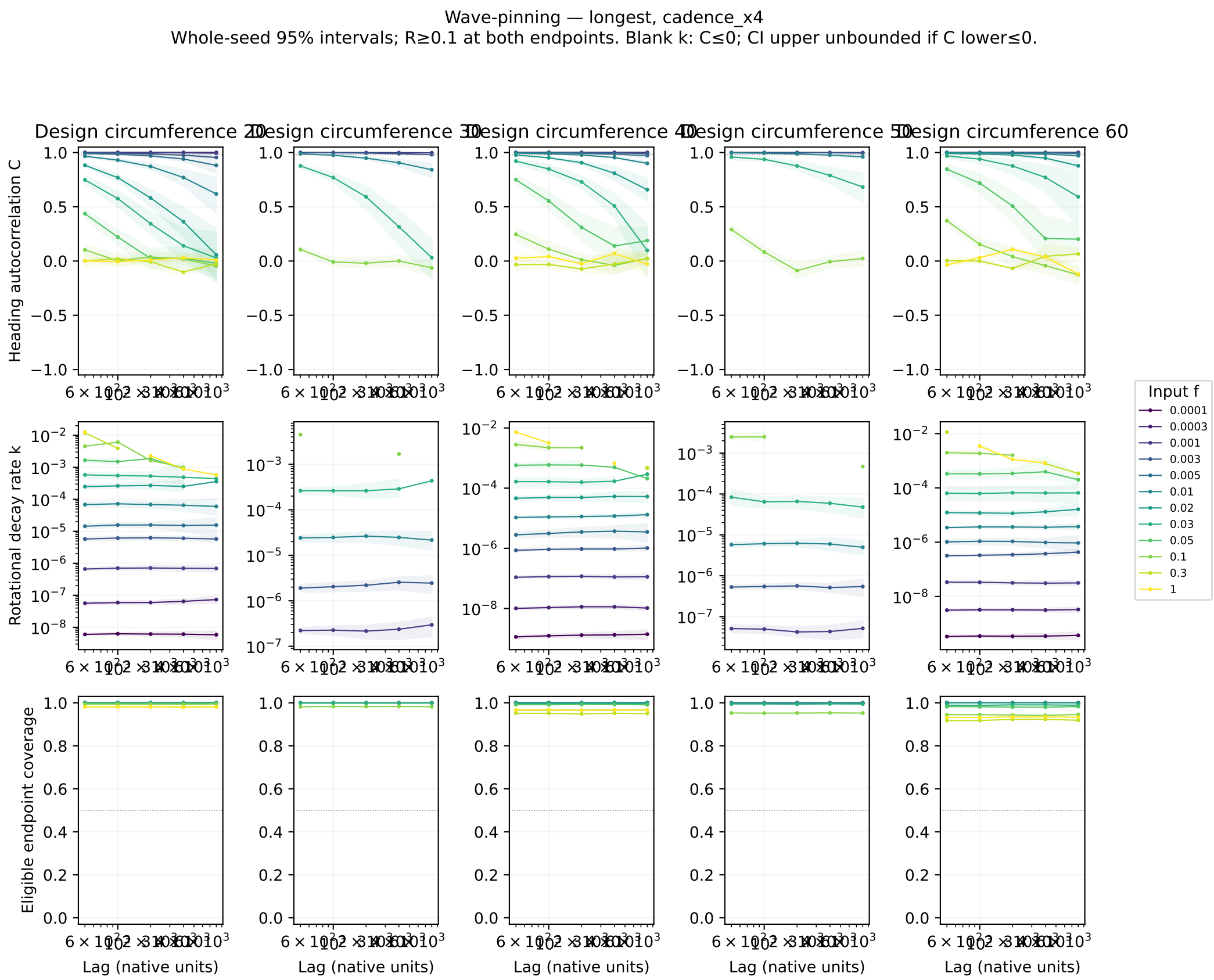

Spring — longest, late_quarter
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

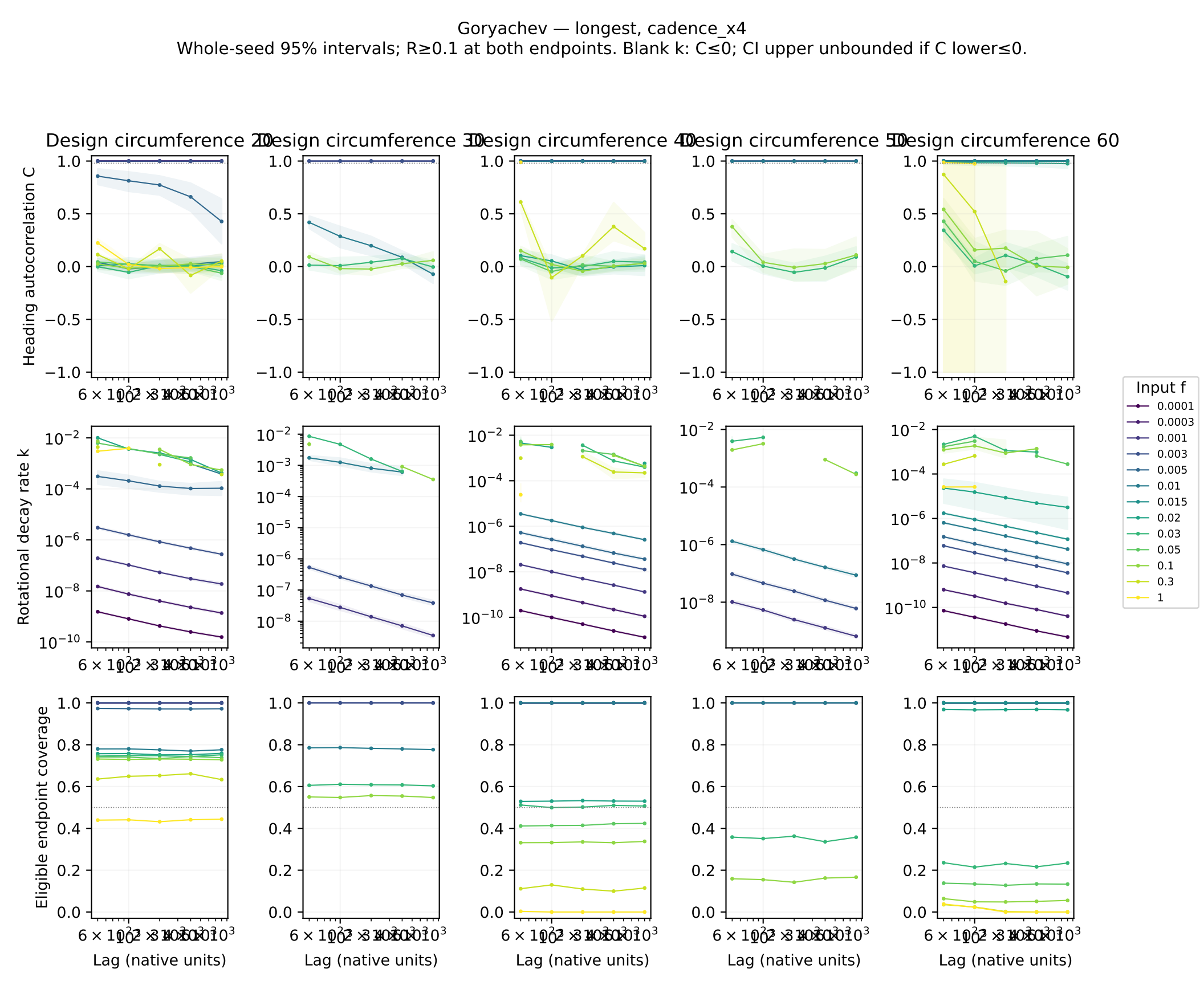

Wave-pinning — longest, cadence_x2
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

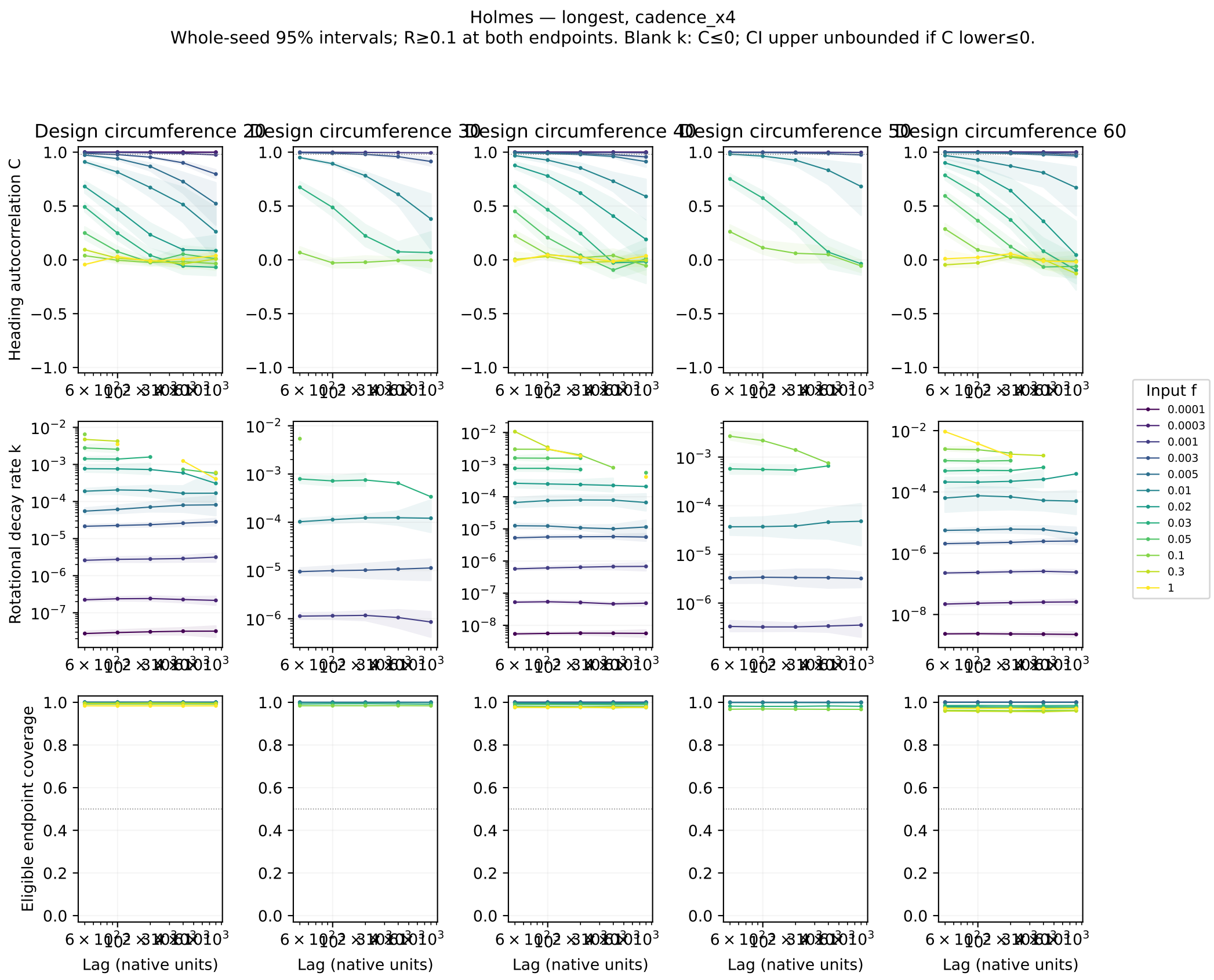

Goryachev — longest, cadence_x2
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1

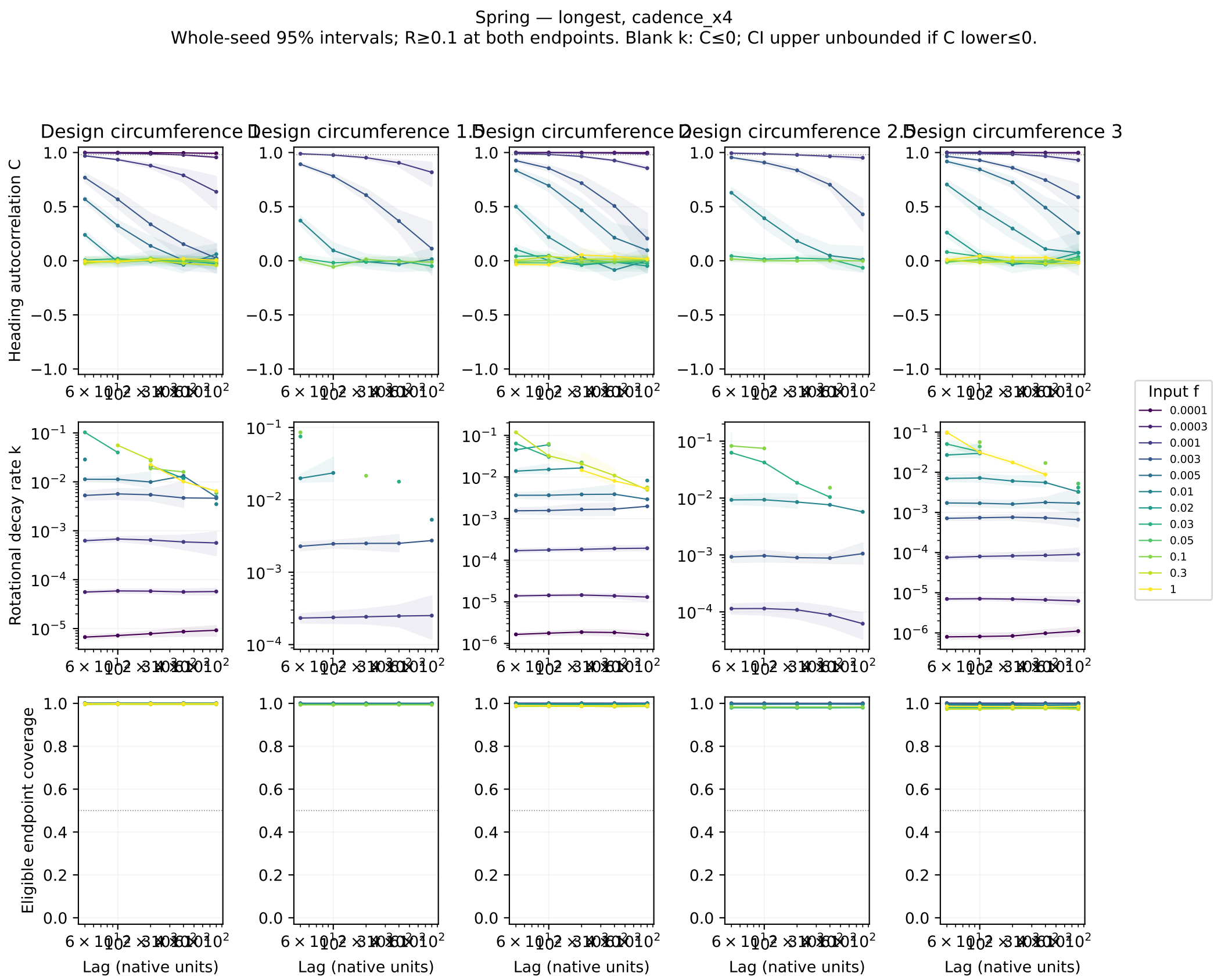

Holmes — longest, cadence_x2
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

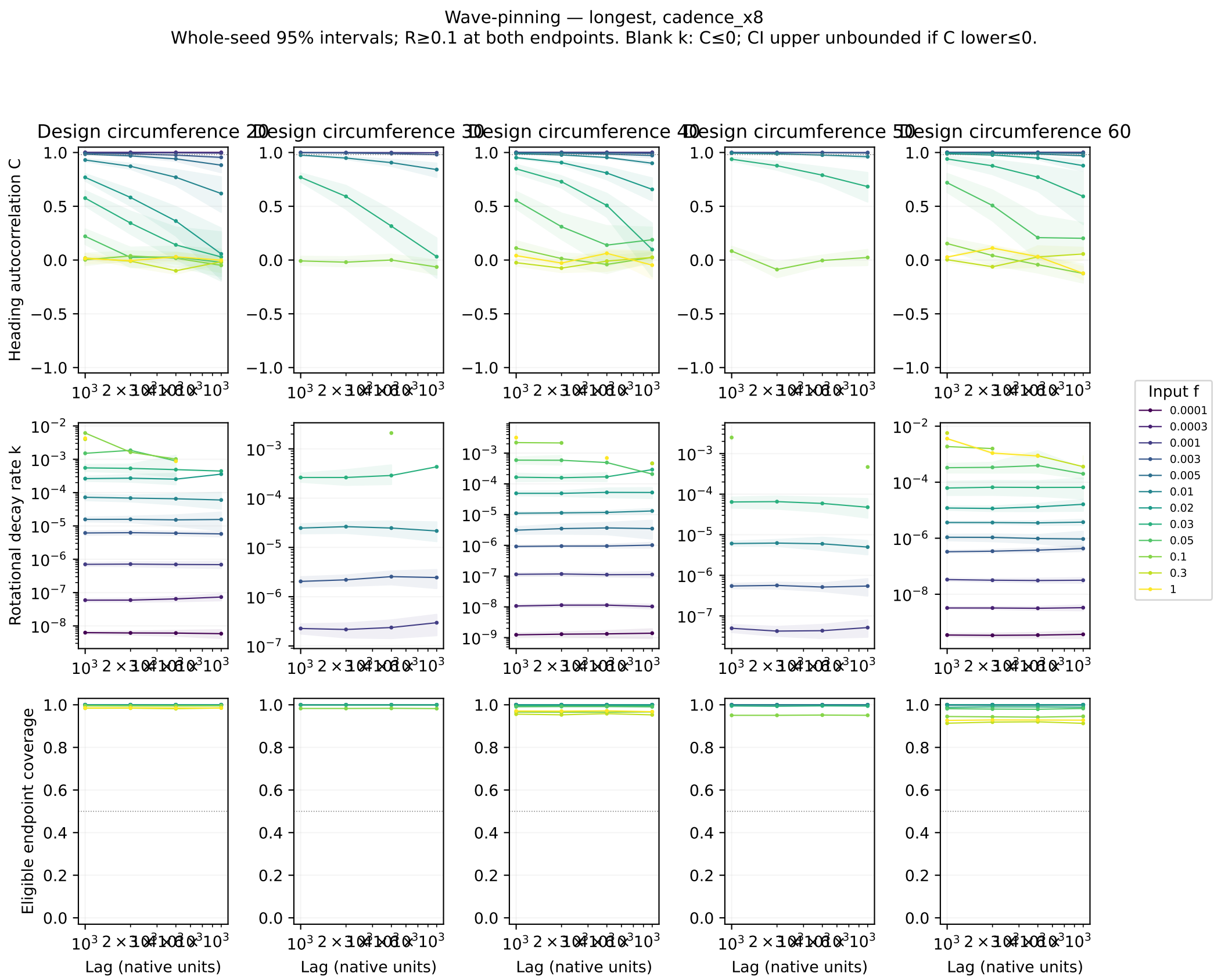

Spring — longest, cadence_x2
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

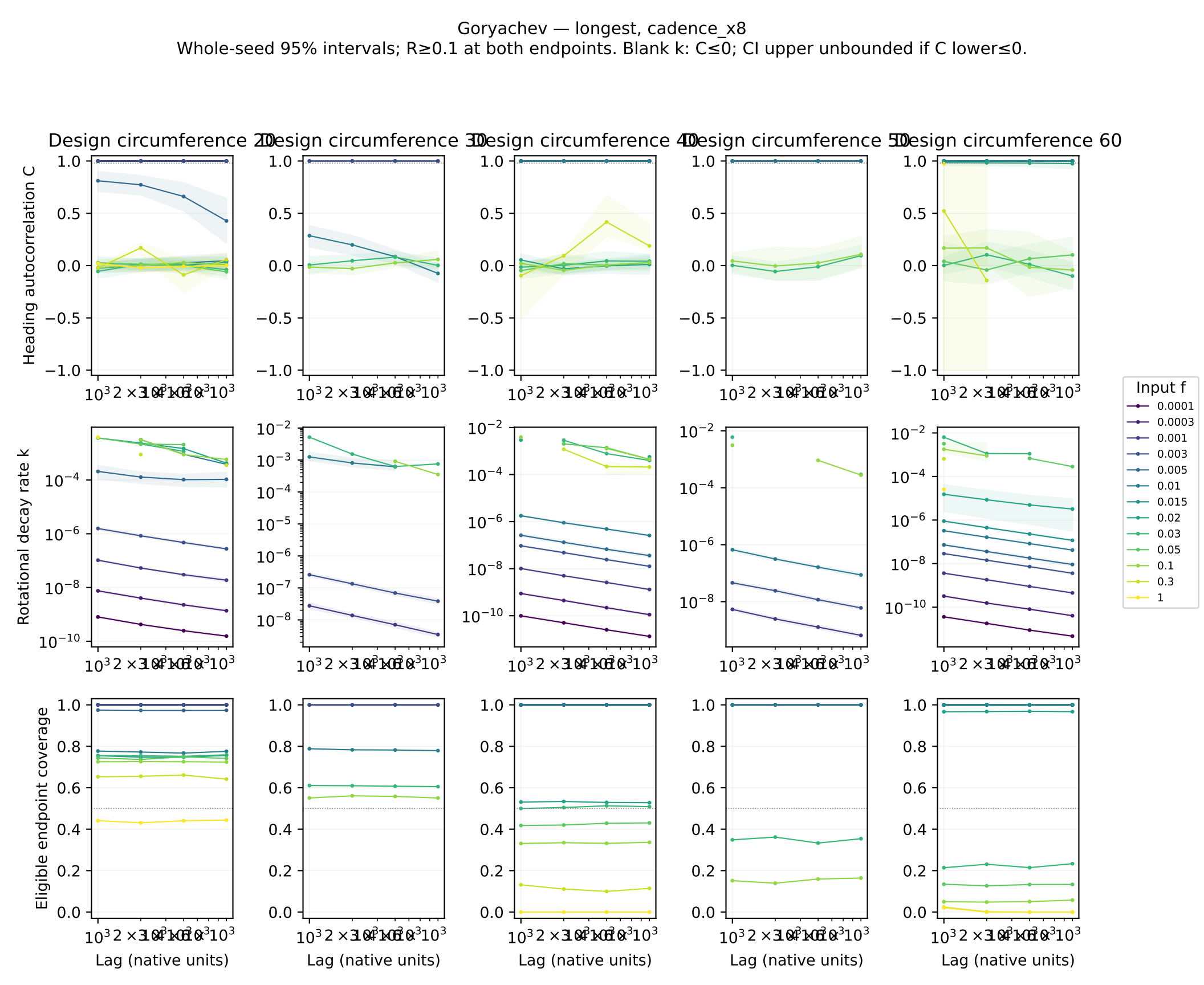

Wave-pinning — longest, cadence_x4
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

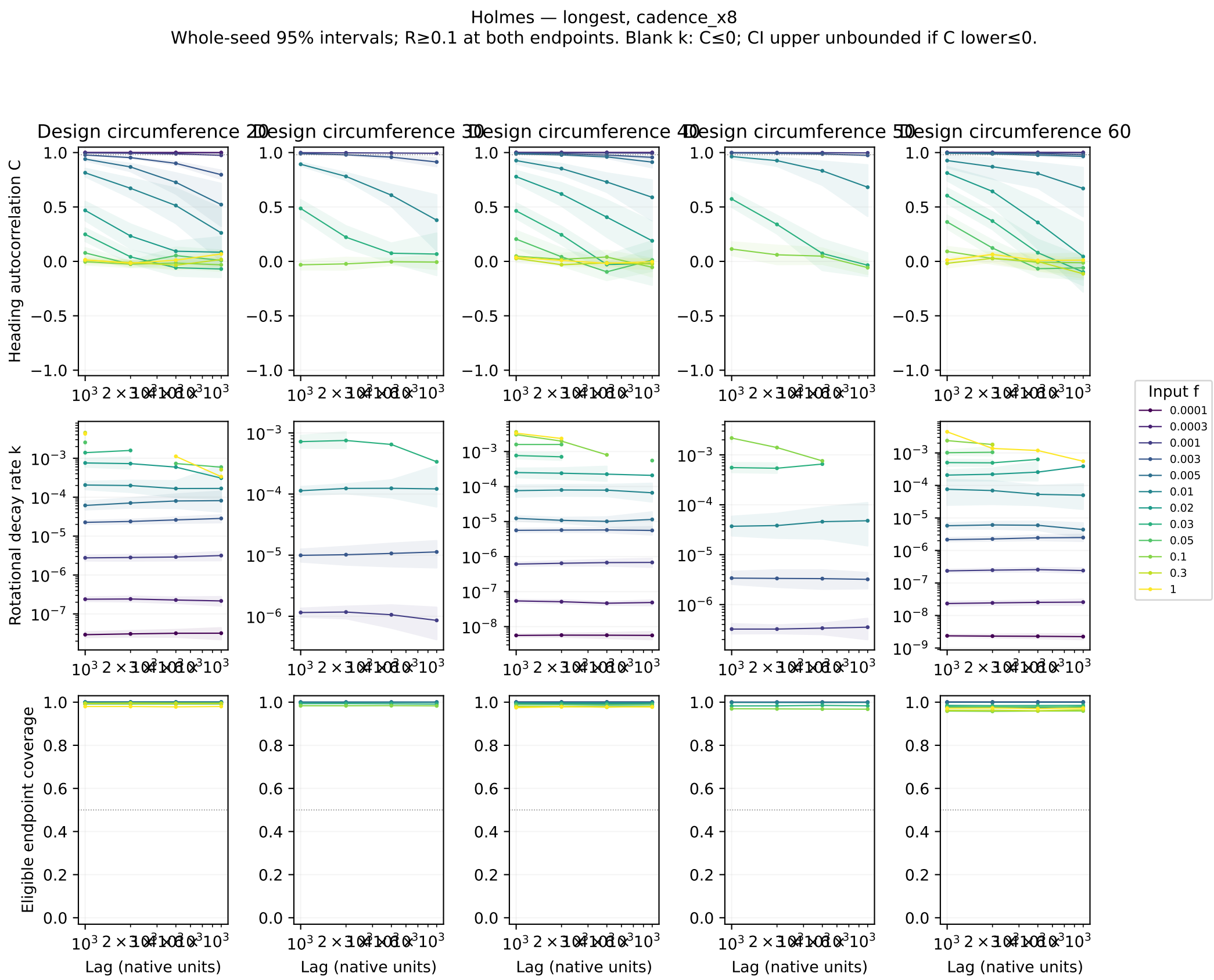

Goryachev — longest, cadence_x4
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1

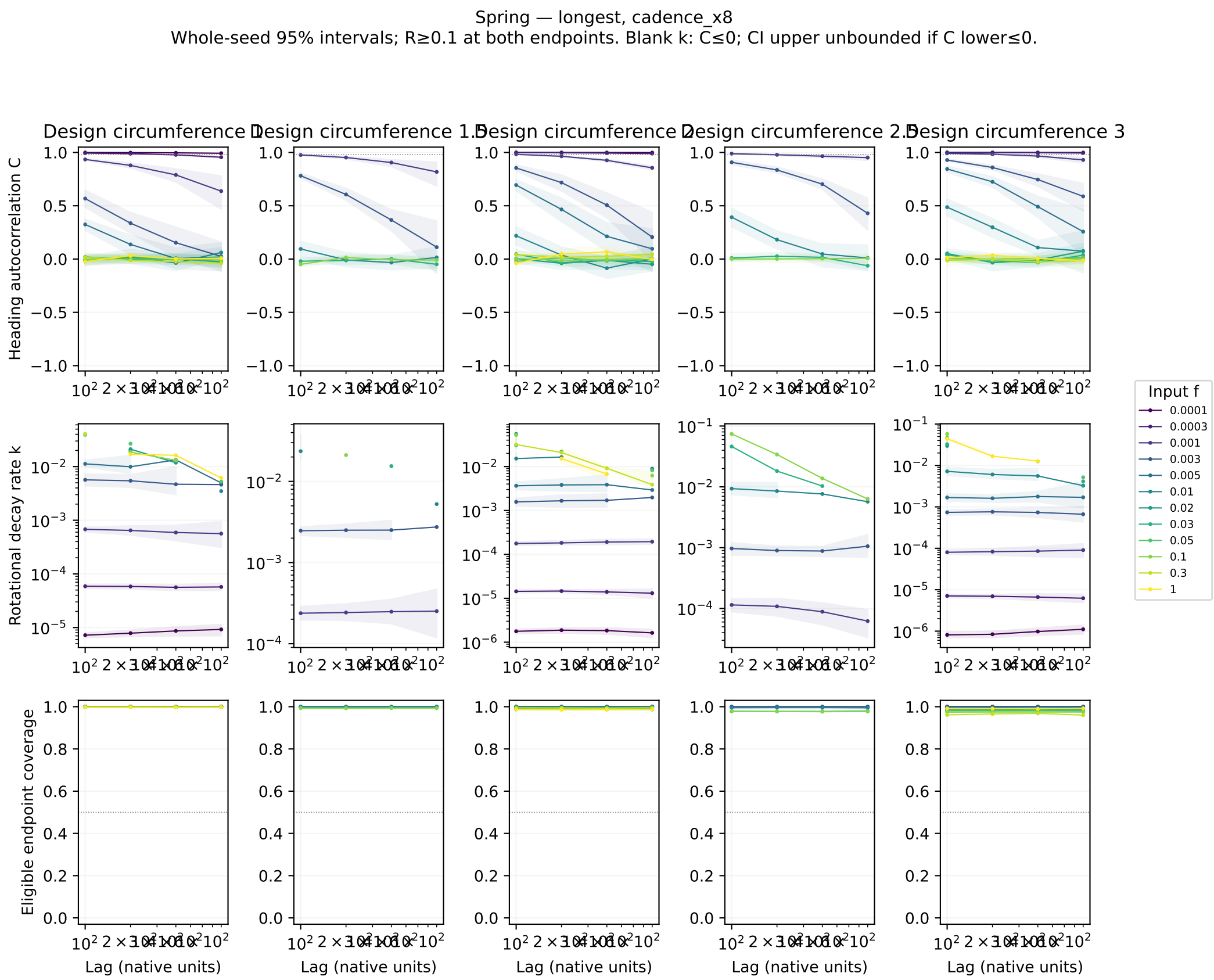

Holmes — longest, cadence_x4
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1

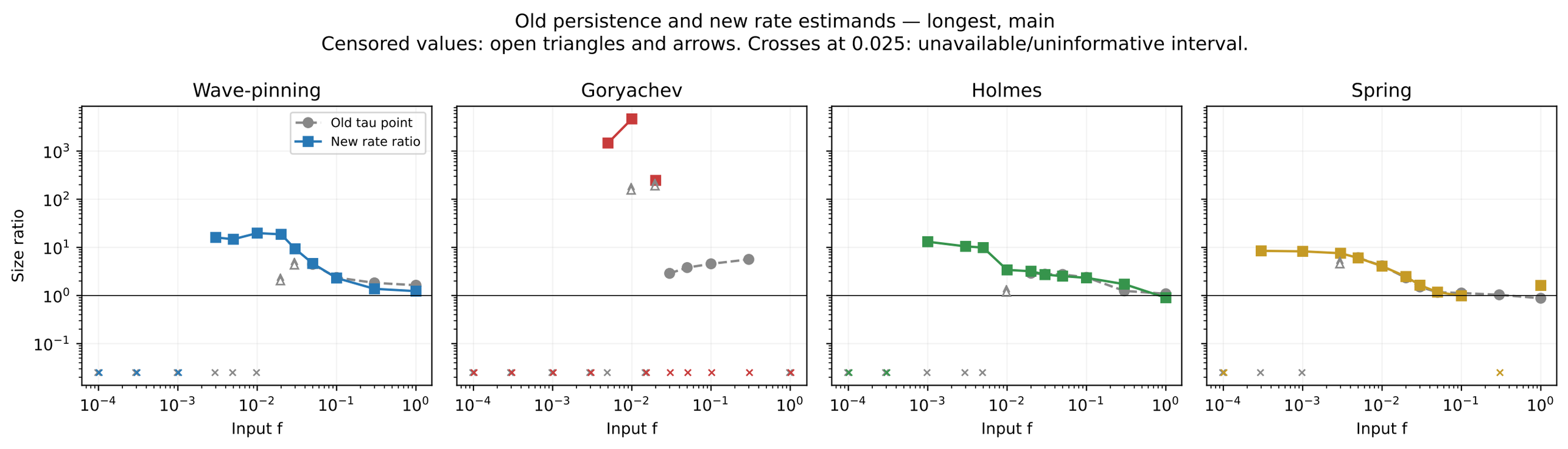

Spring — longest, cadence_x4
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1


Wave-pinning — longest, cadence_x8
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1


Goryachev — longest, cadence_x8
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.015
0.02
0.03
0.05
0.1
0.3
1


Holmes — longest, cadence_x8
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 20
Design circumference 30
Design circumference 40
Design circumference 50
Design circumference 60
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1


Spring — longest, cadence_x8
Whole-seed 95% intervals; R≥0.1 at both endpoints. Blank k: C≤0; CI upper unbounded if C lower≤0.
Design circumference 1
Design circumference 1.5
Design circumference 2
Design circumference 2.5
Design circumference 3
Heading autocorrelation C
Rotational decay rate k
Eligible endpoint coverage
Lag (native units)
Input f
0.0001
0.0003
0.001
0.003
0.005
0.01
0.02
0.03
0.05
0.1
0.3
1


Old persistence and new rate estimands — longest, main
Censored values: open triangles and arrows. Crosses at 0.025: unavailable/uninformative interval.
Wave-pinning
Goryachev
Holmes
Spring
Old tau point
New rate ratio
Size ratio
Input f

Heading persistence · f = 0.05

